Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\boldsymbol{x}_i\}_{i=1}^n$ where $\boldsymbol{x}_i \in \mathbb{R}^p$ for all $i = 1, \ldots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\boldsymbol{x}_i$'s as realizations of independent random variables $\boldsymbol{X}_i = \boldsymbol{W}\boldsymbol{Z}_i + \boldsymbol{\epsilon_i}$, where $\boldsymbol{W}$ is an unknown $p \times q$ loading matrix of full rank, $\boldsymbol{Z}_i \sim N_q(\mathbf{0}, \boldsymbol{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon_i} \sim N_p(\mathbf{0}, \sigma^2 \boldsymbol{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n, \boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\boldsymbol{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ. Θ is thus the collection of all tuples $(\boldsymbol{W}, \sigma^2)$ such that $\boldsymbol{W}$ is a real $p \times q$ matrix with rank$(\boldsymbol{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \left\{ N_p\left(\mathbf{0}, \boldsymbol{W}_\theta \boldsymbol{W}_\theta^T + \sigma_\theta^2 \boldsymbol{I}_p\right) \right\}_{\theta \in \Theta}. \tag{1.1}$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\boldsymbol{A}, s) \in \Theta$ and $(\boldsymbol{B}, t) \in \Theta$, we define $(\boldsymbol{A}, s) \sim (\boldsymbol{B}, t)$ to mean that they give rise to the same covariance matrix,namely when $\boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p = \boldsymbol{B}\boldsymbol{B}^T + t\boldsymbol{I}_p$. The equivalence classes are denoted by $[(\boldsymbol{A}, s)]$ or more briefly $[\boldsymbol{A}, s]$. Unless otherwise mentioned, $(\boldsymbol{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\boldsymbol{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\boldsymbol{A}, s] = \{(\boldsymbol{A}\boldsymbol{Q}, s) \mid \boldsymbol{Q} \in O(q)\}$. Therefore, $M := \Theta/\sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see Lee, 2013, Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension

$$d = pq + 1 - \frac{q(q-1)}{2} \tag{1.2}$$

and the map $\pi : \Theta \to M$ defined by

$$\pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s] \tag{1.3}$$

is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomertic with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see Lee, 2018, Theorem 2.28). In more concrete terms, if g^Θ is the standard Euclidean metric given by

$$g^\Theta_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right) = \operatorname{tr}\left(\boldsymbol{V}\boldsymbol{U}^T\right) + cb, \tag{1.4}$$

whenever $(\boldsymbol{A}, s) \in \Theta$, $(\boldsymbol{V}, c)$, $(\boldsymbol{U}, b) \in T_{(\boldsymbol{A},s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$, then

$$g_{[\boldsymbol{A},s]}(\boldsymbol{v}, \boldsymbol{h}) = g_{[\boldsymbol{A},s]}\left(d\pi_{(A,s)}((\boldsymbol{V}, c)), d\pi_{(A,s)}((\boldsymbol{U}, b))\right) = g^\Theta_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right), \tag{1.5}$$

whenever $\boldsymbol{v}$, $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ lifting to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ respectively. For later convenience, let us define two functions $\Phi : \Theta \to \mathbb{R}^{p\times p}$ and $\boldsymbol{\Sigma} : M \to \mathbb{R}^{p\times p}$ by setting, for every $[\boldsymbol{A}, s] \in M$:

$$\boldsymbol{\Sigma}([\boldsymbol{A}, s]) := \Phi((\boldsymbol{A}, s)) := \boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p. \tag{1.6}$$

In other words, $\boldsymbol{\Sigma} \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))\}_{[\boldsymbol{A},s]\in M} \tag{1.7}$$

and this is the identifiable model we shall work with.

Finally, thoughout the entire document, our default norm for a matrix $\boldsymbol{A}$ will be the spectral norm, denoted by $\|\boldsymbol{A}\|$. We will also make use of the Frobenius norm $\|\boldsymbol{A}\|_F$ as well. The eigenvalues and the singular values of a matrix will always be labelled in decreasing order. In symbols, if $\boldsymbol{A}$ is a matrix of size $p \times q$, then we denote its q singular values as

$$\|\boldsymbol{A}\| = \sigma_1(\boldsymbol{A}) \geqslant \sigma_2(\boldsymbol{A}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{A}) = \sigma_{\min}(\boldsymbol{A}) \geqslant 0$$

and, if $\boldsymbol{S}$ is a symmetric matrix of size $p \times p$, then we denote its p eigenvalues as

$$\lambda_1(\boldsymbol{S}) \geqslant \lambda_2(\boldsymbol{S}) \geqslant \cdots \geqslant \lambda_p(\boldsymbol{S}).$$

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces.

Proposition 1.

1. The map $\boldsymbol{\Sigma}$ defined in (1.6) is a smooth injective immersion.

2. At every $(\boldsymbol{A}, s) \in \Theta$, the verticle tangent space

$$V_{(\boldsymbol{A},s)} := \ker(\mathrm{d}\pi_{(\boldsymbol{A},s)}) = \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)}) \tag{1.8}$$

   equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

3. At every $(\boldsymbol{A}, s) \in \Theta$, the horizontal tangent space $H_{(\boldsymbol{A},s)} := V^{\perp}_{(\boldsymbol{A},s)}$ equals to the collection of all tangent vectors of the form $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $p \times q$ matrices $\boldsymbol{V}$ satisfying $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$.

Proof. Fix a point $(\boldsymbol{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$, where Φ is given in (1.6). For $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ by setting $\gamma(t) = (\boldsymbol{A} + t\boldsymbol{V}, s + tc)$. Then:

$$\mathrm{d}\Phi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c)) = \frac{d}{dt}\bigg|_{t=0} \Phi(\gamma(t)) = \boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p.$$

If $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$, then

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p = \boldsymbol{0}. \tag{1.9}$$

If $\boldsymbol{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\boldsymbol{A}$, then $\boldsymbol{A}^T\boldsymbol{u} = \boldsymbol{0}$ and

$$0 = \boldsymbol{u}^T(\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p)\boldsymbol{u} = c\,\boldsymbol{u}^T\boldsymbol{u}.$$

Given that $\boldsymbol{u} \neq \boldsymbol{0}$, we must have $c = 0$ and (1.9) reduces to

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T = \boldsymbol{0}. \tag{1.10}$$

Apply $\boldsymbol{u}$ to the right to get:

$$\boldsymbol{A}\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Given that $\boldsymbol{A}$ has full rank, rank nullity theorem forces

$$\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Therefore, we have proven that every vector $\boldsymbol{u}$ orthgonal to columns vectors of $\boldsymbol{A}$ will also be orthgonal to column vectors of $\boldsymbol{V}$. This means $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ for some $q \times q$ matrix $\boldsymbol{\Omega}$. Substitute this back to (1.10) gives

$$\boldsymbol{A}(\boldsymbol{\Omega}^T + \boldsymbol{\Omega})\boldsymbol{A}^T = \boldsymbol{0}.$$

Applying $(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T$ to the left and $\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A})^{-1}$ to the right yields $\boldsymbol{\Omega}^T + \boldsymbol{\Omega} = \boldsymbol{0}$. Therefore, $\boldsymbol{\Omega}$ is skew-symmetric. Conversely, (1.10) is satisfied by $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ with any skew-symmetric matrix $\boldsymbol{\Omega}$. We have proven that $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$ equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.8). Clearly, $\boldsymbol{\Sigma}$ is well-defined and is injective. The smoothness of $\boldsymbol{\Sigma}$ is guaranteed by relation $\boldsymbol{\Sigma} \circ \pi = \Phi$ (see for example Lee, 2013, Theorem 4.29). By the chain rule, we have

$$\mathrm{d}\Phi_{(\boldsymbol{A},s)} = \mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]} \circ \mathrm{d}\pi_{(\boldsymbol{A},s)}. \tag{1.11}$$

It follows immediately that $\ker(\mathrm{d}\pi_{(\boldsymbol{A},s)}) \subseteq \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$. The other direction will follow if we can prove that $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective.

Suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ is such that $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = 0$. We prove that $\boldsymbol{v} = \boldsymbol{0}$. Let $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ be the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$, meaning that $(\boldsymbol{V}, c)$ is the unique horizontal tangent vector at $(\boldsymbol{A}, s)$ such that $\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$. (1.11) then implies that $(\boldsymbol{V}, c) \in \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ for some skew-symmetric $\boldsymbol{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ defined by $\gamma(t) = (\boldsymbol{A}e^{t\boldsymbol{\Omega}}, s)$. Clearly $\gamma(0) = (\boldsymbol{A}, s)$ and $\gamma'(0) = (\boldsymbol{A}\boldsymbol{\Omega}, 0)$. Moreover, the matrix $e^{t\boldsymbol{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\boldsymbol{\Omega}$ so that $\pi(\gamma(t)) = [\boldsymbol{A}, s]$ for all t. We have

$$\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{A}\boldsymbol{\Omega}, 0)) = \mathrm{d}\pi_{(\boldsymbol{A},s)}(\gamma'(0)) = \left.\frac{d}{dt}\right|_{t=0} \pi(\gamma(t)) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M.$$

Therefore, $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective and the verticle tangent space is the same as $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$. Since $[\boldsymbol{A}, s]$ is arbitrary, $\boldsymbol{\Sigma}$ is an immersion.

To compute the horizontal tangent space, any $(\boldsymbol{V}, c) \in V^{\perp}_{(\boldsymbol{A},s)}$ will satisfy

$$0 = g^{\Theta}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{A}\boldsymbol{\Omega}, 0)\right) = \mathrm{tr}\big(\boldsymbol{A}^T\boldsymbol{V}\boldsymbol{\Omega}\big) \tag{1.12}$$

for all skew-symmetric $\boldsymbol{\Omega}$. Decomposing $\boldsymbol{A}^T\boldsymbol{V} = \boldsymbol{S} + \boldsymbol{K}$, where $\boldsymbol{S}$ is the symmetric part and $\boldsymbol{K}$ is the skew-symmetric part, we find out that

$$0 = \mathrm{tr}(\boldsymbol{S}\boldsymbol{\Omega}) + \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}) = 0 + \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}) = \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}).$$

Setting $\boldsymbol{\Omega} = \boldsymbol{K}^T$ will yield that $\boldsymbol{K} = \boldsymbol{0}$. Therefore, $\boldsymbol{A}^T\boldsymbol{V}$ is a symmetric matrix. Conversely, any $\boldsymbol{V}$ such that $\boldsymbol{A}^T\boldsymbol{V}$ is symmetric will satisfy (1.12). The proof is complete. $\square$

We now move to compute the exponential map and the logarithmic map. The following lemma is crucial for computing the orthogonal matrix based on given points in M.

Lemma 2. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix and $\boldsymbol{Q} \in O(q)$ are such that $\boldsymbol{LQ}$ is symmetric, then there exists matrices $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$, a $q \times q$ diagonal matrix $\boldsymbol{D}$ whose diagonal entries are 1 or -1, and a $q \times q$ diagonal matrix $\boldsymbol{S}$ with diagonal entries $\sigma_1(\boldsymbol{L}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{L}) > 0$ such that $\boldsymbol{L} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$.

If we add furthermore the assumption that $\boldsymbol{L}$ is symmetric positive definite, then we can achieve the results mentioned above with $\boldsymbol{P} = \boldsymbol{R}$. In particular, this implies $\boldsymbol{Q} = \boldsymbol{Q}^T$ and $\|\boldsymbol{Q} - \boldsymbol{I}_q\| = \|\boldsymbol{D} - \boldsymbol{I}_q\|$ can take only 0 or 2 as values.

Proof. Let us start by applying the singular value decomposition of $\boldsymbol{L}$ to get matrices $\boldsymbol{R}' \in O(q)$, $\boldsymbol{P}' \in O(q)$ and a diagonal matrix $\boldsymbol{S}$ with diagonal entries $\sigma_1(\boldsymbol{L}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{L}) > 0$ such that $\boldsymbol{L} = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T$. Note that all singular values are positive because $\boldsymbol{L}$ is invertible. Substituting the given condition that $\boldsymbol{LQ}$ is symmetric will yield that $\boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$, where $\boldsymbol{J} := (\boldsymbol{P}')^T\boldsymbol{Q}\boldsymbol{R}'$.

Clearly $\boldsymbol{J} \in O(q)$. The equation

$$\boldsymbol{S}^2 = \boldsymbol{SJJ}^T\boldsymbol{S} = (\boldsymbol{SJ})(\boldsymbol{SJ})^T = (\boldsymbol{J}^T\boldsymbol{S})(\boldsymbol{J}^T\boldsymbol{S})^T = \boldsymbol{J}^T\boldsymbol{S}^2\boldsymbol{J}$$

shows that

$$\boldsymbol{JS}^2 = \boldsymbol{S}^2\boldsymbol{J}.$$

Looking at each entry of both side, we end up with

$$\boldsymbol{J}_{ij}(\sigma_j(\boldsymbol{L})^2 - \sigma_i(\boldsymbol{L})^2) = 0$$

for $i, j = 1, \ldots q$. Given that all singular values are positive, we in fact have

$$\boldsymbol{J}_{ij}(\sigma_j(\boldsymbol{L}) - \sigma_i(\boldsymbol{L})) = 0 \tag{1.13}$$

for all i and all j so that $\boldsymbol{JS} = \boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$. Inverting $\boldsymbol{S}$ yields that $\boldsymbol{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\boldsymbol{J}^2 = \boldsymbol{I}_q$ and the eigenvalues of $\boldsymbol{J}$ must be 1 or -1. Another look at (1.13) shows that $\boldsymbol{J}$ is a block-diagonal matrix, diagonal if and only if all singular values are distinct. Furthermore, each block of $\boldsymbol{J}$ must still be orthogonal and symmetric.

We now diagonalize $\boldsymbol{J}$ based on the above information. Suppose that $\boldsymbol{S} = \oplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{I}_{m_i}$, where $\boldsymbol{s}_1 > \ldots > \boldsymbol{s}_k$ are the unique diagonal entries of $\boldsymbol{S}$ and each repeats for m_i times. Thus, $\boldsymbol{J}$ has k blocks. Write the ith block by $\boldsymbol{J}_i$ so that $\boldsymbol{J} = \oplus_{i=1}^k \boldsymbol{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\boldsymbol{J}_i = \boldsymbol{K}_i\boldsymbol{D}_i\boldsymbol{K}_i^T$, where $\boldsymbol{K}_i \in O(m_i)$ and $\boldsymbol{D}_i$ is an $m_i \times m_i$ diagonal matrix with all diagonal entries being 1 or -1. Put $\boldsymbol{K} = \oplus_{i=1}^k \boldsymbol{K}_i$ and $\boldsymbol{D} = \oplus_{i=1}^k \boldsymbol{D}_i$. Then $\boldsymbol{K} \in O(q)$, $\boldsymbol{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\boldsymbol{J} = \boldsymbol{KDK}^T$. This involved construction leads us to the outcome that $\boldsymbol{K}$ in fact commutes with $\boldsymbol{S}$:

$$\boldsymbol{KS} = \bigoplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{K}_i = \boldsymbol{SK}.$$

Put $\boldsymbol{P} = \boldsymbol{P}'\boldsymbol{K}$, $\boldsymbol{R} = \boldsymbol{R}'\boldsymbol{K}$. We have

$$\boldsymbol{Q} = \boldsymbol{P}'\boldsymbol{J}(\boldsymbol{R}')^T = \boldsymbol{P}'\boldsymbol{KDK}^T(\boldsymbol{R}')^T = \boldsymbol{PDR}^T$$

and

$$\boldsymbol{RSP}^T = \boldsymbol{R}'\boldsymbol{KSK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{SKK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T = \boldsymbol{L}.$$

This completes the proof.

Finally, let us just note that, in case $\boldsymbol{L}$ is symmetric positive definite, the spectral theorem allows us to take $\boldsymbol{R}' = \boldsymbol{P}'$ from the start and it will yield in the end $\boldsymbol{L} = \boldsymbol{PSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDP}^T$. It then follows easily that $\boldsymbol{Q}^2 = \boldsymbol{I}_q$ and $\boldsymbol{Q}^T = \boldsymbol{Q}$. If $\boldsymbol{D} = \boldsymbol{I}_q$, then $\|\boldsymbol{Q} - \boldsymbol{I}_q\| = \|\boldsymbol{D} - \boldsymbol{I}_q\| = 0$. If any entry of $\boldsymbol{D}$ is -1, $\|\boldsymbol{D} - \boldsymbol{I}_q\| = 2$. $\square$

We now state and prove all the results about the exponential and the logarithmic map in one single proposition.

Proposition 3. Given $[\boldsymbol{A}, s] \in M$ and $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$, geodesics in M starting at $[\boldsymbol{A}, s]$ with initial velocity $\boldsymbol{v}$ takes the format of curves $\gamma : I \to M$, where I is an open interval containing 0, defined by $\gamma(t) = [\boldsymbol{A} + t\boldsymbol{V}, s + tc]$, where $(\boldsymbol{V}, c) \in H_{(\boldsymbol{A},s)}$ is the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$. Consequently, the exponential map is computed as

$$\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A} + \boldsymbol{V}, s + c]. \tag{1.14}$$

Let us use $\mathcal{E}_{[\boldsymbol{A},s]}$ to denote the open ball in $T_{[\boldsymbol{A},s]}M \cong \mathbb{R}^d$ centred at $\mathbf{0}$ of radius $\min(\sigma_{\min}(\boldsymbol{A}), s)$. Given that (1.14) is well-defined at least over all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, we from now on treat $\mathcal{E}_{[\boldsymbol{A},s]}$ as the domain of the exponential map. That is, $\operatorname{Exp}_{[\boldsymbol{A},s]} : \mathcal{E}_{[\boldsymbol{A},s]} \to U_{[\boldsymbol{A},s]}$, where $U_{[\boldsymbol{A},s]} := \operatorname{Exp}_{[\boldsymbol{A},s]}(\mathcal{E}_{[\boldsymbol{A},s]})$ is an open subset of M, is defined by (1.14). Note the following facts:

1. For every $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, $\boldsymbol{A}^T(\boldsymbol{A} + \boldsymbol{V})$ is symmetric positive definite.

2. $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is a diffeomorphism.

3. If $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$, then for every representative $(\boldsymbol{A}, s)$ of $[\boldsymbol{A}, s]$ and every representative $(\boldsymbol{B}, t)$ of $[\boldsymbol{B}, t]$, the matrix $\boldsymbol{A}^T\boldsymbol{B}$ will always be invertible. Given representative $(\boldsymbol{A}, s)$ of $[\boldsymbol{A}, s]$, *some* representative $(\boldsymbol{B}, t)$ of $[\boldsymbol{B}, t]$ will furthermore make $\boldsymbol{A}^T\boldsymbol{B}$ symmetric positive definite.

4. The logarithmic map $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]} : U_{[\boldsymbol{A},s]} \to \mathcal{E}_{[\boldsymbol{A},s]}$ is computed as follows: for every $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$,

$$\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}, t - s)), \tag{1.15}$$

   where

$$\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) = (\boldsymbol{B}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A} \in O(q). \tag{1.16}$$

5. $U_{[\boldsymbol{A},s]}$ equals to the set of all $[\boldsymbol{B}, t] \in M$ such that

$$\|\boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}\|_F^2 + (t - s)^2 < \min\left(\sigma_{\min}(\boldsymbol{A}), s\right)^2. \tag{1.17}$$

Proof. Before we start the proof, let us first check that all the statements are indeed well-defined. First of all, the curve γ is well defined. If $(\boldsymbol{A}\boldsymbol{Q}, s)$, where $\boldsymbol{Q} \in O(q)$, is another representative of $[\boldsymbol{A}, s]$, let $(\boldsymbol{V}', c')$ be the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{A}\boldsymbol{Q}, s)$. With the function $R_{\boldsymbol{Q}} : \Theta \to \Theta$ defined by $R_{\boldsymbol{Q}}((\boldsymbol{A}, s)) = (\boldsymbol{A}\boldsymbol{Q}, s)$, we have

$$\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}(\pi \circ R_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}\pi_{(\boldsymbol{A}\boldsymbol{Q},s)}(dR_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}\pi_{(\boldsymbol{A}\boldsymbol{Q},s)}(\boldsymbol{V}\boldsymbol{Q}, c).$$

Thus, $\boldsymbol{V}' = \boldsymbol{V}\boldsymbol{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve. Secondly, if $\boldsymbol{R} \in O(q)$ and $\boldsymbol{P} \in O(q)$, then

$$\boldsymbol{Q}(\boldsymbol{A}\boldsymbol{P}, \boldsymbol{B}\boldsymbol{R}) = \boldsymbol{R}^T\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B})\boldsymbol{P}. \tag{1.18}$$

Therefore, the same compuatation will show that (1.15) is indeed well-defined. Also, the contents of the inequality (1.17) is independent of representatives and the description of the set $U_{[\boldsymbol{A},s]}$ makes sense.

A well-known property of geodesics in case of Riemannian submersion (see for example O'Neill, 1967, Section 2 Corollary 2 or Valencia, 2021, Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M. Therefore, (1.14) is valid for all vectors $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ of small enough norm making sure that $(\boldsymbol{A} + \boldsymbol{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we use the classical result of Weyl's perturbation inequality (see for example Chapter 7.3 Problem 16 equation (7.3.15) of Horn and Johnson, 2013) to get

$$\sigma_{\min}(\boldsymbol{A} + \boldsymbol{V}) \geqslant \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\| \geqslant \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_F \geqslant \sigma_{\min}(\boldsymbol{A}) - |\boldsymbol{v}| > 0,$$

whenever $|\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$ or equivalently $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$. We have made use of the two facts that $\|\boldsymbol{V}\| \leqslant \|\boldsymbol{V}\|_F$ for any matrix $\boldsymbol{V}$ and that $|\boldsymbol{v}| = \sqrt{\|\boldsymbol{V}\|_F^2 + c^2}$ by isometry. Thus, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{V}) = q$ and the exponential map is well-defined.

We now start the proof of the five facts of the function $\operatorname{Exp}_{[\boldsymbol{A},s]} : \mathcal{E}_{[\boldsymbol{A},s]} \to U_{[\boldsymbol{A},s]}$.

The first fact can be quickly settled. Let $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ with horizontal lift $(\boldsymbol{V}, c)$ to $(\boldsymbol{A}, s)$ be given. As proved in Proposition 1, $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$ and thus $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is symmetric. If $\boldsymbol{x} \in \mathbb{R}^q$ is any unit vector, then

$$\boldsymbol{x}^T\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{x} = |\boldsymbol{A}\boldsymbol{x}|^2 + (\boldsymbol{A}\boldsymbol{x})^T\boldsymbol{V}\boldsymbol{x} \geqslant |\boldsymbol{A}\boldsymbol{x}|(|\boldsymbol{A}\boldsymbol{x}| - |\boldsymbol{V}\boldsymbol{x}|) \geqslant \sigma_{\min}(\boldsymbol{A})(\sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_F) > 0$$

because $\sigma_{\min}(\boldsymbol{A}) > |\boldsymbol{v}| \geqslant \|\boldsymbol{V}\|_F$.

The second fact has a longer proof. Being a well-defined exponential map, $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is automatically a smooth map. To prove that it is actually a diffeomorphism in $\mathcal{E}_{[\boldsymbol{A},s]}$, we must first show that it is injective in $\mathcal{E}_{[\boldsymbol{A},s]}$ so that the inverse function, the logarithmic map $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}$, exists. Once this is done, we prove that $U_{[\boldsymbol{A},s]}$ is open and that $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}$ is smooth in one shot by proving that its differential $\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}$ is an isomorphism for every given $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$.

We thus move to show that $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is injective in $\mathcal{E}_{[\boldsymbol{A},s]}$. Suppose that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are two vectors in $\mathcal{E}_{[\boldsymbol{A},s]}$ who lifts to $(\boldsymbol{V}_1, c_1)$ and $(\boldsymbol{V}_2, c_2)$ respectively at $(\boldsymbol{A}, s)$ such that $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}_1) = \operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}_2)$. We prove that $\boldsymbol{v}_1 = \boldsymbol{v}_2$. (1.14) immediately shows that $c_1 = c_2$ and

$$\boldsymbol{A} + \boldsymbol{V}_1 = (\boldsymbol{A} + \boldsymbol{V}_2)\boldsymbol{Q}$$

for some $\boldsymbol{Q} \in O(q)$. Lemma 2 with the symmetric positive definite matrix $\boldsymbol{L} = \boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}_2)$ shows that $\|\boldsymbol{Q} - \boldsymbol{I}_q\|$ can only equal to 0 or 2. Given that

$$\sigma_{\min}(\boldsymbol{A})\|\boldsymbol{Q} - \boldsymbol{I}_q\| \leqslant \|\boldsymbol{A}\boldsymbol{Q} - \boldsymbol{A}\| = \|\boldsymbol{V}_1 - \boldsymbol{V}_2\boldsymbol{Q}\| \leqslant \|\boldsymbol{V}_1\| + \|\boldsymbol{V}_2\| < 2\sigma_{\min}(\boldsymbol{A}),$$

we have proved that $\|\boldsymbol{Q} - \boldsymbol{I}_q\| < 2$ and $\boldsymbol{Q} = \boldsymbol{I}_q$. Therefore, $\boldsymbol{V}_1 = \boldsymbol{V}_2$ and $\boldsymbol{v}_1 = \boldsymbol{v}_2$. The exponential map is indeed injective in $\mathcal{E}_{[\boldsymbol{A},s]}$.

Next, let $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ be given. We prove that $\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}$ is an isomorphism because $\ker\left(\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}\right)$ consists only of the zero vector. Suppose that $\boldsymbol{v}$ lifts to $(\boldsymbol{V}, c)$ at $(\boldsymbol{A}, s)$. To make the task notationally feasible, we implicitly canonically identify the Euclidean spaces together. Let us write $l : T_{[\boldsymbol{A},s]}M \to H_{(\boldsymbol{A},s)}$ for the horizontal lift map: $l(\boldsymbol{v}) = (\boldsymbol{V}, c)$, which is a linear isometry. Define a map $F : l(\mathcal{E}_{[\boldsymbol{A},s]}) \to M$ by setting, for any $(\boldsymbol{U}, b) \in l(\mathcal{E}_{[\boldsymbol{A},s]}) \subseteq H_{(\boldsymbol{A},s)}$, that $F((\boldsymbol{U}, b)) = \pi((\boldsymbol{A}+\boldsymbol{U}, s+b))$. In this notation, $\operatorname{Exp}_{[\boldsymbol{A},s]} = F \circ l$ and the chain rule shows that

$$\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}} = \mathrm{d}F_{(\boldsymbol{V},c)} \circ \mathrm{d}l_{\boldsymbol{v}}.$$

Therefore, the only thing left to show is that $\ker\left(\mathrm{d}F_{(\boldsymbol{V},c)}\right)$ contains only the zero vector. For any $(\boldsymbol{U}, b) \in l(\mathcal{E}_{[\boldsymbol{A},s]})$, we always have

$$\mathrm{d}F_{(\boldsymbol{V},c)}((\boldsymbol{U}, b)) = \left.\frac{d}{dt}\right|_{t=0} \pi((\boldsymbol{A}+\boldsymbol{V}+t\boldsymbol{U}, s+c+tb)) = \mathrm{d}\pi_{(\boldsymbol{A}+\boldsymbol{V},s+c)}((\boldsymbol{U}, b)).$$

Therefore, if $(\boldsymbol{U}, b) \in \ker\left(\mathrm{d}F_{(\boldsymbol{V},c)}\right)$, then $(\boldsymbol{U}, b) \in V_{(\boldsymbol{A}+\boldsymbol{V},s+c)}$ so that $b = 0$ and $\boldsymbol{U} = (\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega}$ for some $q \times q$ skew-symmetric matrix $\boldsymbol{\Omega}$. But $(\boldsymbol{U}, b) \in H_{(\boldsymbol{A},s)}$ as well. Thus, $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega} = \boldsymbol{\Omega}^T(\boldsymbol{A}+\boldsymbol{V})^T\boldsymbol{A}$. Given that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is also symmetric, we arrive at

$$\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}) = \boldsymbol{0}.$$

By the spectral theorem, choose $\boldsymbol{P} \in O(q)$ such that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}) = \boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T$ where $\boldsymbol{S}$ is the $q \times q$ diagonal matrix with diagonal entries $\lambda_1(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})), \ldots, \lambda_q(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}))$. Substitute this yields

$$\boldsymbol{S}(\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P}) + (\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P})\boldsymbol{S} = \boldsymbol{0}.$$

Looking at each entry to get

$$(\lambda_i(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})) + \lambda_j(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})))(\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P})_{ij} = 0,$$

for all i and all $j \in \{1,\dots,q\}$. Given that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is positive definite, this forces $\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P} = \boldsymbol{0}$ and hence $\boldsymbol{\Omega} = \boldsymbol{0}$. We have proven that $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a diffeomorphism in $\mathcal{E}_{[\boldsymbol{A},s]}$ and that $U_{[\boldsymbol{A},s]}$ is an open subset of M.

Now that the logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]} : U_{[\boldsymbol{A},s]} \to \mathcal{E}_{[\boldsymbol{A},s]}$ is well-defined and is smooth, we will compute it and show the three last facts in one shot. Let $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$ be given. By (1.14), $[\boldsymbol{B},t] = [\boldsymbol{A}+\boldsymbol{V}, s+c]$, where $(\boldsymbol{V},c)$ is the horizontal lift of $\boldsymbol{v} = \mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t])$ to $(\boldsymbol{A},s)$. It immediately follows that $c = t-s$. The first fact shows that, with $\boldsymbol{B}_0 := \boldsymbol{A}+\boldsymbol{V}$, $\boldsymbol{A}^T\boldsymbol{B}_0$ is symmetric positive definite. Given that the value of $\det(\boldsymbol{A}^T\boldsymbol{B})$ is independent of representatives, $\boldsymbol{A}^T\boldsymbol{B}$ is always invertible for any representative. The third fact is therefore established. Given that $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B}_0) = \boldsymbol{I}_q$, we immediately see that (1.15) holds with $\boldsymbol{B}_0$ in place of $\boldsymbol{B}$. However, (1.15) is independent of representatives so it is valid for every representative $\boldsymbol{B}$. Finally, let us write C for the set of all points $[\boldsymbol{B},t] \in M$ satisfying (1.17). By using $\boldsymbol{B}_0$ in place of $\boldsymbol{B}$ in (1.17), it is clear that $U_{[\boldsymbol{A},s]} \subseteq C$. Conversely, let $[\boldsymbol{B},t] \in C$ be given. Put $\boldsymbol{B}_0 := \boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$. Then $\boldsymbol{A}^T\boldsymbol{B}_0$ is again symmetric positive definite. In particular, $\boldsymbol{A}^T(\boldsymbol{B}_0 - \boldsymbol{A})$ is symmetric and $(\boldsymbol{V},c) := (\boldsymbol{B}_0 - \boldsymbol{A}, t-s)$ is a horizontal vector at $(\boldsymbol{A},s)$. If $\boldsymbol{v} := \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V},c))$, then (1.17) shows that $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ and $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$. We conclude that $U_{[\boldsymbol{A},s]} = C$. The proof is complete. □

Remark 4. In the above proof, it might seem that the proof of (1.15) is trivialized while the amazing matrix (1.16) seems to come from nowhere. This is not at all the case. The matrix $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ comes from considering the cases for a generic representative whose matrix $\boldsymbol{A}^T\boldsymbol{B}$ is only invertible, but not necessarily symmetric or positive definite. Given $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$, (1.14) shows that

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t]) = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q} - \boldsymbol{A}, t-s))$$

for some $\boldsymbol{Q} \in O(q)$. Given that $\boldsymbol{A}^T(\boldsymbol{B}\boldsymbol{Q} - \boldsymbol{A})$ is symmetric, it follows that $\boldsymbol{A}^T\boldsymbol{B}\boldsymbol{Q}$ is symmetric and Lemma 2 shows that there exists $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$ such that $\boldsymbol{A}^T\boldsymbol{B} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T$ and $\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$. Thus, by continuity of the logarithmic map, if $[\boldsymbol{B},t] \to [\boldsymbol{A},s]$ and for some representative $(\boldsymbol{B},t) \to (\boldsymbol{A},s)$, then it must be the case that $\boldsymbol{B}\boldsymbol{Q} - \boldsymbol{A} \to \boldsymbol{0}$ or equivalently $\boldsymbol{Q} \to \boldsymbol{I}_q$. Therefore, heurestically, the only reasonable choice of $\boldsymbol{D}$ is $\boldsymbol{I}_q$. This argument is not discarded: it is still in the proof of the injectivity of the exponential map but simplified by symmetric positive definiteness. So $\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{R}^T$. This value is actually uniquely determined by $\boldsymbol{A}$ and $\boldsymbol{B}$ and gives exactly $\boldsymbol{Q} = \boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ with $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ as in (1.16), despite that neither $\boldsymbol{P}$ nor $\boldsymbol{R}$ is unique. This is shown in the following lemma:

Lemma 5. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix with singular value decomposition $\boldsymbol{L} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T$, then

$$\boldsymbol{P}\boldsymbol{R}^T = (\boldsymbol{L}^T\boldsymbol{L})^{-\frac{1}{2}}\boldsymbol{L}^T.$$

Proof. Given that

$$\boldsymbol{L}^T\boldsymbol{L} = \boldsymbol{P}\boldsymbol{S}^2\boldsymbol{P}^T,$$

we have

$$(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}} = \boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T.$$

Therefore:

$$\boldsymbol{L} = \boldsymbol{R}\boldsymbol{P}^T\boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T = \boldsymbol{R}\boldsymbol{P}^T(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}.$$

The result follows from inverting $(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}$ and take transpose. □

We end this section with a result concerning the limit of a sequence.

Proposition 6. Given a sequence $\{\boldsymbol{A}_n\}_{n=1}^{\infty}$ of real $p \times q$ matrices with the property $\boldsymbol{A}_n\boldsymbol{A}_n^T \to \boldsymbol{A}\boldsymbol{A}^T$ as $n \to \infty$ for a $p \times q$ matrix $\boldsymbol{A}$ with $\operatorname{rank}(\boldsymbol{A}) = q$, then

$$\lim_{n\to\infty} \min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_n\boldsymbol{Q} - \boldsymbol{A}\|_F = 0.$$

In partciular, if $\{(\boldsymbol{A}_n, s_n)\}_{n=1}^{\infty}$ is a sequence in Θ such that, for some $(\boldsymbol{A}, s) \in \Theta$, we have both $s_n \to s$ and $\Phi((\boldsymbol{A}_n, s_n)) \to \Phi((\boldsymbol{A}, s))$ as $n \to \infty$, then $[\boldsymbol{A}_n, s_n] \to [\boldsymbol{A}, s]$ in M as $n \to \infty$.

Proof. Assume, for the sake of contradiction, that there exists an $\epsilon > 0$ and a subsequence $\{n_k\}_{k=1}^{\infty}$ such that, for all k,

$$\min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_{n_k}\boldsymbol{Q} - \boldsymbol{A}\|_F \geqslant \epsilon.$$

Given that $\|\boldsymbol{A}_n\|_F^2 = \operatorname{tr}\big(\boldsymbol{A}_n\boldsymbol{A}_n^T\big)$ is convergent, in particular bounded, choose subsubsequence $\{n_{k_l}\}_{l=1}^{\infty}$ such that $\boldsymbol{A}_{n_{k_l}} \to \boldsymbol{B}$ as $l \to \infty$ for some $p\times q$ matrix $\boldsymbol{B}$. It follows that $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{A}\boldsymbol{A}^T$. Therefore, $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{B}) = q$ and $\boldsymbol{B} = \boldsymbol{A}\boldsymbol{P}$ for some $\boldsymbol{P} \in O(q)$. We have

$$\min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{Q} - \boldsymbol{A}\|_F \leqslant \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{P}^T - \boldsymbol{A}\|_F = \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{P}^T - \boldsymbol{B}\boldsymbol{P}^T\|_F = \|\boldsymbol{A}_{n_{k_l}} - \boldsymbol{B}\|_F \to 0$$

as $l \to \infty$, a contradiction.

Now suppose that $\{(\boldsymbol{A}_n, s_n)\}_{n=1}^{\infty}$ is a sequence in Θ such that, for some $(\boldsymbol{A}, s) \in \Theta$, we have both $s_n \to s$ and $\Phi((\boldsymbol{A}_n, s_n)) \to \Phi((\boldsymbol{A}, s))$ as $n \to \infty$. It follows that $\boldsymbol{A}_n\boldsymbol{A}_n^T \to \boldsymbol{A}\boldsymbol{A}^T$ and the previous paragraph shows that there exists a sequence $\{\boldsymbol{Q}_n\}_{n=1}^{\infty}$ of $q \times q$ orthogonal matrices such that $\boldsymbol{A}_n\boldsymbol{Q}_n \to \boldsymbol{A}$ as $n \to \infty$. Thus,

$$\lim_{n\to\infty} [\boldsymbol{A}_n, s_n] = \lim_{n\to\infty} \pi((\boldsymbol{A}_n\boldsymbol{Q}_n, s_n)) = \pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s] .$$

The proof is complete. □

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.7) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in Sun et al. (2026), definition III.1 and III.2.

Let us introduce a few notations. For each $[\boldsymbol{W}, \sigma^2] \in M$, let

$$p(\boldsymbol{x}; [\boldsymbol{W}, \sigma^2]) = (2\pi)^{-\frac{p}{2}} \det{}^{-\frac{1}{2}}(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])) \exp\left(-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]))^{-1}\boldsymbol{x}\right) \tag{1.19}$$

be the density of $N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]))$. For $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]} \subset T_{[\boldsymbol{W},\sigma^2]}M \cong \mathbb{R}^d$, write $l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ for the log-likelihood function of the local experiment

$$l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = \log\left(p(\boldsymbol{x}; \operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right). \tag{1.20}$$

Define the square root likelihood function $r_{[\boldsymbol{W},\sigma^2]} : \mathcal{E}_{[\boldsymbol{W},\sigma^2]} \to L^2(\mathbb{R}^p)$ by setting, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$, that

$$r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = \sqrt{p(\boldsymbol{x}; \operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))}. \tag{1.21}$$

Theorem 7. *$r_{[\boldsymbol{W},\sigma^2]}$ is Fréchet differentiable at any $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score*

$$s(\boldsymbol{x};[\boldsymbol{W},\sigma^2])(\boldsymbol{h}) = \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right]\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right\}, \tag{1.22}$$

where

$$D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) := \left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(Exp_{[\boldsymbol{W},\sigma^2]}(t\boldsymbol{h})) = \boldsymbol{U}\boldsymbol{W}^T + \boldsymbol{W}\boldsymbol{U}^T + b\boldsymbol{I}_p, \tag{1.23}$$

if $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ lifts to $(\boldsymbol{U}, b)$ at $(\boldsymbol{W},\sigma^2)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h}_1,\boldsymbol{h}_2) = \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1} D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1} D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right\} \tag{1.24}$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\boldsymbol{W},\sigma^2]}$ at $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$ in a given direction $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\boldsymbol{v} = \boldsymbol{0}$ to prove (1.22) and (1.24).

Given $\boldsymbol{v}$ and $\boldsymbol{h}$, there exists an $\epsilon > 0$ such that $\boldsymbol{v} + t\boldsymbol{h} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$ whenever $t \in (-\epsilon,\epsilon)$. If they respectively lift to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ at $(\boldsymbol{W},\sigma^2)$, then $\boldsymbol{v} + t\boldsymbol{h}$ lifts to $(\boldsymbol{V} + t\boldsymbol{U}, c + tb)$ at $(\boldsymbol{W},\sigma^2)$. First of all:

$$\begin{aligned} D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})) &:= \left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})) \\ &= \left.\frac{d}{dt}\right|_{t=0}\left\{(\boldsymbol{W}+\boldsymbol{V}+t\boldsymbol{U})(\boldsymbol{W}+\boldsymbol{V}+t\boldsymbol{U})^T + (\sigma^2+c+tb)\boldsymbol{I}_p\right\} \\ &= \boldsymbol{U}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{U}^T + \boldsymbol{U}\boldsymbol{W}^T + \boldsymbol{W}\boldsymbol{U}^T + b\boldsymbol{I}_p, \end{aligned} \tag{1.25}$$

proving (1.23) with special case $\boldsymbol{v} = \boldsymbol{0}$. Secondly, for all $\boldsymbol{x} \in \mathbb{R}^p$:

$$\begin{aligned} &D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) := \left.\frac{d}{dt}\right|_{t=0} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x}) \\ &= \left.\frac{d}{dt}\right|_{t=0}\left\{-\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right) - \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)^{-1}\boldsymbol{x}\right\} \\ &= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(\left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\right] \\ &+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(\left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\ &= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right] \\ &+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\ &= \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right]\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right\}. \end{aligned} \tag{1.26}$$

Finally:

$$\begin{aligned} D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) &= D_{\boldsymbol{h}}\exp\left(\frac{1}{2}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right) = \frac{1}{2}\exp\left(\frac{1}{2}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) \\ &= \frac{1}{2}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}). \end{aligned} \tag{1.27}$$

To prove that (1.27) indeed defines a valid Gâteaux derivative, we must prove that $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in Gupta and Nagar (2000). For $\boldsymbol{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, Theorem 7.7.1(i) in Gupta and Nagar (2000) states that

$$\mathbb{E}\left[\boldsymbol{X}^T \boldsymbol{Z} \boldsymbol{X}\right] = \operatorname{tr}\left(\boldsymbol{Z}\boldsymbol{\Sigma}\right)$$

for any $p \times p$ matrix $\boldsymbol{Z}$ and Theorem 7.7.2 in Gupta and Nagar (2000) states that

$$\operatorname{Cov}\left(\boldsymbol{X}^T \boldsymbol{Z} \boldsymbol{X}, \boldsymbol{X}^T \boldsymbol{Y} \boldsymbol{X}\right) = \operatorname{tr}\big(\boldsymbol{Z}\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big) + \operatorname{tr}\big(\boldsymbol{Z}^T\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big)$$

for any two $p \times p$ matrices $\boldsymbol{Y}$ and $\boldsymbol{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right] \\
&+ \frac{1}{2}\int_{\mathbb{R}^p}\left[\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x}\right]p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = 0
\end{aligned}$$

and, whenever $\boldsymbol{h}_1$ and $\boldsymbol{h}_2 \in T_{[\boldsymbol{W},\sigma^2]}M$:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}_1} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) D_{\boldsymbol{h}_2} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= \frac{1}{4}\operatorname{Cov}\left(\boldsymbol{X}^T\boldsymbol{Z}_1\boldsymbol{X},\ \boldsymbol{X}^T\boldsymbol{Z}_2\boldsymbol{X}\right) = \frac{1}{4}\cdot 2\operatorname{tr}\left(\boldsymbol{Z}_1\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\boldsymbol{Z}_2\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right) \\
&= \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right\},
\end{aligned} \tag{1.28}$$

where $\boldsymbol{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))$ and

$$\boldsymbol{Z}_j := \left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}_j}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} \tag{1.29}$$

is symmetric for $j = 1, 2$, so that the two terms in Theorem 7.7.2 of Gupta and Nagar (2000) coincide. Thus, in the particular case $\boldsymbol{h} = \boldsymbol{h}_1 = \boldsymbol{h}_2$:

$$\begin{aligned}
&\|D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|^2 = \int_{\mathbb{R}^p}\left[D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right]^2\mathrm{d}\boldsymbol{x} \\
&= \frac{1}{4}\int_{\mathbb{R}^p}\left(D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)^2 p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = \frac{1}{8}\operatorname{tr}\left\{\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right]^2\right\}.
\end{aligned} \tag{1.30}$$

Therefore, $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$ and (1.27) is indeed a valid Gâteaux derivative at $\boldsymbol{v}$ in the direction $\boldsymbol{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\boldsymbol{v}, \boldsymbol{h})(\boldsymbol{x}) := r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v} + \boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) - D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})$$

and prove that

$$\lim_{\boldsymbol{h}\to\mathbf{0}} \frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v}, \boldsymbol{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+\boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = \int_0^1 \frac{d}{dt} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t$$
$$= \int_0^1 \lim_{s\to 0} \frac{r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+(t+s)\boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})}{s}\mathrm{d}t = \int_0^1 D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t.$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$\|R(\boldsymbol{v},\boldsymbol{h}); L^2(\mathbb{R}^p)\| = \|\int_0^1 \left(D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x}) - D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)\mathrm{d}t; L^2(\mathbb{R}^p)\|$$
$$\leqslant \int_0^1 \|D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t = |\boldsymbol{h}| \int_0^1 \|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t$$

and

$$\frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v},\boldsymbol{h}); L^2(\mathbb{R}^p)\|^2 \leqslant \left\{\int_0^1 \|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2$$
$$\leqslant \left\{\int_0^1 \sup_{|\boldsymbol{u}|=1} \|D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2.$$

The proof will be complete once we can show that

$$\lim_{\boldsymbol{h}\to\boldsymbol{0}} \int_0^1 \sup_{|\boldsymbol{u}|=1} \|D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t = 0. \tag{1.31}$$

We mus therefore examine the continuity of $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$, which is easy to see if we revisit the formulas. (1.25) shows that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Thus, (1.27) shows that, for each fixed $\boldsymbol{x} \in \mathbb{R}^p$, $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$ and (1.30) shows that $\|D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Scheffé's lemma shows that $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\boldsymbol{h}$ and in $\boldsymbol{v}$. The integrand inside (1.31) is thus bounded uniformly over all $t \in [0,1]$ and all $\boldsymbol{h}$ in a compact neighborhood of $\boldsymbol{0}$ and it suffices to show that, for every $t \in [0,1]$, that

$$\lim_{\boldsymbol{h}\to\boldsymbol{0}} \sup_{|\boldsymbol{u}|=1} \|D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| = 0, \tag{1.32}$$

thanks to the bounded convergence theorem. To prove (1.32), let $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ be a $g_{[\boldsymbol{W},\sigma^2]}$ orthonormal basis. Every unit vector $\boldsymbol{u}$ can be expressed as $\boldsymbol{u} = \sum_{i=1}^d u_i\boldsymbol{e}_i$ with $\sum_{i=1}^d u_i^2 = 1$. By linearity of $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$\|D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| \leqslant \sum_{i=1}^d |u_i|\|D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|$$
$$\leqslant \sqrt{d} \max_{i=1,\ldots,d} \|D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|.$$

Taking supremum yields

$$\sup_{|\boldsymbol{u}|=1} \|D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| \leqslant \sqrt{d} \max_{i=1,\ldots,d} \|D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|.$$

Given that $D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}) \to D_{\boldsymbol{e}_i} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $L^2(\mathbb{R}^p)$ as $\boldsymbol{h} \to \boldsymbol{0}$ for every i, every t, and every $\boldsymbol{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.25), (1.26), and (1.27) simplifies when $\boldsymbol{v} = \mathbf{0}$:

$$
\begin{aligned}
&D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\mathbf{0})(\boldsymbol{x}) = \frac{1}{2} r_{[\boldsymbol{W},\sigma^2]}(\mathbf{0})(\boldsymbol{x}) D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\mathbf{0})(\boldsymbol{x}) \\
&= \frac{1}{4}\sqrt{p(\boldsymbol{x};[\boldsymbol{W},\sigma^2])}\,\mathrm{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right]\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right\}.
\end{aligned}
$$

The score $s(\boldsymbol{x};[\boldsymbol{W},\sigma^2])(\boldsymbol{h})$ therefore equals to $D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\mathbf{0})(\boldsymbol{x})$ and is indeed given by (1.22). The Fisher information operator (1.24) is obtained by taking $\boldsymbol{v} = \mathbf{0}$ in (1.28). To show that $G_{[\boldsymbol{W},\sigma^2]}$ is positive definite, let $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ be given. We show that $G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) \geqslant 0$ and equality holds only when $\boldsymbol{h} = \mathbf{0}$. The non-negative part is clear from (1.24):

$$
G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\left\{\left[\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1} D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right]^2\right\} \geqslant 0.
$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) = \mathbf{0}$. We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$
\boldsymbol{N} := \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-\frac{1}{2}} D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-\frac{1}{2}}
$$

such that

$$
0 = G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\big(\boldsymbol{N}^2\big).
$$

Therefore, $\boldsymbol{N} = \mathbf{0}$ and $D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) = \mathbf{0}$. Now (1.23) reads the same as (1.9) in the proof of Proposition 1. We therefore get $(\boldsymbol{U},b) \in \ker(d\Phi_{(\boldsymbol{W},\sigma^2)}) = V_{(\boldsymbol{W},\sigma^2)}$. However, being lifted from $\boldsymbol{h}$, $(\boldsymbol{U},b)$ must also be a horizontal vector. Therefore, $\boldsymbol{h} = \mathbf{0}$. We have proven that $G_{[\boldsymbol{W},\sigma^2]}$ is positive definite. □

Proposition 8. Suppose that $[\boldsymbol{W},\sigma^2] \in M$ is the true parameter so that each $\boldsymbol{X}_i$ has density $p(\,\cdot\,;[\boldsymbol{W},\sigma^2])$ as in (1.19). We have the local asymptotic normality expansion

$$
\sum_{i=1}^{n}\log\frac{p\left(\boldsymbol{X}_i;\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])} = \sum_{i=1}^{n} s(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) - \frac{1}{2}G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) + o_{\mathbb{P}}(1) \tag{1.33}
$$

whenever n large enough that $\frac{\boldsymbol{h}}{\sqrt{n}} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$. In particular,

$$
\sum_{i=1}^{n}\log\frac{p\left(\boldsymbol{X}_i;\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])} \to N\left(-\frac{1}{2}G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}),\, G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h})\right) \tag{1.34}
$$

in distribution as $n \to \infty$.

Proof. Given that $\mathcal{P}$ is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in Sun et al. (2026). □

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

In this section, we assume that there is a true parameter $[\boldsymbol{W}_0,\sigma_0^2] \in M$ for the model $\mathcal{P}$ given in (1.7) and we investigate the consistency and asymptotic normality of the maximum likelihood estimator towards this true parameter.

More concretely, given data points $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n \in \mathbb{R}^p$, we consider them as realizations of independent identically distributed random variables $\boldsymbol{X}_1, \dots, \boldsymbol{X}_n$ whose true density is $p(\boldsymbol{x}; [\boldsymbol{W}_0, \sigma_0^2])$, where p is given in (1.19). In what follows, we denote the true covariance matrix by

$$\boldsymbol{\Sigma}_0 := \boldsymbol{\Sigma}([\boldsymbol{W}_0, \sigma_0^2]) = \boldsymbol{W}_0 \boldsymbol{W}_0^T + \sigma_0^2 \boldsymbol{I}_p. \tag{1.35}$$

The eigenvalues of the true covariance matrix are

$$\lambda_j(\boldsymbol{\Sigma}_0) = \begin{cases} \sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2 & j = 1, \dots, q \\ \sigma_0^2 & j = q+1, \dots, p. \end{cases} \tag{1.36}$$

The sample covariance matrix is defined by

$$\boldsymbol{S}_n = \frac{1}{n} \sum_{i=1}^n \boldsymbol{x}_i \boldsymbol{x}_i^T. \tag{1.37}$$

Given that we will discuss $\boldsymbol{S}_n$ a lot, we shall write λ_j in place of $\lambda_j(\boldsymbol{S}_n)$. We need also to fix one choice of orthonormal basis of eigenvectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_p$ where $\boldsymbol{v}_j = \boldsymbol{v}_j(\boldsymbol{S}_n)$ corresponds to the eigenvalue λ_j. The facts we will use repeatedly is that, when p and q are both fixed, $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ almost surely as $n \to \infty$ and, due to the fact that eigenvalues depends continuously on the matrices, $\lambda_j \to \lambda_j(\boldsymbol{\Sigma}_0)$ almost surely as $n \to \infty$.

We now formulate the maximum likelihood estimator as computed in Tipping and Bishop (1999). Put $\boldsymbol{U} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_q]$, a $p \times q$ matrix, and Δ_q to be a $q \times q$ diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_q$. Then, the collection of all choices of $(\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2)$ maximizing the likelihood is given by

$$\hat{\boldsymbol{W}}_n = \boldsymbol{U} \left(\Delta_q - \hat{\sigma}_n^2 \boldsymbol{I}_q\right)^{\frac{1}{2}} \boldsymbol{R} \tag{1.38}$$

and

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \tag{1.39}$$

ranged over all rotation matrices $\boldsymbol{R} \in O(q)$.

It is immediately clear from (1.36) that, almost surely, for all sufficiently large n, $\lambda_j > 0$ for all j and $\hat{\sigma}_n^2 > 0$ so that $\hat{\sigma}_n^2$ is well-defined. Also, $\hat{\sigma}_n^2 \to \hat{\sigma}_0^2$ almost surely as $n \to \infty$. Another look at (1.36) shows that, almost surely, for all sufficiently large n, $\lambda_j - \hat{\sigma}_n^2 > 0$ for all $j \in \{1, \dots, q\}$ and $\operatorname{rank}(\hat{\boldsymbol{W}}_n) = q$ so that $\hat{\boldsymbol{W}}_n$ is well-defined. Thus, almost surely, for all sufficiently large n, the same model parametrized by M has a unique maximum likelihood estimator $\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right]$.

First of all, I would like to argue that this quotient manifold parameter space allows us to settle consistency in an (arguably more straightforward) way that does not use Datta and Chakrabarty (2023) or Redner (1981).

Theorem 9. *The maximum likelihood estimator is consistent. That is,* $\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right] \to [\boldsymbol{W}_0, \sigma_0^2]$ *almost surely as* $n \to \infty$.

Proof. Put

$$\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}\left(\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right]\right) = \hat{\boldsymbol{W}}_n \hat{\boldsymbol{W}}_n^T + \hat{\sigma}_n^2 \boldsymbol{I}_p. \tag{1.40}$$

In light of Proposition 6 and the fact that $\hat{\sigma}_n^2$ is consistent, it suffices to prove that $\hat{\boldsymbol{\Sigma}}_n \to \boldsymbol{\Sigma}_0$ almost surely as $n \to \infty$. The fact that $\sum_{j=1}^p \boldsymbol{v}_j \boldsymbol{v}_j^T = \boldsymbol{I}_p$ allows us to write

$$\hat{\boldsymbol{\Sigma}}_n = \boldsymbol{U}\left(\Delta_q - \hat{\sigma}_n^2 \boldsymbol{I}_q\right) \boldsymbol{U}^T + \hat{\sigma}_n^2 \boldsymbol{I}_p = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \boldsymbol{v}_j \boldsymbol{v}_j^T + \hat{\sigma}_n^2 \boldsymbol{I}_p = \sum_{j=1}^q \lambda_j \boldsymbol{v}_j \boldsymbol{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \boldsymbol{v}_j \boldsymbol{v}_j^T$$

and

$$\boldsymbol{S}_n = \boldsymbol{S}_n \sum_{j=1}^{p} \boldsymbol{v}_j \boldsymbol{v}_j^T = \sum_{j=1}^{p} \lambda_j \boldsymbol{v}_j \boldsymbol{v}_j^T.$$

Thus,

$$\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n = \sum_{j=q+1}^{p} \left(\hat{\sigma}_n^2 - \lambda_j\right) \boldsymbol{v}_j \boldsymbol{v}_j^T$$

and

$$\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| \leqslant \sum_{j=q+1}^{p} \left|\hat{\sigma}_n^2 - \lambda_j\right| \|\boldsymbol{v}_j \boldsymbol{v}_j^T\| = \sum_{j=q+1}^{p} \left|\hat{\sigma}_n^2 - \lambda_j\right|.$$

Therefore, given that both $\hat{\sigma}_n^2$ and λ_j will converge almost surely to σ_0^2 when $j \in \{q+1, \ldots, p\}$, $\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| \to 0$ almost surely as $n \to \infty$. The proof is complete. □

We now move to prove that, locally at the true parameter, the maximum likelihood estimator is asymptotically normal in the tangent space of the true parameter by local linearity. Given that the manifold is locally Euclidean, this result can be attacked with the classical approach, namely by verifying conditions of Theorem 5.23 in van der Vaart (1998), without too much use of geometric analysis. We start with computational lemmas.

Lemma 10. Given $[\boldsymbol{W}, \sigma^2] \in M$, there exists a function $m : \mathbb{R}^p \to \mathbb{R}$ such that, whenever $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{E}_{[\boldsymbol{W}, \sigma^2]}$ and $\boldsymbol{x} \in \mathbb{R}^p$, we always have

$$\left|l_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}_1)(\boldsymbol{x}) - l_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}_2)(\boldsymbol{x})\right| \leqslant m(\boldsymbol{x})|\boldsymbol{v}_1 - \boldsymbol{v}_2|.$$

This m has its second moment under any distribution μ on $\mathbb{R}^p$ that has the fourth moment. That is,

$$\int_{\mathbb{R}^p} (m(\boldsymbol{x}))^2 \mathrm{d}\mu(\boldsymbol{x}) < \infty$$

whenever

$$\int_{\mathbb{R}^p} |\boldsymbol{x}|^4 \mathrm{d}\mu(\boldsymbol{x}) < \infty.$$

Proof. Given $[\boldsymbol{W}, \sigma^2] \in M$, the result will follow by the mean value inequality if we can find a function m such that

$$\left|D_{\boldsymbol{h}} l_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right| \leqslant m(\boldsymbol{x})$$

for all unit vector $\boldsymbol{h} \in T_{[\boldsymbol{W}, \sigma^2]}M$, all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}, \sigma^2]}$, and all $\boldsymbol{x} \in \mathbb{R}^p$. Given that we have proved in Proposition 7 that $\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)^{-1}$ is continuous in $\boldsymbol{v}$ and that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))$ is continuous in both $\boldsymbol{h}$ and $\boldsymbol{v}$, we choose constants $C_1 \in (0, \infty)$ and $C_2 \in (0, \infty)$ such that

$$\frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)^{-1}\right\|_F \leqslant C_1$$

and

$$\frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W}, \sigma^2]}(\boldsymbol{v}))\right)\right\|_F \leqslant C_2$$

for all all unit vector $\boldsymbol{h} \in T_{[\boldsymbol{W}, \sigma^2]}M$ and all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}, \sigma^2]}$, thanks to the fact that $|\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{W}), \sigma^2)$ for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}, \sigma^2]}$ from Proposition 3. By (1.26), we have

$$\left|D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right| \leqslant \frac{1}{2}\|\boldsymbol{x}\boldsymbol{x}^T\|_F \left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\right\|_F$$
$$+\frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right\|_F \leqslant C_1\boldsymbol{x}^T\boldsymbol{x}+C_2$$

Setting $m(\boldsymbol{x}) = C_1\boldsymbol{x}^T\boldsymbol{x}+C_2$ will finish the proof. □

Lemma 11. Define function $L : \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]} \to \mathbb{R}$ by setting, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$,

$$L(\boldsymbol{v}) = \int_{\mathbb{R}^p} l_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})(\boldsymbol{x})p(\boldsymbol{x};\boldsymbol{\Sigma}_0)\mathrm{d}\boldsymbol{x}.$$

Then $\mathbf{0}$ is the unique maximizer of L and, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$:

$$L(\boldsymbol{v}) = L(\mathbf{0}) - \frac{1}{2}G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v},\boldsymbol{v}) + R(\boldsymbol{v})$$

with the remainder of the second order Taylor exapnsion satisfying

$$\lim_{|\boldsymbol{v}|\to 0}\frac{R(\boldsymbol{v})}{|\boldsymbol{v}|^2} = 0.$$

Proof. One should immediately recognize that $L(\mathbf{0}) - L(\boldsymbol{v})$ equals to the Kullback-Leibler divergence of the true model $N_p(\mathbf{0},\boldsymbol{\Sigma}_0)$ with respect to the model $N_p\left(\mathbf{0},\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})\right)\right)$. Therefore, $L(\boldsymbol{v})$ is maximized at all $\boldsymbol{v}$ with $\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})\right) = \boldsymbol{\Sigma}_0$. Given that $\boldsymbol{\Sigma}$ is injective and that we have proven in Proposition 3 that $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}$ is injective in $\mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$, this only happens when $\boldsymbol{v} = \mathbf{0}$ and $\mathbf{0}$ is the unique maximizer of L.

To obtain the Taylor expansion, we basically have to repeat the computation part in the proof of Proposition 7 on a slightly different problem with the second derivative at $\mathbf{0}$ being computed as well. The first order derivative at $\mathbf{0}$ is automatically 0 because $\mathbf{0}$ is the maximizer.

Let $\boldsymbol{h} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ be given. We can easily compute that

$$L(t\boldsymbol{h}) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right) - \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}.$$

Suppose that $\boldsymbol{h}$ lifts to $(\boldsymbol{U},b)$ at $(\boldsymbol{W}_0,\sigma_0^2)$. (1.25) immediately shows that

$$\left.\frac{d^2}{dt^2}\right|_{t=0}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right) = 2\boldsymbol{U}\boldsymbol{U}^T.$$

We now compute the second derivative L term by term. For the log determinant term, we have

$$\frac{d}{dt}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right) = -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right],$$

$$\frac{d^2}{dt^2}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right) = -\frac{1}{2}\frac{d}{dt}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right]$$
$$= -\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d^2}{dt^2}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right\}$$
$$+\frac{1}{2}\operatorname{tr}\left\{\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right]^2\right\},$$

and therefore

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right)=-\operatorname{tr}\left\{\boldsymbol{\Sigma}_0^{-1}\boldsymbol{U}\boldsymbol{U}^T\right\}+\frac{1}{2}\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}.$$

Next, for the trace term, we have

$$\begin{aligned}
&\frac{d}{dt}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)\\
&=\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\},
\end{aligned}$$

$$\begin{aligned}
&\frac{d^2}{dt^2}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)\\
&=\frac{1}{2}\operatorname{tr}\bigg\{-2\bigg\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\\
&\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\bigg\}\bigg\}\\
&+\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d^2}{dt^2}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\},
\end{aligned}$$

and therefore

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)=-\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}+\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{U}\boldsymbol{U}^T\right).$$

We end up the negative of the Fisher information operator, as computed in (1.24).

$$\frac{d^2}{dt^2}\bigg|_{t=0}L(t\boldsymbol{h})=-\frac{1}{2}\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}=-G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{h},\boldsymbol{h}).$$

This completes the proof. □

Lemma 12. Fix a $g_{[\boldsymbol{W}_0,\sigma_0^2]}$ orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ of $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$. We identify $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ canonically as $\mathbb{R}^d$ through this basis. For $i,\ j\in\{1,\ldots,d\}$, define the matrix version $\boldsymbol{G}\in\mathbb{R}^{d\times d}$ of the bilinear form $G_{[\boldsymbol{W}_0,\sigma_0^2]}$ by setting $\boldsymbol{G}_{ij}=G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{e}_i,\boldsymbol{e}_j)$ and the gradient version $\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])\in\mathbb{R}^d$ of the score by setting $(\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2]))_i=s(\boldsymbol{X}_i;[\boldsymbol{W}_0,\sigma_0^2])(\boldsymbol{e}_i)$. Then

$$\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)-\frac{1}{\sqrt{n}}\sum_{i=1}^n\boldsymbol{G}^{-1}\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])\to 0$$

in probability as $n\to\infty$. In particular, $\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)$ converges in distribution to $N_d(\boldsymbol{0},\boldsymbol{G}^{-1})$ as $n\to\infty$.

Proof. This is immdiate from Theorem 5.23 in van der Vaart (1998) because the two previous lemmas filled all the requirements of the theorem. □

Chapter 2

Draft

From here on we change the names of the coordinates on Θ and write a point of Θ as $(\boldsymbol{W}, \sigma^2)$, with $\boldsymbol{W}$ the loading matrix and σ^2 the noise variance of (1.1), and a point of M as $[\boldsymbol{W}, \sigma^2]$, following the notation of Datta and Chakrabarty (2023) and Datta (2026). This is nothing but a change of names — the generic representative called $(\boldsymbol{A}, s)$ in the previous sections is now called $(\boldsymbol{W}, \sigma^2)$, so that $\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]) = \boldsymbol{W}\boldsymbol{W}^T + \sigma^2\boldsymbol{I}_p$ by (1.6), and every statement proved so far applies verbatim — but it is the notation in which the statistical results of this section are usually quoted, and it frees the letter s for the score (1.22). Orthonormal eigenvector matrices, which in this section are extracted from the sample covariance matrix rather than from the parameter, are correspondingly denoted by $\boldsymbol{K}$ and $\boldsymbol{K}_q$ below. Singular values keep the notation $\sigma_j(\,\cdot\,)$ and $\sigma_{\min}(\,\cdot\,)$ of Section 1.1; they always carry an explicit matrix argument, so they are never confused with the noise variance σ^2.

Throughout this section we fix a *true parameter* $[\boldsymbol{W}_0, \sigma_0^2] \in M$, we assume that $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ are independent and identically distributed random vectors with common density $p(\,\cdot\,; [\boldsymbol{W}_0, \sigma_0^2])$ as in (1.19), and we abbreviate

$$\boldsymbol{\Sigma}_0 := \boldsymbol{\Sigma}([\boldsymbol{W}_0, \sigma_0^2]) = \boldsymbol{W}_0\boldsymbol{W}_0^T + \sigma_0^2\boldsymbol{I}_p, \qquad \boldsymbol{S}_n := \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{X}_i\boldsymbol{X}_i^T, \qquad d := pq + 1 - \frac{q(q-1)}{2} = \dim M. \tag{2.1}$$

We write $\rightsquigarrow$ for convergence in distribution and, following (1.5), we abbreviate $|\boldsymbol{v}| := \sqrt{g_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}, \boldsymbol{v})}$ for $\boldsymbol{v} \in T_{[\boldsymbol{W},\sigma^2]}M$.

The whole point of the construction of Section 1.1 is that the naive statement "$\sqrt{n}(\hat{\boldsymbol{W}}_n - \boldsymbol{W}_0)$ is asymptotically normal" is not merely false but meaningless: by non-identifiability the estimator $\hat{\boldsymbol{W}}_n$ produced by any algorithm is only determined up to a right multiplication by an element of $O(q)$, so that $\hat{\boldsymbol{W}}_n$ need not converge to $\boldsymbol{W}_0$ at all, and only its orbit can. The correct object to look at is the intrinsic estimation error $\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]) \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ of Sun et al. (2026), which by Proposition **??** is the ordinary error *after optimal rotational alignment*. The goal of this section is to prove that this tangent vector is asymptotically linear in the sense of Sun et al. (2026), Lemma III.3, with influence operator $G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}s(\,\cdot\,; [\boldsymbol{W}_0, \sigma_0^2])$, and consequently asymptotically normal with covariance operator $G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}$.

It will be convenient to name the horizontal lift. By Proposition 1 and the definition (1.5) of g, the restriction of $d\pi_{(\boldsymbol{W},\sigma^2)}$ to $H_{(\boldsymbol{W},\sigma^2)}$ is a linear isometry onto $T_{[\boldsymbol{W},\sigma^2]}M$; we denote its inverse by

$$\iota_{(\boldsymbol{W},\sigma^2)} := \left(d\pi_{(\boldsymbol{W},\sigma^2)}\big|_{H_{(\boldsymbol{W},\sigma^2)}} \right)^{-1} : T_{[\boldsymbol{W},\sigma^2]}M \longrightarrow H_{(\boldsymbol{W},\sigma^2)}, \qquad \iota_{(\boldsymbol{W},\sigma^2)}(\boldsymbol{h}) =: (\boldsymbol{U}_{\boldsymbol{h}}, b_{\boldsymbol{h}}), \tag{2.2}$$

so that $\iota_{(\boldsymbol{W},\sigma^2)}$ is a *linear* isometry and $|\boldsymbol{h}|^2 = \mathrm{tr}\big(\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}_{\boldsymbol{h}}^T\big) + b_{\boldsymbol{h}}^2$. Finally, since $G_{[\boldsymbol{W},\sigma^2]}$ is a positive definite bilinear form on $T_{[\boldsymbol{W},\sigma^2]}M$ by Theorem 7, the map $\boldsymbol{h} \mapsto G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h}, \,\cdot\,)$ is a linear isomorphism from $T_{[\boldsymbol{W},\sigma^2]}M$ onto its dual $T^*_{[\boldsymbol{W},\sigma^2]}M$; we write $G^{-1}_{[\boldsymbol{W},\sigma^2]}$ for its inverse, characterized by

$$G_{[\boldsymbol{W},\sigma^2]}\left(G^{-1}_{[\boldsymbol{W},\sigma^2]}(\varphi), \boldsymbol{h} \right) = \varphi(\boldsymbol{h}) \qquad \text{for all } \varphi \in T^*_{[\boldsymbol{W},\sigma^2]}M \text{ and all } \boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M. \tag{2.3}$$

Since the score $s(\boldsymbol{x}; [\boldsymbol{W}, \sigma^2])$ of (1.22) is a linear functional on $T_{[\boldsymbol{W},\sigma^2]}M$, the composition $G^{-1}_{[\boldsymbol{W},\sigma^2]} s(\boldsymbol{x}; [\boldsymbol{W}, \sigma^2])$ is a genuine tangent vector.

Preliminaries on the sample covariance matrix

Before anything else we collect the elementary properties of the sample covariance matrix $\boldsymbol{S}_n$ of (2.1) that will be used over and over below. The reason for gathering them here is that the sample enters the problem exclusively through $\boldsymbol{S}_n$ — the log-likelihood (2.16) is a function of $\boldsymbol{S}_n$ alone — and that the maximum likelihood estimator produced by Proposition 19 is a function of nothing but the eigenvalues of $\boldsymbol{S}_n$ and the eigenspace attached to the q largest of them. Everything in this subsection is classical and the proofs are short; they are given so that the section remains self-contained and so that the statements are available in the notation of this document.

We write $\mathrm{Sym}(p)$ for the vector space of real symmetric $p \times p$ matrices, which we equip with the Frobenius inner product $\langle \boldsymbol{B}, \boldsymbol{C} \rangle := \mathrm{tr}(\boldsymbol{BC})$ and the associated norm $\|\cdot\|_F$, so that $\mathrm{Sym}(p)$ becomes a Euclidean space of dimension $\frac{p(p+1)}{2}$; as in Section 1.1, $\|\cdot\|$ continues to denote the spectral norm. For $\boldsymbol{B} \in \mathrm{Sym}(p)$ we write $\lambda_1(\boldsymbol{B}) \geqslant \ldots \geqslant \lambda_p(\boldsymbol{B})$ for the eigenvalues of $\boldsymbol{B}$, repeated according to multiplicity and arranged in non-increasing order, and we write $\boldsymbol{B} \succeq \boldsymbol{0}$, respectively $\boldsymbol{B} \succ \boldsymbol{0}$, to say that $\boldsymbol{B}$ is positive semi-definite, respectively positive definite. The first lemma is pure linear algebra and applies to any symmetric matrix.

Lemma 13. Let $\boldsymbol{S} \in \mathrm{Sym}(p)$ and write $\lambda_j := \lambda_j(\boldsymbol{S})$.

1. (Spectral decomposition and its ambiguity.) There is $\boldsymbol{K} = (\boldsymbol{k}_1, \ldots, \boldsymbol{k}_p) \in O(p)$ with

$$\boldsymbol{S} = \boldsymbol{K} \mathrm{diag}(\lambda_1, \ldots, \lambda_p) \boldsymbol{K}^T = \sum_{j=1}^{p} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T. \tag{2.4}$$

   The eigenvalues are uniquely determined by $\boldsymbol{S}$, and so are the orthogonal projectors $\boldsymbol{\Pi}_1, \ldots, \boldsymbol{\Pi}_r$ onto the eigenspaces $E_1, \ldots, E_r$ of the distinct eigenvalues $\mu_1 > \ldots > \mu_r$ of $\boldsymbol{S}$, since

$$\boldsymbol{\Pi}_k = \prod_{l \neq k} \frac{\boldsymbol{S} - \mu_l \boldsymbol{I}_p}{\mu_k - \mu_l}, \qquad k = 1, \ldots, r. \tag{2.5}$$

   The matrix $\boldsymbol{K}$, by contrast, is *not* determined by $\boldsymbol{S}$: writing $m_k := \dim E_k$, a matrix $\boldsymbol{K}' \in O(p)$ satisfies (2.4) if and only if $\boldsymbol{K}' = \boldsymbol{K}\,\mathrm{diag}(\boldsymbol{R}_1, \ldots, \boldsymbol{R}_r)$ for some $\boldsymbol{R}_k \in O(m_k)$.

2. (Variational characterization.) For every $k \in \{1, \ldots, p\}$,

$$\lambda_k = \max_{\substack{V \subseteq \mathbb{R}^p \\ \dim V = k}} \min_{\substack{\boldsymbol{v} \in V \\ \boldsymbol{v}^T\boldsymbol{v}=1}} \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v} = \min_{\substack{V \subseteq \mathbb{R}^p \\ \dim V = p-k+1}} \max_{\substack{\boldsymbol{v} \in V \\ \boldsymbol{v}^T\boldsymbol{v}=1}} \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v}; \tag{2.6}$$

   in particular $\lambda_1 = \max_{\boldsymbol{v}^T\boldsymbol{v}=1} \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v}$ is a convex function of $\boldsymbol{S}$ and $\lambda_p = \min_{\boldsymbol{v}^T\boldsymbol{v}=1} \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v}$ is a concave function of $\boldsymbol{S}$.

3. (Ky Fan.) Let $1 \leqslant q < p$. For every $\boldsymbol{U} \in \mathbb{R}^{p \times q}$ with $\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}_q$,

$$\mathrm{tr}\left(\boldsymbol{U}^T \boldsymbol{S} \boldsymbol{U}\right) \leqslant \sum_{j=1}^{q} \lambda_j. \tag{2.7}$$

   If moreover $\lambda_q > \lambda_{q+1}$, then equality holds in (2.7) if and only if the column space of $\boldsymbol{U}$ equals $\mathrm{span}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_q)$, a subspace which by 1 does not depend on the choice of $\boldsymbol{K}$ in (2.4).

4. (Weyl.) For every $\boldsymbol{T} \in \mathrm{Sym}(p)$,

$$\max_{1 \leqslant k \leqslant p} |\lambda_k(\boldsymbol{S}) - \lambda_k(\boldsymbol{T})| \leqslant \|\boldsymbol{S} - \boldsymbol{T}\|. \tag{2.8}$$

In particular each $\lambda_k : \mathrm{Sym}(p) \to \mathbb{R}$ is 1-Lipschitz, hence continuous.

Proof. 1. The existence of (2.4) is the spectral theorem. Formula (2.5) is checked on eigenvectors: if $\boldsymbol{v} \in E_m$ then the right hand side maps $\boldsymbol{v}$ to $\left(\prod_{l \neq k} \frac{\mu_m - \mu_l}{\mu_k - \mu_l}\right) \boldsymbol{v}$, which is $\boldsymbol{v}$ when $m = k$ and $\mathbf{0}$ when $m \neq k$, the factor indexed by $l = m$ vanishing; since $\mathbb{R}^p = E_1 \oplus \ldots \oplus E_r$ orthogonally, this determines both sides. As for $\boldsymbol{K}$, the columns of a matrix satisfying (2.4) are exactly orthonormal eigenvectors listed so that the associated eigenvalues are non-increasing, that is, consecutive blocks of $m_1, \ldots, m_r$ columns forming orthonormal bases of $E_1, \ldots, E_r$; two orthonormal bases of the same m_k-dimensional space differ precisely by right multiplication by an element of $O(m_k)$.

2. This is the Courant–Fischer min-max theorem (Horn and Johnson, 2013, Section 4.2). The convexity of λ_1 and the concavity of λ_p follow because $\boldsymbol{S} \mapsto \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v}$ is linear for each fixed $\boldsymbol{v}$, and a pointwise supremum, respectively infimum, of linear functions is convex, respectively concave.

3. Put $\boldsymbol{M} := \boldsymbol{K}^T \boldsymbol{U}$, which again satisfies $\boldsymbol{M}^T \boldsymbol{M} = \boldsymbol{I}_q$, and let $c_j := \sum_{k=1}^q M_{jk}^2$ be the squared norm of its j-th row, that is the j-th diagonal entry of the rank q orthogonal projector $\boldsymbol{M}\boldsymbol{M}^T$; hence $c_j \in [0, 1]$ and $\sum_{j=1}^p c_j = \mathrm{tr}\big(\boldsymbol{M}\boldsymbol{M}^T\big) = q$. By (2.4), $\mathrm{tr}\big(\boldsymbol{U}^T \boldsymbol{S} \boldsymbol{U}\big) = \mathrm{tr}\left(\boldsymbol{M}^T \mathrm{diag}(\lambda_1, \ldots, \lambda_p) \boldsymbol{M}\right) = \sum_{j=1}^p \lambda_j c_j$. Using $c_j - 1 \leqslant 0$ and $\lambda_j \geqslant \lambda_q$ for $j \leqslant q$, and $c_j \geqslant 0$ and $\lambda_j \leqslant \lambda_{q+1}$ for $j > q$, we get

$$\sum_{j=1}^p \lambda_j c_j - \sum_{j=1}^q \lambda_j = \sum_{j \leqslant q} \lambda_j (c_j - 1) + \sum_{j > q} \lambda_j c_j \leqslant \lambda_q \sum_{j \leqslant q} (c_j - 1) + \lambda_{q+1} \sum_{j > q} c_j = -(\lambda_q - \lambda_{q+1}) \sum_{j > q} c_j \leqslant 0,$$

where the last equality used $\sum_{j \leqslant q} (c_j - 1) = -\sum_{j > q} c_j$. This is (2.7). If $\lambda_q > \lambda_{q+1}$, the displayed chain shows that equality forces $c_j = 0$ for every $j > q$, that is, the last $p - q$ rows of $\boldsymbol{M}$ vanish, that is, the column space of $\boldsymbol{U} = \boldsymbol{K}\boldsymbol{M}$ is contained in $\mathrm{span}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_q)$, hence equal to it by equality of dimensions. Conversely, if the two spaces coincide then $\boldsymbol{U} = (\boldsymbol{k}_1, \ldots, \boldsymbol{k}_q) \boldsymbol{R}$ for some $\boldsymbol{R} \in O(q)$ and $\mathrm{tr}\big(\boldsymbol{U}^T \boldsymbol{S} \boldsymbol{U}\big) = \mathrm{tr}\left(\boldsymbol{R}^T \mathrm{diag}(\lambda_1, \ldots, \lambda_q) \boldsymbol{R}\right) = \sum_{j \leqslant q} \lambda_j$.

4. Write $\boldsymbol{E} := \boldsymbol{T} - \boldsymbol{S}$. For every unit vector $\boldsymbol{v}$ we have $\boldsymbol{v}^T \boldsymbol{T} \boldsymbol{v} \leqslant \boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v} + \lambda_1(\boldsymbol{E})$, so taking minima over a k-dimensional subspace and then the maximum over such subspaces in (2.6) gives $\lambda_k(\boldsymbol{T}) \leqslant \lambda_k(\boldsymbol{S}) + \lambda_1(\boldsymbol{E})$; exchanging the roles of $\boldsymbol{S}$ and $\boldsymbol{T}$ gives $\lambda_k(\boldsymbol{S}) \leqslant \lambda_k(\boldsymbol{T}) - \lambda_p(\boldsymbol{E})$. Hence $|\lambda_k(\boldsymbol{T}) - \lambda_k(\boldsymbol{S})| \leqslant \max\left(|\lambda_1(\boldsymbol{E})|, |\lambda_p(\boldsymbol{E})|\right) = \|\boldsymbol{E}\|$, the spectral norm of a symmetric matrix being the largest modulus of its eigenvalues. □

Remark 14. Part 3 of Lemma 13 is the reason the word *principal* appears in the name of the model. For any $\boldsymbol{U} \in \mathbb{R}^{p \times q}$ with $\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}_q$ the matrix $\boldsymbol{U}\boldsymbol{U}^T$ is the orthogonal projector onto the column space of $\boldsymbol{U}$, so that

$$\frac{1}{n} \sum_{i=1}^n \|\boldsymbol{X}_i - \boldsymbol{U}\boldsymbol{U}^T \boldsymbol{X}_i\|^2 = \frac{1}{n} \sum_{i=1}^n \boldsymbol{X}_i^T \left(\boldsymbol{I}_p - \boldsymbol{U}\boldsymbol{U}^T\right) \boldsymbol{X}_i = \mathrm{tr}(\boldsymbol{S}_n) - \mathrm{tr}\left(\boldsymbol{U}^T \boldsymbol{S}_n \boldsymbol{U}\right) : \tag{2.9}$$

minimizing the mean squared reconstruction error over q-dimensional subspaces is the same as maximizing the retained empirical variance $\mathrm{tr}\big(\boldsymbol{U}^T \boldsymbol{S}_n \boldsymbol{U}\big)$, and by (2.7) the optimum is $\sum_{j \leqslant q} \lambda_j(\boldsymbol{S}_n)$, attained exactly at the span of the q leading eigenvectors of $\boldsymbol{S}_n$ as soon as $\lambda_q(\boldsymbol{S}_n) > \lambda_{q+1}(\boldsymbol{S}_n)$. Note also that part 1 already contains, in embryo, the non-identifiability that forced the quotient construction of Section 1.1: what the data determine is the leading eigen*space*, whereas an orthonormal basis $\boldsymbol{K}_q$ of it is determined only up to right multiplication by an element of $O(q)$. This is exactly the ambiguity $\hat{\boldsymbol{W}} = \boldsymbol{K}_q(\boldsymbol{\Lambda}_q - \hat{\sigma}^2 \boldsymbol{I}_q)^{\frac{1}{2}} \boldsymbol{R}$ recorded in Proposition 19, and it disappears on $\mathcal{M}$. Finally, when a group of eigenvalues is separated from the rest, the projector onto the corresponding invariant subspace is not merely well defined but real analytic in $\boldsymbol{S}$; this is taken up in Lemma 25 on the set $\mathcal{O}$ of (2.17), which by part 4 above is open.

We now turn to the properties of $\boldsymbol{S}_n$ as a random matrix under the sampling scheme fixed at the beginning of this section.

Lemma 15. Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \dots$ be independent and identically distributed $N_p(\mathbf{0}, \boldsymbol{\Sigma}_0)$ random vectors and let $\boldsymbol{S}_n$ be as in (2.1).

1. $\boldsymbol{S}_n \in \mathrm{Sym}(p)$ and $\boldsymbol{S}_n \succeq \mathbf{0}$, with $\operatorname{rank}(\boldsymbol{S}_n) \leqslant \min(n, p)$; in particular $\boldsymbol{S}_n \succ \mathbf{0}$ is impossible when $n < p$.

2. $n\boldsymbol{S}_n$ follows the Wishart law $W_p(n, \boldsymbol{\Sigma}_0)$, and for $n \geqslant p$ its distribution has a density with respect to the Lebesgue measure of $\mathrm{Sym}(p)$ (Gupta and Nagar, 2000, Chapter 3).

3. $\boldsymbol{S}_n$ is an unbiased estimator of $\boldsymbol{\Sigma}_0$, and for all $\boldsymbol{B}, \boldsymbol{C} \in \mathrm{Sym}(p)$ and all $i, j, k, l \in \{1, \dots, p\}$,
$$\begin{gathered} \mathbb{E}\left[\boldsymbol{S}_n\right] = \boldsymbol{\Sigma}_0, \qquad \operatorname{Cov}\left(\operatorname{tr}(\boldsymbol{B}\boldsymbol{S}_n), \operatorname{tr}(\boldsymbol{C}\boldsymbol{S}_n)\right) = \frac{2}{n} \operatorname{tr}\left(\boldsymbol{B}\boldsymbol{\Sigma}_0\boldsymbol{C}\boldsymbol{\Sigma}_0\right), \\ \operatorname{Cov}\left((\boldsymbol{S}_n)_{ij}, (\boldsymbol{S}_n)_{kl}\right) = \frac{1}{n}\left((\boldsymbol{\Sigma}_0)_{ik}(\boldsymbol{\Sigma}_0)_{jl} + (\boldsymbol{\Sigma}_0)_{il}(\boldsymbol{\Sigma}_0)_{jk}\right). \end{gathered} \tag{2.10}$$

4. $\boldsymbol{S}_n$ is a sufficient statistic for $\mathcal{P}$: the joint density of $(\boldsymbol{X}_1, \dots, \boldsymbol{X}_n)$ at $[\boldsymbol{W}, \sigma^2] \in M$ is
$$\begin{gathered} \prod_{i=1}^{n} p(\boldsymbol{x}_i; [\boldsymbol{W}, \sigma^2]) = (2\pi)^{-\frac{np}{2}} \left(\det \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])\right)^{-\frac{n}{2}} \exp\left(-\frac{n}{2} \operatorname{tr}\left(\left(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])\right)^{-1} \boldsymbol{s}_n\right)\right), \\ \boldsymbol{s}_n := \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i^T. \end{gathered} \tag{2.11}$$

5. $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ almost surely, and
$$\sqrt{n}\left(\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\right) \rightsquigarrow \boldsymbol{G} \qquad \text{in } \mathrm{Sym}(p), \tag{2.12}$$
where $\boldsymbol{G}$ is the centred Gaussian random element of $\mathrm{Sym}(p)$ with $\operatorname{Cov}\left(\langle \boldsymbol{B}, \boldsymbol{G}\rangle, \langle \boldsymbol{C}, \boldsymbol{G}\rangle\right) = 2\operatorname{tr}(\boldsymbol{B}\boldsymbol{\Sigma}_0\boldsymbol{C}\boldsymbol{\Sigma}_0)$ for $\boldsymbol{B}, \boldsymbol{C} \in \mathrm{Sym}(p)$. In particular $\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$.

6. Consequently $\max_{1 \leqslant k \leqslant p} |\lambda_k(\boldsymbol{S}_n) - \lambda_k(\boldsymbol{\Sigma}_0)| \to 0$ almost surely, and this quantity is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$. The convergence of the eigenvalues is not monotone in any useful sense, but it is biased outwards in the mean: $\mathbb{E}\left[\lambda_1(\boldsymbol{S}_n)\right] \geqslant \lambda_1(\boldsymbol{\Sigma}_0)$ and $\mathbb{E}\left[\lambda_p(\boldsymbol{S}_n)\right] \leqslant \lambda_p(\boldsymbol{\Sigma}_0)$.

Proof. 1. Symmetry is clear and $\boldsymbol{v}^T\boldsymbol{S}_n\boldsymbol{v} = \frac{1}{n}\sum_{i=1}^n (\boldsymbol{X}_i^T\boldsymbol{v})^2 \geqslant 0$; the column space of $\boldsymbol{S}_n$ is contained in the span of $\boldsymbol{X}_1, \dots, \boldsymbol{X}_n$, whence the bound on the rank.

2. By definition, a sum of n independent $\boldsymbol{X}_i\boldsymbol{X}_i^T$ with $\boldsymbol{X}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}_0)$ is $W_p(n, \boldsymbol{\Sigma}_0)$ distributed, and this law is non-singular, with the usual density on the cone of positive definite matrices, as soon as $n \geqslant p$ and $\boldsymbol{\Sigma}_0 \succ \mathbf{0}$; the latter holds by Lemma 16 below. The absolute continuity of the law of $\boldsymbol{S}_n$ is re-derived by a more elementary route, avoiding any Wishart computation, in the proof of Lemma 27.

3. Since $\mathbb{E}\left[\boldsymbol{X}_1\right] = \mathbf{0}$ we have $\mathbb{E}\left[\boldsymbol{X}_1\boldsymbol{X}_1^T\right] = \boldsymbol{\Sigma}_0$, and averaging gives the first identity of (2.10). For the second, note that $\operatorname{tr}(\boldsymbol{B}\boldsymbol{S}_n) = \frac{1}{n}\sum_{i=1}^n \boldsymbol{X}_i^T\boldsymbol{B}\boldsymbol{X}_i$ is an average of independent quadratic forms, so that, by Theorem 7.7.2 of Gupta and Nagar (2000) already used in the proof of Theorem 7,
$$\operatorname{Cov}\left(\operatorname{tr}(\boldsymbol{B}\boldsymbol{S}_n), \operatorname{tr}(\boldsymbol{C}\boldsymbol{S}_n)\right) = \frac{1}{n}\operatorname{Cov}\left(\boldsymbol{X}_1^T\boldsymbol{B}\boldsymbol{X}_1, \boldsymbol{X}_1^T\boldsymbol{C}\boldsymbol{X}_1\right) = \frac{2}{n}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{\Sigma}_0\boldsymbol{C}\boldsymbol{\Sigma}_0\right).$$
The entrywise formula is the special case $\boldsymbol{B} = \frac{1}{2}\left(\boldsymbol{e}_i\boldsymbol{e}_j^T + \boldsymbol{e}_j\boldsymbol{e}_i^T\right)$ and $\boldsymbol{C} = \frac{1}{2}\left(\boldsymbol{e}_k\boldsymbol{e}_l^T + \boldsymbol{e}_l\boldsymbol{e}_k^T\right)$, for which $\operatorname{tr}(\boldsymbol{B}\boldsymbol{S}_n) = (\boldsymbol{S}_n)_{ij}$, $\operatorname{tr}(\boldsymbol{C}\boldsymbol{S}_n) = (\boldsymbol{S}_n)_{kl}$ and, expanding the four resulting terms and using the symmetry of $\boldsymbol{\Sigma}_0$, $\operatorname{tr}(\boldsymbol{B}\boldsymbol{\Sigma}_0\boldsymbol{C}\boldsymbol{\Sigma}_0) = \frac{1}{2}\left((\boldsymbol{\Sigma}_0)_{ik}(\boldsymbol{\Sigma}_0)_{jl} + (\boldsymbol{\Sigma}_0)_{il}(\boldsymbol{\Sigma}_0)_{jk}\right)$.

4. Immediate from (1.19) and the identity $\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x} = \operatorname{tr}\left(\boldsymbol{B}\boldsymbol{x}\boldsymbol{x}^T\right)$, which turns $\sum_{i=1}^n \boldsymbol{x}_i^T \left(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])\right)^{-1} \boldsymbol{x}_i$ into $n \operatorname{tr}\left(\left(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])\right)^{-1} \boldsymbol{s}_n\right)$; sufficiency then follows from the Fisher–Neyman factorization criterion, the right hand side of (2.11) depending on the sample only through $\boldsymbol{s}_n$.

5. The entries of $\boldsymbol{X}_1\boldsymbol{X}_1^T$ are products of two jointly Gaussian variables, hence integrable with finite second moments, so the strong law of large numbers applied to each of the finitely many entries gives $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ almost surely, and the multivariate central limit theorem (van der Vaart, 1998, Chapter 2) applied to the independent and identically distributed random elements $\boldsymbol{X}_i\boldsymbol{X}_i^T$ of the $\frac{p(p+1)}{2}$-dimensional Euclidean space $\mathrm{Sym}(p)$ gives (2.12), the covariance of the limit being that of $\boldsymbol{X}_1\boldsymbol{X}_1^T$, computed in 3. The last assertion follows because all norms on $\mathrm{Sym}(p)$ are equivalent and a weakly convergent sequence is uniformly tight.

6. The first two claims follow from (2.8) and 5. For the last two, λ_1 is convex and λ_p is concave on $\mathrm{Sym}(p)$ by part 2 of Lemma 13, so Jensen's inequality and $\mathbb{E}\left[\boldsymbol{S}_n\right] = \boldsymbol{\Sigma}_0$ give $\mathbb{E}\left[\lambda_1(\boldsymbol{S}_n)\right] \geqslant \lambda_1\left(\mathbb{E}\left[\boldsymbol{S}_n\right]\right) = \lambda_1(\boldsymbol{\Sigma}_0)$ and the reverse inequality for λ_p; the expectations are finite because $|\lambda_k(\boldsymbol{S}_n)| \leqslant \operatorname{tr}(\boldsymbol{S}_n)$, which is integrable. □

It remains to record the spectrum of the matrix that $\boldsymbol{S}_n$ approximates. The point of the next lemma is that $\boldsymbol{\Sigma}_0$ has exactly the *spiked* shape — q eigenvalues above a flat floor of height σ_0^2 — which the model postulates, and that the corresponding spectral gap is strictly positive; by Lemma 15 the sample covariance matrix inherits both features for n large.

Lemma 16. Let $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{D}_0\boldsymbol{V}_0^T$ be a singular value decomposition of $\boldsymbol{W}_0$, with $\boldsymbol{U}_0 \in \mathbb{R}^{p\times q}$ having orthonormal columns, $\boldsymbol{D}_0 = \operatorname{diag}(\sigma_1(\boldsymbol{W}_0), \ldots, \sigma_q(\boldsymbol{W}_0))$ and $\boldsymbol{V}_0 \in O(q)$, and let $\boldsymbol{P}_0 := \boldsymbol{U}_0\boldsymbol{U}_0^T$ be the orthogonal projector onto the column space of $\boldsymbol{W}_0$. Then

$$\boldsymbol{\Sigma}_0 = \boldsymbol{U}_0\left(\boldsymbol{D}_0^2 + \sigma_0^2\boldsymbol{I}_q\right)\boldsymbol{U}_0^T + \sigma_0^2\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right), \qquad \boldsymbol{\Sigma}_0^{-1} = \boldsymbol{U}_0\left(\boldsymbol{D}_0^2 + \sigma_0^2\boldsymbol{I}_q\right)^{-1}\boldsymbol{U}_0^T + \frac{1}{\sigma_0^2}\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right), \tag{2.13}$$

so that the eigenvalues of $\boldsymbol{\Sigma}_0$ are $\lambda_j(\boldsymbol{\Sigma}_0) = \sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2$ for $j = 1, \ldots, q$ and $\lambda_j(\boldsymbol{\Sigma}_0) = \sigma_0^2$ for $j = q+1, \ldots, p$. Consequently $\boldsymbol{\Sigma}_0 \succeq \sigma_0^2\boldsymbol{I}_p \succ \boldsymbol{0}$, $\|\boldsymbol{\Sigma}_0^{-1}\| = \sigma_0^{-2}$, $\det\boldsymbol{\Sigma}_0 = \sigma_0^{2(p-q)}\prod_{j=1}^q(\sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2)$, the eigenspace of the q largest eigenvalues is the column space of $\boldsymbol{W}_0$, and

$$\lambda_q(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) = \sigma_{\min}(\boldsymbol{W}_0)^2 =: \delta_0 > 0. \tag{2.14}$$

Moreover, almost surely there is a (random) N such that for all $n \geqslant N$,

$$\boldsymbol{S}_n \succeq \frac{\sigma_0^2}{2}\boldsymbol{I}_p \succ \boldsymbol{0} \qquad \text{and} \qquad \lambda_q(\boldsymbol{S}_n) - \lambda_{q+1}(\boldsymbol{S}_n) \geqslant \frac{\delta_0}{2}. \tag{2.15}$$

Proof. Since $\boldsymbol{W}_0\boldsymbol{W}_0^T = \boldsymbol{U}_0\boldsymbol{D}_0^2\boldsymbol{U}_0^T$ and $\boldsymbol{I}_p = \boldsymbol{P}_0 + (\boldsymbol{I}_p - \boldsymbol{P}_0)$, the first identity of (2.13) follows from $\boldsymbol{\Sigma}_0 = \boldsymbol{W}_0\boldsymbol{W}_0^T + \sigma_0^2\boldsymbol{I}_p$; completing the columns of $\boldsymbol{U}_0$ to a basis of $\mathbb{R}^p$ by an orthonormal basis of the orthogonal complement of the column space of $\boldsymbol{W}_0$ exhibits (2.13) as a spectral decomposition (2.4), whence the list of eigenvalues, the identification of the eigenspace of the q largest ones with the column space of $\boldsymbol{W}_0$, and the formula for the determinant. Since $\operatorname{rank}(\boldsymbol{W}_0) = q$ we have $\sigma_j(\boldsymbol{W}_0) > 0$ for all $j \leqslant q$, so all q leading eigenvalues are strictly larger than σ_0^2, which is therefore the smallest eigenvalue; this gives $\boldsymbol{\Sigma}_0 \succeq \sigma_0^2\boldsymbol{I}_p$, $\|\boldsymbol{\Sigma}_0^{-1}\| = \lambda_p(\boldsymbol{\Sigma}_0)^{-1} = \sigma_0^{-2}$ and (2.14). The second identity of (2.13) is checked by multiplying the two expressions and using $\boldsymbol{U}_0^T\boldsymbol{U}_0 = \boldsymbol{I}_q$ together with $\boldsymbol{U}_0^T(\boldsymbol{I}_p - \boldsymbol{P}_0) = \boldsymbol{0}$.

For (2.15), let $\epsilon := \frac{1}{4}\min(\sigma_0^2, \delta_0) > 0$. By part 5 of Lemma 15 we have almost surely $\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| \leqslant \epsilon$ for all large n, and then (2.8) gives $\lambda_p(\boldsymbol{S}_n) \geqslant \lambda_p(\boldsymbol{\Sigma}_0) - \epsilon = \sigma_0^2 - \epsilon \geqslant \frac{\sigma_0^2}{2}$ as well as $\lambda_q(\boldsymbol{S}_n) - \lambda_{q+1}(\boldsymbol{S}_n) \geqslant \lambda_q(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) - 2\epsilon \geqslant \frac{\delta_0}{2}$. □

The maximum likelihood estimator

The log-likelihood of the model (1.7) is the function on M given by

$$\mathbb{L}_n([\boldsymbol{W},\sigma^2]) := \sum_{i=1}^{n} \log p(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2]) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log\det\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) - \frac{n}{2}\operatorname{tr}\left(\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\boldsymbol{S}_n\right); \tag{2.16}$$

it depends on the sample only through $\boldsymbol{S}_n$ and it is genuinely a function on the quotient M, since the right hand side depends on $(\boldsymbol{W},\sigma^2)$ only through $\boldsymbol{\Sigma}$. This is precisely the reason why the model had to be quotiented in the first place: $\mathbb{L}_n$ is exactly constant along the fibers of π, so no maximizer of $\mathbb{L}_n$ on Θ can ever be unique, whereas on M it can. The following classical computation identifies the maximizer. We write $\lambda_1(\boldsymbol{S}) \geqslant \ldots \geqslant \lambda_p(\boldsymbol{S})$ for the eigenvalues of $\boldsymbol{S} \in \mathrm{Sym}(p)$, the space of real symmetric $p \times p$ matrices, repeated according to multiplicity, and we let

$$\mathcal{O} := \{\boldsymbol{S} \in \mathrm{Sym}(p) : \lambda_q(\boldsymbol{S}) > \lambda_{q+1}(\boldsymbol{S})\}, \tag{2.17}$$

which is an open subset of $\mathrm{Sym}(p)$ because the eigenvalues depend continuously on $\boldsymbol{S}$. For $\boldsymbol{S} \in \mathcal{O}$ we write $\boldsymbol{P}_q(\boldsymbol{S})$ for the orthogonal projector of $\mathbb{R}^p$ onto the sum of the eigenspaces of $\lambda_1(\boldsymbol{S}),\ldots,\lambda_q(\boldsymbol{S})$; this is well defined precisely because of the spectral gap in (2.17).

The next lemma collects the identities that let one pass between the projector form and the eigenvector form of the maximizer below. They are less innocent than they look: the second one uses that the columns of $\boldsymbol{K}_q$ are eigenvectors and not merely an orthonormal basis of the range of $\boldsymbol{P}_q$, as Remark 18 shows.

Lemma 17. Let $\boldsymbol{S} \in \mathcal{O}$, put $\lambda_j := \lambda_j(\boldsymbol{S})$, $\boldsymbol{P}_q := \boldsymbol{P}_q(\boldsymbol{S})$ and $\boldsymbol{\Lambda}_q := \mathrm{diag}(\lambda_1,\ldots,\lambda_q)$, and let $\boldsymbol{K}_q \in \mathbb{R}^{p\times q}$ be a matrix whose columns are orthonormal eigenvectors of $\boldsymbol{S}$ associated with $\lambda_1,\ldots,\lambda_q$ in this order, that is

$$\boldsymbol{K}_q^T\boldsymbol{K}_q = \boldsymbol{I}_q \qquad \text{and} \qquad \boldsymbol{S}\boldsymbol{K}_q = \boldsymbol{K}_q\boldsymbol{\Lambda}_q. \tag{2.18}$$

Then the column space of $\boldsymbol{K}_q$ is the range of $\boldsymbol{P}_q$, the projector $\boldsymbol{P}_q$ commutes with $\boldsymbol{S}$, and

$$\boldsymbol{P}_q = \boldsymbol{K}_q\boldsymbol{K}_q^T, \qquad \boldsymbol{S}\boldsymbol{P}_q = \boldsymbol{P}_q\boldsymbol{S} = \boldsymbol{P}_q\boldsymbol{S}\boldsymbol{P}_q = \boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{K}_q^T, \qquad \boldsymbol{P}_q\boldsymbol{S}\boldsymbol{P}_q - c\,\boldsymbol{P}_q = \boldsymbol{K}_q\left(\boldsymbol{\Lambda}_q - c\,\boldsymbol{I}_q\right)\boldsymbol{K}_q^T \tag{2.19}$$

for every $c \in \mathbb{R}$. Finally, a matrix $\boldsymbol{K}_q'$ satisfies (2.18) if and only if $\boldsymbol{K}_q' = \boldsymbol{K}_q\boldsymbol{R}$ for some $\boldsymbol{R} \in O(q)$ commuting with $\boldsymbol{\Lambda}_q$, that is block diagonal along the blocks of equal values in $(\lambda_1,\ldots,\lambda_q)$; in particular none of the matrices in (2.19) depends on the choice of $\boldsymbol{K}_q$.

Proof. Let E denote the sum of the eigenspaces of $\lambda_1,\ldots,\lambda_q$, so that $\boldsymbol{P}_q$ is by definition the orthogonal projector onto E. Because $\lambda_q > \lambda_{q+1}$, no eigenvalue λ_j with $j > q$ is equal to any of $\lambda_1,\ldots,\lambda_q$, so the total multiplicity of the latter is exactly q and $\dim E = q$. The columns of $\boldsymbol{K}_q$ are eigenvectors associated with those eigenvalues, hence lie in E, and they are orthonormal, hence span a q-dimensional subspace of E; therefore the column space of $\boldsymbol{K}_q$ is E. Since $\boldsymbol{K}_q^T\boldsymbol{K}_q = \boldsymbol{I}_q$, the matrix $\boldsymbol{K}_q\boldsymbol{K}_q^T$ is symmetric and idempotent with range E, that is $\boldsymbol{P}_q = \boldsymbol{K}_q\boldsymbol{K}_q^T$, which is the first identity.

For the second, $\boldsymbol{S}\boldsymbol{P}_q = \boldsymbol{S}\boldsymbol{K}_q\boldsymbol{K}_q^T = \boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{K}_q^T$ by (2.18); this matrix is symmetric, so $\boldsymbol{P}_q\boldsymbol{S} = (\boldsymbol{S}\boldsymbol{P}_q)^T = \boldsymbol{S}\boldsymbol{P}_q$, which is the asserted commutation, and $\boldsymbol{P}_q\boldsymbol{S}\boldsymbol{P}_q = \boldsymbol{K}_q\boldsymbol{K}_q^T\boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{K}_q^T = \boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{K}_q^T$ again by $\boldsymbol{K}_q^T\boldsymbol{K}_q = \boldsymbol{I}_q$. Subtracting $c\,\boldsymbol{P}_q = c\,\boldsymbol{K}_q\boldsymbol{K}_q^T$ gives the third identity.

Suppose now that $\boldsymbol{K}_q'$ also satisfies (2.18). By the first part both matrices have column space E, so $\boldsymbol{K}_q' = \boldsymbol{K}_q\boldsymbol{R}$ with $\boldsymbol{R} := \boldsymbol{K}_q^T\boldsymbol{K}_q'$, and $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{K}_q'^T\boldsymbol{K}_q\boldsymbol{K}_q^T\boldsymbol{K}_q' = \boldsymbol{K}_q'^T\boldsymbol{P}_q\boldsymbol{K}_q' = \boldsymbol{I}_q$ because the columns of $\boldsymbol{K}_q'$ lie in E; thus $\boldsymbol{R} \in O(q)$. Applying $\boldsymbol{S}$ gives $\boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{R} = \boldsymbol{S}\boldsymbol{K}_q\boldsymbol{R} = \boldsymbol{S}\boldsymbol{K}_q' = \boldsymbol{K}_q'\boldsymbol{\Lambda}_q = \boldsymbol{K}_q\boldsymbol{R}\boldsymbol{\Lambda}_q$, and multiplying on the left by $\boldsymbol{K}_q^T$ yields $\boldsymbol{\Lambda}_q\boldsymbol{R} = \boldsymbol{R}\boldsymbol{\Lambda}_q$; comparing entries, $(\lambda_i - \lambda_j)R_{ij} = 0$, so $R_{ij} = 0$ whenever $\lambda_i \neq \lambda_j$, which is the asserted block structure. Conversely such an $\boldsymbol{R}$ clearly returns a matrix satisfying (2.18). Independence of the choice is then immediate, since by (2.19) every matrix occurring there is expressed through $\boldsymbol{S}$ and $\boldsymbol{P}_q$ alone. $\square$

Remark 18. The hypothesis that the columns of $\boldsymbol{K}_q$ be eigenvectors, and not merely an orthonormal basis of the range of $\boldsymbol{P}_q$, cannot be dropped from the second identity of (2.19), even though it is irrelevant for the first. If $\boldsymbol{U} = \boldsymbol{K}_q\boldsymbol{R}$ with $\boldsymbol{R} \in O(q)$ arbitrary then $\boldsymbol{U}\boldsymbol{U}^T = \boldsymbol{P}_q$ still, whereas $\boldsymbol{U}\boldsymbol{\Lambda}_q\boldsymbol{U}^T = \boldsymbol{K}_q\boldsymbol{R}\boldsymbol{\Lambda}_q\boldsymbol{R}^T\boldsymbol{K}_q^T$ equals $\boldsymbol{K}_q\boldsymbol{\Lambda}_q\boldsymbol{K}_q^T$ only when $\boldsymbol{R}$ commutes with $\boldsymbol{\Lambda}_q$. For instance, for $p = 3$, $q = 2$ and $\boldsymbol{S} = \operatorname{diag}(2, 1, \frac{1}{2}) \in \mathcal{O}$, the vectors $\frac{1}{\sqrt{2}}(1, 1, 0)^T$ and $\frac{1}{\sqrt{2}}(1, -1, 0)^T$ form an orthonormal basis of the range of $\boldsymbol{P}_2 = \operatorname{diag}(1, 1, 0)$, but the corresponding matrix $\boldsymbol{U}\boldsymbol{\Lambda}_2\boldsymbol{U}^T$ has upper left 2×2 block $\begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$, whereas $\boldsymbol{P}_2\boldsymbol{S}\boldsymbol{P}_2$ has upper left block $\operatorname{diag}(2, 1)$. This is also the reason why the ordering requirement in (2.18) matters: the eigenvalues must be attached to the eigenvectors that carry them.

Proposition 19 (Tipping–Bishop)**.** Let $\boldsymbol{S} \in \mathcal{O}$ be positive definite and put $\lambda_j := \lambda_j(\boldsymbol{S})$ and $\boldsymbol{P}_q := \boldsymbol{P}_q(\boldsymbol{S})$. Then the function

$$[\boldsymbol{W}, \sigma^2] \longmapsto -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]) - \frac{1}{2}\operatorname{tr}\left(\left(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])\right)^{-1}\boldsymbol{S}\right)$$

attains its maximum over M at one and only one point $[\hat{\boldsymbol{W}}, \hat\sigma^2] \in M$, determined by

$$\hat\sigma^2 = \frac{1}{p-q}\sum_{j=q+1}^{p}\lambda_j, \qquad \hat{\boldsymbol{W}}\hat{\boldsymbol{W}}^T = \boldsymbol{P}_q\boldsymbol{S}\boldsymbol{P}_q - \hat\sigma^2\,\boldsymbol{P}_q. \tag{2.20}$$

Concretely, $\hat{\boldsymbol{W}} = \boldsymbol{K}_q(\boldsymbol{\Lambda}_q - \hat\sigma^2\boldsymbol{I}_q)^{\frac{1}{2}}\boldsymbol{R}$ for an arbitrary $\boldsymbol{R} \in O(q)$, where the columns of $\boldsymbol{K}_q \in \mathbb{R}^{p\times q}$ are orthonormal eigenvectors of $\boldsymbol{S}$ associated with $\lambda_1, \dots, \lambda_q$ and $\boldsymbol{\Lambda}_q = \operatorname{diag}(\lambda_1, \dots, \lambda_q)$.

Proof. This is Section 3.2 together with Appendix A of Tipping and Bishop (1999): equations (7) and (8) there give the stationary points and equations (17)–(20) of their Appendix A.2 show that the global maximum is attained exactly when the q *largest* eigenvalues of $\boldsymbol{S}$ are retained. Since $\lambda_q > \lambda_{q+1}$ the set of retained eigenvalues, and hence the projector $\boldsymbol{P}_q$, is unambiguous, so the maximizer is a single point of M. The two descriptions agree by Lemma 17: the last identity of (2.19) with $c = \hat\sigma^2$ gives

$$\boldsymbol{P}_q\boldsymbol{S}\boldsymbol{P}_q - \hat\sigma^2\boldsymbol{P}_q = \boldsymbol{K}_q\left(\boldsymbol{\Lambda}_q - \hat\sigma^2\boldsymbol{I}_q\right)\boldsymbol{K}_q^T = \left(\boldsymbol{K}_q\left(\boldsymbol{\Lambda}_q - \hat\sigma^2\boldsymbol{I}_q\right)^{\frac{1}{2}}\boldsymbol{R}\right)\left(\boldsymbol{K}_q\left(\boldsymbol{\Lambda}_q - \hat\sigma^2\boldsymbol{I}_q\right)^{\frac{1}{2}}\boldsymbol{R}\right)^T = \hat{\boldsymbol{W}}\hat{\boldsymbol{W}}^T$$

for every $\boldsymbol{R} \in O(q)$. Finally $[\hat{\boldsymbol{W}}, \hat\sigma^2]$ is really a point of M: $\hat\sigma^2 > 0$ because $\boldsymbol{S}$ is positive definite, and $\operatorname{rank}(\hat{\boldsymbol{W}}) = q$ because $\hat\sigma^2 \leqslant \lambda_{q+1} < \lambda_q$ forces every diagonal entry of $\boldsymbol{\Lambda}_q - \hat\sigma^2\boldsymbol{I}_q$ to be strictly positive. ☐

Definition 20. Whenever $\boldsymbol{S}_n$ is positive definite and lies in $\mathcal{O}$, we let $[\hat{\boldsymbol{W}}_n, \hat\sigma_n^2] \in M$ denote the maximum likelihood estimator, that is the unique maximizer of (2.16) over M supplied by Proposition 19 applied to $\boldsymbol{S} = \boldsymbol{S}_n$. On the complementary event we set $[\hat{\boldsymbol{W}}_n, \hat\sigma_n^2] := [\boldsymbol{W}_0, \sigma_0^2]$; Lemma 27 below shows that this convention is invoked only on a null set as soon as $n \geqslant p$.

The fitted covariance matrix and the exact fit equation

It is worth recording what the maximum likelihood estimator does to the covariance matrix rather than to the parameter, if only because (2.16) immediately suggests a competitor: instead of maximizing the likelihood, why not simply ask the model to reproduce the sample covariance matrix, that is, why not solve

$$\boldsymbol{S} = \boldsymbol{W}\boldsymbol{W}^T + \sigma^2\boldsymbol{I}_p, \qquad [\boldsymbol{W}, \sigma^2] \in M? \tag{2.21}$$

The next two propositions answer this: the maximum likelihood fit is the sample covariance matrix with its trailing eigenvalues flattened, (2.21) is almost surely unsolvable, and its least squares relaxation returns the maximum likelihood estimator again.

Proposition 21. Let $\boldsymbol{S} \in \mathcal{O}$ be positive definite, let $[\hat{\boldsymbol{W}}, \hat{\sigma}^2]$ be the maximizer supplied by Proposition 19 and let

$$\hat{\boldsymbol{\Sigma}} := \boldsymbol{\Sigma}\left([\hat{\boldsymbol{W}}, \hat{\sigma}^2]\right) = \hat{\boldsymbol{W}}\hat{\boldsymbol{W}}^T + \hat{\sigma}^2 \boldsymbol{I}_p \tag{2.22}$$

be the *fitted covariance matrix*. Write $\lambda_j := \lambda_j(\boldsymbol{S})$, let $\boldsymbol{k}_1, \ldots, \boldsymbol{k}_p$ be orthonormal eigenvectors of $\boldsymbol{S}$ ordered as in (2.4) and put $\boldsymbol{P}_q := \boldsymbol{P}_q(\boldsymbol{S})$. Then

1. $\hat{\boldsymbol{\Sigma}}$ has the same eigenvectors as $\boldsymbol{S}$, keeps its q largest eigenvalues and replaces the remaining $p-q$ by their average $\hat{\sigma}^2$:
$$\hat{\boldsymbol{\Sigma}} = \sum_{j=1}^{q} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T + \hat{\sigma}^2 \sum_{j=q+1}^{p} \boldsymbol{k}_j \boldsymbol{k}_j^T, \qquad \hat{\boldsymbol{\Sigma}} - \boldsymbol{S} = \sum_{j=q+1}^{p} \left(\hat{\sigma}^2 - \lambda_j\right) \boldsymbol{k}_j \boldsymbol{k}_j^T. \tag{2.23}$$

2. $\operatorname{tr} \hat{\boldsymbol{\Sigma}} = \operatorname{tr} \boldsymbol{S}$ and $\boldsymbol{P}_q \hat{\boldsymbol{\Sigma}} \boldsymbol{P}_q = \boldsymbol{P}_q \boldsymbol{S} \boldsymbol{P}_q$; moreover $\hat{\boldsymbol{\Sigma}}$ is positive definite, lies in $\mathcal{O}$, satisfies $\boldsymbol{P}_q(\hat{\boldsymbol{\Sigma}}) = \boldsymbol{P}_q(\boldsymbol{S})$, and the map $\boldsymbol{S} \mapsto \hat{\boldsymbol{\Sigma}}$ is idempotent.

3. The fit is exact if and only if the trailing spectrum of $\boldsymbol{S}$ is flat:
$$\hat{\boldsymbol{\Sigma}} = \boldsymbol{S} \iff \lambda_{q+1} = \lambda_{q+2} = \ldots = \lambda_p \iff \boldsymbol{S} \in \boldsymbol{\Sigma}(M), \tag{2.24}$$
that is, if and only if (2.21) is solvable, in which case $[\hat{\boldsymbol{W}}, \hat{\sigma}^2]$ is its unique solution in M.

Proof. 1. Lemma 17 applied to $\boldsymbol{K}_q := (\boldsymbol{k}_1, \ldots, \boldsymbol{k}_q)$, which satisfies (2.18) by (2.4), gives $\boldsymbol{P}_q = \sum_{j \leqslant q} \boldsymbol{k}_j \boldsymbol{k}_j^T$ and $\boldsymbol{P}_q \boldsymbol{S} \boldsymbol{P}_q = \sum_{j \leqslant q} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T$, so (2.20) gives

$$\hat{\boldsymbol{\Sigma}} = \left(\boldsymbol{P}_q \boldsymbol{S} \boldsymbol{P}_q - \hat{\sigma}^2 \boldsymbol{P}_q\right) + \hat{\sigma}^2 \left(\boldsymbol{P}_q + (\boldsymbol{I}_p - \boldsymbol{P}_q)\right) = \sum_{j \leqslant q} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T + \hat{\sigma}^2 \sum_{j > q} \boldsymbol{k}_j \boldsymbol{k}_j^T,$$

and subtracting $\boldsymbol{S} = \sum_{j=1}^{p} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T$ gives the second identity of (2.23).

2. Taking traces in (2.23) and using $\hat{\sigma}^2 = \frac{1}{p-q} \sum_{j>q} \lambda_j$ gives $\operatorname{tr} \hat{\boldsymbol{\Sigma}} = \sum_{j \leqslant q} \lambda_j + (p-q)\hat{\sigma}^2 = \operatorname{tr} \boldsymbol{S}$, and $\boldsymbol{P}_q \hat{\boldsymbol{\Sigma}} \boldsymbol{P}_q = \sum_{j \leqslant q} \lambda_j \boldsymbol{k}_j \boldsymbol{k}_j^T = \boldsymbol{P}_q \boldsymbol{S} \boldsymbol{P}_q$ because the $\boldsymbol{k}_j$ are orthonormal. The eigenvalues of $\hat{\boldsymbol{\Sigma}}$ are $\lambda_1, \ldots, \lambda_q$ and $\hat{\sigma}^2$ with multiplicity $p-q$; they are all strictly positive, and $\hat{\sigma}^2 \leqslant \lambda_{q+1} < \lambda_q$ shows both that $\hat{\boldsymbol{\Sigma}} \in \mathcal{O}$ and that its leading q-dimensional eigenspace is $\operatorname{span}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_q)$, that is $\boldsymbol{P}_q(\hat{\boldsymbol{\Sigma}}) = \boldsymbol{P}_q(\boldsymbol{S})$. Idempotence follows: the trailing eigenvalues of $\hat{\boldsymbol{\Sigma}}$ are already all equal to $\hat{\sigma}^2$, so applying (2.23) to $\hat{\boldsymbol{\Sigma}}$ reproduces it.

3. By the second identity of (2.23) the difference $\hat{\boldsymbol{\Sigma}} - \boldsymbol{S}$ is symmetric with eigenvalues $\hat{\sigma}^2 - \lambda_j$, $j > q$, in the directions $\boldsymbol{k}_{q+1}, \ldots, \boldsymbol{k}_p$, so it vanishes if and only if $\lambda_j = \hat{\sigma}^2$ for every $j > q$, that is if and only if the trailing eigenvalues are all equal, in which case their common value is their average $\hat{\sigma}^2$. If this holds then $\boldsymbol{S} = \hat{\boldsymbol{\Sigma}} \in \boldsymbol{\Sigma}(M)$; conversely, if $\boldsymbol{S} = \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])$ for some $[\boldsymbol{W}, \sigma^2] \in M$, then Lemma 16 applied to $(\boldsymbol{W}, \sigma^2)$ in place of $(\boldsymbol{W}_0, \sigma_0^2)$ shows that the $p-q$ smallest eigenvalues of $\boldsymbol{S}$ all equal σ^2. Uniqueness of the solution is uniqueness of the maximizer in Proposition 19, since any solution of (2.21) satisfies $\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]) = \boldsymbol{S} = \hat{\boldsymbol{\Sigma}}$ and $\boldsymbol{\Sigma}$ is injective on M by construction. □

Remark 22. Two consequences of (2.24) deserve to be separated. If $q = p-1$ the trailing spectrum consists of the single eigenvalue λ_p, the flatness condition is vacuous, and the fit is always exact: $\hat{\boldsymbol{\Sigma}} = \boldsymbol{S}$, and (2.21) is always solvable. This is not an accident, since $\dim M = p(p-1) + 1 - \frac{(p-1)(p-2)}{2} = \frac{p(p+1)}{2} = \dim \operatorname{Sym}(p)$ when $q = p-1$: the model is then saturated and imposes no constraint on $\boldsymbol{\Sigma}$ beyond positive definiteness and a simple smallest eigenvalue. If $q < p-1$, by contrast, an exact fit forces $\boldsymbol{S}$ to have a repeated eigenvalue. Matrices with a repeated eigenvalue form a Lebesgue null subset of $\operatorname{Sym}(p)$, being the zero set of the discriminant of the characteristic polynomial as in the proof of Lemma 25, and the law of $\boldsymbol{S}_n$ is absolutely continuous for $n \geqslant p$ by Lemma 27; hence almost surely $\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]) \neq \boldsymbol{S}_n$ and (2.21) has no solution at all. The two matrices are nevertheless close: by (2.23), (2.8) and part 5 of Lemma 15,

$$\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| = \max_{j>q} \left|\hat{\sigma}_n^2 - \lambda_j(\boldsymbol{S}_n)\right| \leqslant 2 \max_{j>q} \left|\lambda_j(\boldsymbol{S}_n) - \sigma_0^2\right| \leqslant 2\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

the middle inequality holding because $\hat{\sigma}_n^2$ is an average of the $\lambda_j(\boldsymbol{S}_n)$, $j > q$. The discrepancy disappears in the limit precisely because the population matrix does have a flat trailing spectrum, $\lambda_{q+1}(\boldsymbol{\Sigma}_0) = \ldots = \lambda_p(\boldsymbol{\Sigma}_0) = \sigma_0^2$, by Lemma 16.

Since (2.21) is almost surely unsolvable, the natural way to rescue the estimator it was meant to define is to require it only approximately, that is to replace it by the minimum distance, or least squares, estimator

$$[\tilde{\boldsymbol{W}}_n, \tilde{\sigma}_n^2] := \underset{[\boldsymbol{W},\sigma^2]\in M}{\arg\min} \left\|\boldsymbol{S}_n - \boldsymbol{W}\boldsymbol{W}^T - \sigma^2 \boldsymbol{I}_p\right\|_F^2. \tag{2.25}$$

This turns out to produce nothing new.

Proposition 23. Let $\boldsymbol{S} \in \mathcal{O}$ be positive definite and keep the notation of Proposition 21. Then

$$\min_{[\boldsymbol{W},\sigma^2]\in M} \left\|\boldsymbol{S} - \boldsymbol{W}\boldsymbol{W}^T - \sigma^2 \boldsymbol{I}_p\right\|_F^2 = \sum_{j=q+1}^{p} \left(\lambda_j - \hat{\sigma}^2\right)^2, \tag{2.26}$$

the minimum being attained at $[\hat{\boldsymbol{W}}, \hat{\sigma}^2]$, and every minimizer has $\sigma^2 = \hat{\sigma}^2$. In particular the least squares estimator (2.25) may be taken to be the maximum likelihood estimator of Definition 20.

Proof. Write $\boldsymbol{B} := \boldsymbol{W}\boldsymbol{W}^T$, which ranges exactly over the positive semi-definite matrices of rank q as $[\boldsymbol{W}, \sigma^2]$ ranges over M with $\sigma^2 > 0$ fixed, put $\boldsymbol{C} := \boldsymbol{S} - \sigma^2 \boldsymbol{I}_p$ and $\boldsymbol{R} := \boldsymbol{C} - \boldsymbol{B}$, and note $\lambda_j(\boldsymbol{C}) = \lambda_j - \sigma^2$. We first bound $\|\boldsymbol{R}\|_F^2$ from below for fixed σ^2. Since $\boldsymbol{B} \succeq \boldsymbol{0}$ we have $\boldsymbol{v}^T\boldsymbol{R}\boldsymbol{v} \leqslant \boldsymbol{v}^T\boldsymbol{C}\boldsymbol{v}$ for every $\boldsymbol{v}$, so (2.6) gives

$$\lambda_j(\boldsymbol{R}) \leqslant \lambda_j(\boldsymbol{C}) \qquad \text{for } j = 1, \ldots, p. \tag{2.27}$$

Moreover $K := \ker \boldsymbol{B}$ has dimension at least $p - q$, and for a subspace V with $\dim V = j + q$ the intersection $V \cap K$ has dimension at least j; choosing $W \subseteq V \cap K$ with $\dim W = j$ and using $\boldsymbol{v}^T\boldsymbol{B}\boldsymbol{v} = 0$ on K,

$$\min_{\boldsymbol{v}\in V,\ \boldsymbol{v}^T\boldsymbol{v}=1} \boldsymbol{v}^T\boldsymbol{C}\boldsymbol{v} \leqslant \min_{\boldsymbol{v}\in W,\ \boldsymbol{v}^T\boldsymbol{v}=1} \boldsymbol{v}^T\boldsymbol{C}\boldsymbol{v} = \min_{\boldsymbol{v}\in W,\ \boldsymbol{v}^T\boldsymbol{v}=1} \boldsymbol{v}^T\boldsymbol{R}\boldsymbol{v} \leqslant \lambda_j(\boldsymbol{R}),$$

and taking the maximum over such V in (2.6),

$$\lambda_{j+q}(\boldsymbol{C}) \leqslant \lambda_j(\boldsymbol{R}) \qquad \text{for } j = 1, \ldots, p - q. \tag{2.28}$$

Let now $n_\sigma := \#\{j : \lambda_j \geqslant \sigma^2\}$. For $1 \leqslant j \leqslant n_\sigma - q$ we have $\lambda_{j+q}(\boldsymbol{C}) \geqslant 0$, so (2.28) gives $\lambda_j(\boldsymbol{R})^2 \geqslant (\lambda_{j+q} - \sigma^2)^2$, while for $j > n_\sigma$ we have $\lambda_j(\boldsymbol{C}) < 0$, so (2.27) gives $\lambda_j(\boldsymbol{R})^2 \geqslant (\sigma^2 - \lambda_j)^2$. These two ranges of indices are disjoint, whence

$$\|\boldsymbol{R}\|_F^2 = \sum_{j=1}^{p} \lambda_j(\boldsymbol{R})^2 \geqslant \sum_{j=q+1}^{n_\sigma} \left(\lambda_j - \sigma^2\right)^2 + \sum_{j>n_\sigma} \left(\sigma^2 - \lambda_j\right)^2 = \sum_{j\leqslant q} \left[\left(\sigma^2 - \lambda_j\right)_+\right]^2 + \sum_{j>q} \left(\lambda_j - \sigma^2\right)^2 =: L(\sigma^2),$$

where $(x)_+ := \max(x, 0)$, the first sum being empty when $n_\sigma \leqslant q$; the last equality is checked separately in the cases $n_\sigma \geqslant q$ and $n_\sigma < q$. The function L is strictly convex and continuously differentiable on $\mathbb{R}$, with $L'(x) = 2\sum_{j\leqslant q}(x - \lambda_j)_+ - 2\sum_{j>q}(\lambda_j - x)$. Since $\hat{\sigma}^2 \leqslant \lambda_{q+1} \leqslant \lambda_j$ for $j \leqslant q$ the first sum vanishes at $x = \hat{\sigma}^2$, and the second vanishes there because $\hat{\sigma}^2$ is the average of $\lambda_{q+1}, \ldots, \lambda_p$; hence $\hat{\sigma}^2$ is the unique minimizer of L, with $L(\hat{\sigma}^2) = \sum_{j>q}(\lambda_j - \hat{\sigma}^2)^2$. Therefore every $[\boldsymbol{W}, \sigma^2] \in M$ satisfies $\|\boldsymbol{S} - \boldsymbol{W}\boldsymbol{W}^T - \sigma^2\boldsymbol{I}_p\|_F^2 \geqslant L(\sigma^2) \geqslant L(\hat{\sigma}^2)$, with the second inequality strict unless $\sigma^2 = \hat{\sigma}^2$. Finally the bound is attained at $[\hat{\boldsymbol{W}}, \hat{\sigma}^2]$: by (2.23), $\boldsymbol{S} - \hat{\boldsymbol{W}}\hat{\boldsymbol{W}}^T - \hat{\sigma}^2\boldsymbol{I}_p = \boldsymbol{S} - \hat{\boldsymbol{\Sigma}} = \sum_{j>q}(\lambda_j - \hat{\sigma}^2)\boldsymbol{k}_j\boldsymbol{k}_j^T$, whose squared Frobenius norm is exactly $\sum_{j>q}(\lambda_j - \hat{\sigma}^2)^2$. □

Remark 24. The coincidence in Proposition 23 is a small piece of good luck rather than a general principle. Maximizing (2.16) is, up to constants not depending on the parameter, the same as minimizing over M the Stein loss $\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_n) - \log\det(\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_n) - p$ with $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])$, which is a very different discrepancy from the squared Frobenius distance minimized in (2.25); that the two are minimized at the same point here reflects the special structure of (2.23), namely that the optimal fit acts only on the eigenvalues of $\boldsymbol{S}_n$ and leaves its eigenvectors alone. The practical conclusion is that the exact fit equation (2.21) defines no new estimator: it is either unsolvable, which is the generic case, or solved by the maximum likelihood estimator, and its least squares relaxation is the maximum likelihood estimator as well.

Genericity of the sample covariance matrix

Both the well-posedness of the estimation error and the consistency of the maximum likelihood estimator rest on the behaviour of the spectral projector $\boldsymbol{P}_q$, which we now record.

Lemma 25.

1. $\mathcal{O}$ is open and connected in $\mathrm{Sym}(p)$.
2. $\boldsymbol{P}_q : \mathcal{O} \to \mathrm{Sym}(p)$ is real analytic.
3. Let $\boldsymbol{U}_0 \in \mathbb{R}^{p\times q}$ have orthonormal columns and define $\phi : \mathcal{O} \to \mathbb{R}$ by $\phi(\boldsymbol{S}) := \det\left(\boldsymbol{U}_0^T \boldsymbol{P}_q(\boldsymbol{S}) \boldsymbol{U}_0\right)$. Then ϕ is real analytic and $\phi^{-1}(\{0\})$ is a Lebesgue null subset of $\mathrm{Sym}(p) \cong \mathbb{R}^{\frac{p(p+1)}{2}}$.
4. Let $\boldsymbol{S}_* \in \mathcal{O}$ and put
$$\delta_* := \lambda_q(\boldsymbol{S}_*) - \lambda_{q+1}(\boldsymbol{S}_*) > 0, \qquad \rho_* := \lambda_1(\boldsymbol{S}_*) - \lambda_{q+1}(\boldsymbol{S}_*) \geqslant \delta_*. \tag{2.29}$$
Then every $\boldsymbol{S} \in \mathrm{Sym}(p)$ with $\|\boldsymbol{S} - \boldsymbol{S}_*\| \leqslant \frac{\delta_*}{8}$ belongs to $\mathcal{O}$ and satisfies
$$\|\boldsymbol{P}_q(\boldsymbol{S}) - \boldsymbol{P}_q(\boldsymbol{S}_*)\| \leqslant \frac{16\rho_*}{\delta_*^2}\,\|\boldsymbol{S} - \boldsymbol{S}_*\|. \tag{2.30}$$
In particular $\boldsymbol{P}_q$ is Lipschitz on a neighbourhood of every point of $\mathcal{O}$, with a constant controlled by the spectral gap at that point alone.
5. $\boldsymbol{P}_q$ is continuous on $\mathcal{O}$; concretely, if $\boldsymbol{S}_* \in \mathcal{O}$ and $\boldsymbol{S}_n \to \boldsymbol{S}_*$ in $\mathrm{Sym}(p)$, then $\boldsymbol{S}_n \in \mathcal{O}$ for all large n and $\boldsymbol{P}_q(\boldsymbol{S}_n) \to \boldsymbol{P}_q(\boldsymbol{S}_*)$.

Proof. 1. Openness was already observed. For connectedness, fix $\boldsymbol{S} \in \mathcal{O}$ and diagonalize $\boldsymbol{S} = \boldsymbol{K}\mathrm{diag}(\boldsymbol{\lambda})\boldsymbol{K}^T$ with $\boldsymbol{K} \in O(p)$ and $\boldsymbol{\lambda} = (\lambda_1(\boldsymbol{S}), \dots, \lambda_p(\boldsymbol{S}))$ non-increasing; replacing one column of $\boldsymbol{K}$ by its negative if necessary, which does not change $\boldsymbol{S}$, we may assume $\det(\boldsymbol{K}) = 1$. Let $\boldsymbol{\mu} := (2, \dots, 2, 1, \dots, 1)$ with the value 2 repeated q times, and consider first the path $t \mapsto \boldsymbol{K}\mathrm{diag}((1-t)\boldsymbol{\lambda} + t\boldsymbol{\mu})\boldsymbol{K}^T$ for $t \in [0, 1]$. The vector $(1-t)\boldsymbol{\lambda} + t\boldsymbol{\mu}$ is non-increasing, being a convex combination of non-increasing vectors, so it lists the ordered eigenvalues, and the gap at position q equals $(1-t)(\lambda_q(\boldsymbol{S}) - \lambda_{q+1}(\boldsymbol{S})) + t > 0$; the path therefore stays inside $\mathcal{O}$ and joins $\boldsymbol{S}$ to $\boldsymbol{K}\mathrm{diag}(\boldsymbol{\mu})\boldsymbol{K}^T$. Next, $SO(p)$ is path connected, so there is a path $u \mapsto \boldsymbol{K}(u)$ in $SO(p)$ from $\boldsymbol{K}$ to $\boldsymbol{I}_p$; the induced path $u \mapsto \boldsymbol{K}(u)\mathrm{diag}(\boldsymbol{\mu})\boldsymbol{K}(u)^T$ has constant spectrum $\boldsymbol{\mu}$, hence constant gap $1 > 0$ at position q, stays in $\mathcal{O}$, and ends at $\mathrm{diag}(\boldsymbol{\mu})$. Every point of $\mathcal{O}$ is thus joined to the fixed point $\mathrm{diag}(\boldsymbol{\mu})$ by a path inside $\mathcal{O}$.

2. Fix $\boldsymbol{S}_* \in \mathcal{O}$, choose $c \in \mathbb{R}$ with $\lambda_{q+1}(\boldsymbol{S}_*) < c < \lambda_q(\boldsymbol{S}_*)$ and let $\Gamma \subset \mathbb{C}$ be the positively oriented circle of centre $\frac{1}{2}(\lambda_1(\boldsymbol{S}_*) + c)$ and radius $\frac{1}{2}(\lambda_1(\boldsymbol{S}_*) - \lambda_{q+1}(\boldsymbol{S}_*))$, which meets the real axis in the interval with endpoints $\frac{1}{2}(c + \lambda_{q+1}(\boldsymbol{S}_*)) \in (\lambda_{q+1}(\boldsymbol{S}_*), c)$ and $\lambda_1(\boldsymbol{S}_*) + \frac{1}{2}(c - \lambda_{q+1}(\boldsymbol{S}_*)) > \lambda_1(\boldsymbol{S}_*)$, so that Γ encircles $\lambda_1(\boldsymbol{S}_*), \dots, \lambda_q(\boldsymbol{S}_*)$ and no other eigenvalue of $\boldsymbol{S}_*$. Put $\delta := \min_{z\in\Gamma} \mathrm{dist}(z, \mathrm{spec}(\boldsymbol{S}_*)) > 0$. If $\boldsymbol{S} \in \mathrm{Sym}(p)$ satisfies $\|\boldsymbol{S} - \boldsymbol{S}_*\|_2 < \delta$ then $z\boldsymbol{I}_p - \boldsymbol{S}$ is invertible for every $z \in \Gamma$, Weyl's inequality keeps exactly q eigenvalues of $\boldsymbol{S}$ inside Γ, and diagonalizing $\boldsymbol{S}$ shows, by the residue theorem applied to each eigenvalue separately, that

$$\boldsymbol{P}_q(\boldsymbol{S}) = \frac{1}{2\pi i}\oint_\Gamma (z\boldsymbol{I}_p - \boldsymbol{S})^{-1}\mathrm{d}z. \tag{2.31}$$

Expanding the resolvent in a Neumann series,

$$(z\boldsymbol{I}_p - \boldsymbol{S})^{-1} = \sum_{k=0}^{\infty}\left[(z\boldsymbol{I}_p - \boldsymbol{S}_*)^{-1}(\boldsymbol{S} - \boldsymbol{S}_*)\right]^k (z\boldsymbol{I}_p - \boldsymbol{S}_*)^{-1},$$

a series which converges absolutely and uniformly in $z \in \Gamma$ whenever $\|\boldsymbol{S} - \boldsymbol{S}_*\|_2 < \delta$, since $\|(z\boldsymbol{I}_p - \boldsymbol{S}_*)^{-1}\|_2 \leqslant \delta^{-1}$ there. Each summand is a homogeneous polynomial of degree k in the entries of $\boldsymbol{S} - \boldsymbol{S}_*$ whose coefficients are continuous in z, so integrating term by term over the compact contour Γ exhibits $\boldsymbol{P}_q$ as a convergent power series in the entries of $\boldsymbol{S} - \boldsymbol{S}_*$ on that ball.

The argument just given is the standard Riesz projection argument of analytic perturbation theory and is not new: it is carried out in Kato (1995), Chapter II, §1.4, where the projection associated with the eigenvalues enclosed by Γ is defined by the contour integral (2.31) and expanded, by term by term integration of the Neumann series of §1.3 there, into a convergent power series in the perturbation, and in §2.1 of the same chapter for a group of eigenvalues, which is the case needed here; §6.1 specializes the discussion to symmetric operators. The use of the contour integral in perturbation theory goes back to Sz.-Nagy and to Kato himself, replacing the earlier method of Rellich, as the footnote to §1.4 of Kato (1995) records. We have reproduced the proof because in the real symmetric finite-dimensional case it is short, and because part 4 below needs the resulting bound with an explicit constant.

3. Real analyticity of ϕ is immediate from 2, the determinant being a polynomial. Moreover ϕ does not vanish identically: taking $\boldsymbol{S}_\star := \boldsymbol{I}_p + \boldsymbol{U}_0\boldsymbol{U}_0^T$, whose eigenvalues are 2 with multiplicity q, with eigenspace the column space of $\boldsymbol{U}_0$, and 1 with multiplicity $p-q$, we get $\boldsymbol{S}_\star \in \mathcal{O}$, $\boldsymbol{P}_q(\boldsymbol{S}_\star) = \boldsymbol{U}_0\boldsymbol{U}_0^T$ and $\phi(\boldsymbol{S}_\star) = \det(\boldsymbol{I}_q) = 1$. By 1 the domain $\mathcal{O}$ is open and connected, so the zero set of the non-trivial real analytic function ϕ is Lebesgue null in $\mathrm{Sym}(p)$ by Mityagin (2020). Finally $\mathrm{Sym}(p) \setminus \mathcal{O}$ is itself Lebesgue null: it is contained in the zero set of the discriminant of the characteristic polynomial of $\boldsymbol{S}$, which is a polynomial in the entries of $\boldsymbol{S}$ and does not vanish at $\mathrm{diag}(1, 2, \ldots, p)$.

4. Continuity is already contained in 2; what has to be produced here is the explicit constant, and for that we specialize the contour of 2 to the midpoint $c := \frac{1}{2}\left(\lambda_q(\boldsymbol{S}_*) + \lambda_{q+1}(\boldsymbol{S}_*)\right)$ of the spectral gap. Then Γ is the circle of centre $a := \frac{1}{2}\left(\lambda_1(\boldsymbol{S}_*) + c\right)$ and radius $r := \frac{\rho_*}{2}$, which meets the real axis at $a - r = \lambda_{q+1}(\boldsymbol{S}_*) + \frac{\delta_*}{4}$ and $a + r = \lambda_1(\boldsymbol{S}_*) + \frac{\delta_*}{4}$. A real number λ is at distance $\left|r - |\lambda - a|\right|$ from Γ, and $\lambda \mapsto r - |\lambda - a|$ is concave, so on the interval $[\lambda_q(\boldsymbol{S}_*), \lambda_1(\boldsymbol{S}_*)]$, which contains $\lambda_1(\boldsymbol{S}_*), \ldots, \lambda_q(\boldsymbol{S}_*)$, it is minimized at an endpoint, where it takes the values $a + r - \lambda_1(\boldsymbol{S}_*) = \frac{\delta_*}{4}$ and $\lambda_q(\boldsymbol{S}_*) - (a - r) = \frac{3\delta_*}{4}$; and every eigenvalue $\lambda_j(\boldsymbol{S}_*)$ with $j > q$ lies to the left of $a - r$ at distance at least $(a - r) - \lambda_{q+1}(\boldsymbol{S}_*) = \frac{\delta_*}{4}$. Hence the quantity δ of part 2 satisfies $\delta \geqslant \frac{\delta_*}{4}$.

Let now $\boldsymbol{E} := \boldsymbol{S} - \boldsymbol{S}_*$ with $\|\boldsymbol{E}\| \leqslant \frac{\delta_*}{8} \leqslant \frac{\delta}{2}$. By (2.8), $\lambda_q(\boldsymbol{S}) - \lambda_{q+1}(\boldsymbol{S}) \geqslant \delta_* - 2\|\boldsymbol{E}\| \geqslant \frac{3\delta_*}{4} > 0$, so $\boldsymbol{S} \in \mathcal{O}$, and every eigenvalue of $\boldsymbol{S}$ is within $\|\boldsymbol{E}\|$ of an eigenvalue of $\boldsymbol{S}_*$, so that $\mathrm{dist}(z, \mathrm{spec}(\boldsymbol{S})) \geqslant \delta - \|\boldsymbol{E}\| \geqslant \frac{\delta}{2}$ for $z \in \Gamma$. Both resolvents are therefore defined on Γ and (2.31) holds for $\boldsymbol{S}$ as well as for $\boldsymbol{S}_*$. Diagonalizing shows that a real symmetric matrix has $\|(z\boldsymbol{I}_p - \boldsymbol{S})^{-1}\| = \mathrm{dist}(z, \mathrm{spec}(\boldsymbol{S}))^{-1}$ for every $z \notin \mathrm{spec}(\boldsymbol{S})$, so subtracting the two copies of (2.31), inserting the resolvent identity

$$(z\boldsymbol{I}_p - \boldsymbol{S})^{-1} - (z\boldsymbol{I}_p - \boldsymbol{S}_*)^{-1} = (z\boldsymbol{I}_p - \boldsymbol{S})^{-1}\boldsymbol{E}\,(z\boldsymbol{I}_p - \boldsymbol{S}_*)^{-1},$$

and estimating the integrand along the contour, of length $2\pi r$, gives

$$\|\boldsymbol{P}_q(\boldsymbol{S}) - \boldsymbol{P}_q(\boldsymbol{S}_*)\| \leqslant \frac{1}{2\pi}\,(2\pi r)\,\frac{2}{\delta}\cdot\frac{1}{\delta}\,\|\boldsymbol{E}\| = \frac{2r}{\delta^2}\|\boldsymbol{E}\| \leqslant \frac{16\rho_*}{\delta_*^2}\|\boldsymbol{E}\|,$$

using $r = \frac{\rho_*}{2}$ and $\delta \geqslant \frac{\delta_*}{4}$.

5. This is contained in 2, and also in 4, but it needs neither and may be quoted directly: it is Theorem 5.1 of Kato (1995), Chapter II, §5.1, a section entitled “Continuity of the eigenvalues and the total projection”, which

asserts that when the operator depends continuously on its argument its eigenvalues are continuous and so is the total projection attached to the group of eigenvalues encircled by a fixed contour; the discussion immediately preceding that theorem states the conclusion in the form used here, for a sequence of operators converging to a limit. One applies it to each of the distinct eigenvalues of $\boldsymbol{S}_*$ among $\lambda_1(\boldsymbol{S}_*), \ldots, \lambda_q(\boldsymbol{S}_*)$, finitely many, whose total projections sum to $\boldsymbol{P}_q$; that $\boldsymbol{S}_n$ eventually lies in $\mathcal{O}$, so that $\boldsymbol{P}_q$ may be evaluated at it at all, is (2.8) once more. □

Remark 26. The factor ρ_* in (2.30) is an artefact of estimating a circular contour that must enclose the whole leading part of the spectrum; the sharp form of the estimate, the Davis–Kahan $\sin\Theta$ theorem, replaces $\frac{16\rho_*}{\delta_*^2}$ by $\frac{2}{\delta_*}$. We do not need the sharp constant anywhere: only the fact that the Lipschitz constant is finite and depends on $\boldsymbol{S}_*$ solely through its spectrum.

Lemma 27. *Suppose $n \geqslant p$. Then, almost surely, $\boldsymbol{S}_n$ is positive definite, $\boldsymbol{S}_n \in \mathcal{O}$, and the $q \times q$ matrix $\boldsymbol{W}_0^T \hat{\boldsymbol{W}}_n$ is invertible. In particular the orthogonal matrix*

$$\boldsymbol{Q}_n := \boldsymbol{Q}(\boldsymbol{W}_0, \hat{\boldsymbol{W}}_n) = \left(\hat{\boldsymbol{W}}_n^T \boldsymbol{W}_0 \boldsymbol{W}_0^T \hat{\boldsymbol{W}}_n\right)^{-\frac{1}{2}} \hat{\boldsymbol{W}}_n^T \boldsymbol{W}_0 \in O(q) \tag{2.32}$$

*of Proposition **??** is almost surely well defined.*

Proof. We first check that the law of $\boldsymbol{S}_n$ is absolutely continuous with respect to the Lebesgue measure of $\mathrm{Sym}(p)$. Let $\boldsymbol{Y}_1 \in \mathbb{R}^{p\times p}$ be the matrix whose rows are $\boldsymbol{X}_1^T, \ldots, \boldsymbol{X}_p^T$ and let $\boldsymbol{Y}_2$ collect the remaining $n-p$ observations, so that $\boldsymbol{S}_n = \frac{1}{n}\left(\boldsymbol{Y}_1^T\boldsymbol{Y}_1 + \boldsymbol{Y}_2^T\boldsymbol{Y}_2\right)$ and $\boldsymbol{Y}_1$ is independent of $\boldsymbol{Y}_2$ with a strictly positive Lebesgue density on $\mathbb{R}^{p\times p}$. Condition on $\boldsymbol{Y}_2$ and consider $T(\boldsymbol{Y}) := \frac{1}{n}(\boldsymbol{Y}^T\boldsymbol{Y} + \boldsymbol{Y}_2^T\boldsymbol{Y}_2)$ on the open set of invertible $\boldsymbol{Y} \in \mathbb{R}^{p\times p}$, whose complement is Lebesgue null. Its differential at $\boldsymbol{Y}$ is $\boldsymbol{H} \mapsto \frac{1}{n}(\boldsymbol{H}^T\boldsymbol{Y} + \boldsymbol{Y}^T\boldsymbol{H})$, which is onto $\mathrm{Sym}(p)$: given $\boldsymbol{B} \in \mathrm{Sym}(p)$, the choice $\boldsymbol{H} = \frac{n}{2}\boldsymbol{Y}^{-T}\boldsymbol{B}$ returns $\boldsymbol{B}$. Thus T is a smooth submersion, and by the rank theorem (Lee, 2013, Theorem 4.12) every point has a neighbourhood on which T is, up to diffeomorphisms, a linear projection $\mathbb{R}^{p^2} \to \mathbb{R}^{\frac{p(p+1)}{2}}$. Since preimages of Lebesgue null sets under such projections are Lebesgue null by Fubini's theorem, and diffeomorphisms preserve Lebesgue null sets, $T^{-1}(N)$ is Lebesgue null for every Lebesgue null $N \subseteq \mathrm{Sym}(p)$, after covering the domain by countably many such neighbourhoods. Hence the conditional law of $\boldsymbol{S}_n$ given $\boldsymbol{Y}_2$ charges no Lebesgue null set, and the same holds for its mixture over $\boldsymbol{Y}_2$.

By Lemma 25, the set of $\boldsymbol{S} \in \mathrm{Sym}(p)$ that either lie outside $\mathcal{O}$ or satisfy $\phi(\boldsymbol{S}) = 0$ is Lebesgue null; here we take $\boldsymbol{U}_0 \in \mathbb{R}^{p\times q}$ to be any matrix with orthonormal columns spanning the column space of $\boldsymbol{W}_0$, which has dimension q because $\operatorname{rank}(\boldsymbol{W}_0) = q$. It follows that almost surely $\boldsymbol{S}_n \in \mathcal{O}$ and $\phi(\boldsymbol{S}_n) \neq 0$. Positive definiteness, by contrast, is not a generic property in $\mathrm{Sym}(p)$ and must come from the structure of $\boldsymbol{S}_n$ instead: the singular matrices form a Lebesgue null subset of $\mathbb{R}^{p\times p}$, so $\boldsymbol{Y}_1$ is almost surely invertible and

$$\boldsymbol{S}_n = \frac{1}{n}\boldsymbol{Y}_1^T\boldsymbol{Y}_1 + \frac{1}{n}\boldsymbol{Y}_2^T\boldsymbol{Y}_2 \succeq \frac{1}{n}\boldsymbol{Y}_1^T\boldsymbol{Y}_1 \succ \boldsymbol{0}$$

almost surely.

It remains to see that $\phi(\boldsymbol{S}_n) \neq 0$ is exactly the invertibility of $\boldsymbol{W}_0^T\hat{\boldsymbol{W}}_n$. Write $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{T}$ with $\boldsymbol{T} \in \mathbb{R}^{q\times q}$ invertible, which is possible since the columns of $\boldsymbol{U}_0$ form a basis of the column space of $\boldsymbol{W}_0$, and use the representative $\hat{\boldsymbol{W}}_n = \boldsymbol{K}_q(\boldsymbol{\Lambda}_q - \hat{\sigma}_n^2\boldsymbol{I}_q)^{\frac{1}{2}}$ of Proposition 19; note that invertibility of $\boldsymbol{W}_0^T\hat{\boldsymbol{W}}_n$ is unchanged if $\hat{\boldsymbol{W}}_n$ is replaced by $\hat{\boldsymbol{W}}_n\boldsymbol{R}$ with $\boldsymbol{R} \in O(q)$, so it is a property of the class $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]$. Since $\boldsymbol{\Lambda}_q - \hat{\sigma}_n^2\boldsymbol{I}_q$ is invertible, we get

$$\boldsymbol{W}_0^T\hat{\boldsymbol{W}}_n \text{ is invertible} \iff \boldsymbol{U}_0^T\boldsymbol{K}_q \text{ is invertible} \iff \det\left(\boldsymbol{U}_0^T\boldsymbol{K}_q\boldsymbol{K}_q^T\boldsymbol{U}_0\right) \neq 0 \iff \phi(\boldsymbol{S}_n) \neq 0,$$

where we used $\det\left(\boldsymbol{U}_0^T\boldsymbol{K}_q\right)^2 = \det\left(\boldsymbol{U}_0^T\boldsymbol{K}_q\boldsymbol{K}_q^T\boldsymbol{U}_0\right)$ and $\boldsymbol{K}_q\boldsymbol{K}_q^T = \boldsymbol{P}_q(\boldsymbol{S}_n)$. □

Consistency and the aligned estimation error

Proposition 28. *Almost surely, $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2] \to [\boldsymbol{W}_0, \sigma_0^2]$ in M.*

Proof. Write $\boldsymbol{P}_0$ for the orthogonal projector onto the column space of $\boldsymbol{W}_0$, so that $\boldsymbol{P}_0\boldsymbol{W}_0 = \boldsymbol{W}_0$, and abbreviate

$$\boldsymbol{P}_n := \boldsymbol{P}_q(\boldsymbol{S}_n), \tag{2.33}$$

the orthogonal projector of $\mathbb{R}^p$ onto the sum of the eigenspaces of the q largest eigenvalues of $\boldsymbol{S}_n$, which is defined as soon as $\boldsymbol{S}_n \in \mathcal{O}$. By Lemma 16 the eigenvalues of $\boldsymbol{\Sigma}_0$ are $\sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2$ for $j = 1, \ldots, q$ and σ_0^2 with multiplicity $p-q$, the eigenspace of the q largest ones is the column space of $\boldsymbol{W}_0$, and the gap of $\boldsymbol{\Sigma}_0$ at position q is $\delta_0 = \sigma_{\min}(\boldsymbol{W}_0)^2 > 0$ by (2.14); its spread, in the sense of (2.29), is

$$\rho_0 := \lambda_1(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) = \sigma_1(\boldsymbol{W}_0)^2. \tag{2.34}$$

In particular $\delta_0 > 0$ says that $\boldsymbol{\Sigma}_0 \in \mathcal{O}$, so that $\boldsymbol{P}_q(\boldsymbol{\Sigma}_0)$ is defined, and we claim that

$$\boldsymbol{P}_q(\boldsymbol{\Sigma}_0) = \boldsymbol{P}_0. \tag{2.35}$$

Indeed, let $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{D}_0\boldsymbol{V}_0^T$ be the singular value decomposition used in Lemma 16, so that the columns of $\boldsymbol{U}_0$ are orthonormal and span the column space of $\boldsymbol{W}_0$, whence $\boldsymbol{P}_0 = \boldsymbol{U}_0\boldsymbol{U}_0^T$, and $\boldsymbol{W}_0\boldsymbol{W}_0^T = \boldsymbol{U}_0\boldsymbol{D}_0^2\boldsymbol{U}_0^T$. Then

$$\boldsymbol{\Sigma}_0\boldsymbol{U}_0 = \boldsymbol{W}_0\boldsymbol{W}_0^T\boldsymbol{U}_0 + \sigma_0^2\boldsymbol{U}_0 = \boldsymbol{U}_0\left(\boldsymbol{D}_0^2 + \sigma_0^2\boldsymbol{I}_q\right) = \boldsymbol{U}_0\boldsymbol{\Lambda}_q, \qquad \boldsymbol{\Lambda}_q := \operatorname{diag}\left(\lambda_1(\boldsymbol{\Sigma}_0), \ldots, \lambda_q(\boldsymbol{\Sigma}_0)\right),$$

the identification of the diagonal entries being that of Lemma 16, and they are listed in non-increasing order because the singular values $\sigma_j(\boldsymbol{W}_0)$ are. Thus $\boldsymbol{U}_0$ satisfies (2.18) for $\boldsymbol{S} = \boldsymbol{\Sigma}_0$, and the first identity of (2.19) in Lemma 17 gives $\boldsymbol{P}_q(\boldsymbol{\Sigma}_0) = \boldsymbol{U}_0\boldsymbol{U}_0^T = \boldsymbol{P}_0$, which is (2.35). The inclusion of the column space of $\boldsymbol{W}_0$ in the range of $\boldsymbol{P}_q(\boldsymbol{\Sigma}_0)$ is the easy half of this; what the gap $\delta_0 > 0$ contributes, through Lemma 17, is that the range is not larger, no eigenvalue of $\boldsymbol{\Sigma}_0$ below the cut being equal to one above it. Equivalently, every $\boldsymbol{v}$ orthogonal to the column space of $\boldsymbol{W}_0$ satisfies $\boldsymbol{W}_0^T\boldsymbol{v} = \boldsymbol{0}$ and hence $\boldsymbol{\Sigma}_0\boldsymbol{v} = \boldsymbol{W}_0\left(\boldsymbol{W}_0^T\boldsymbol{v}\right) + \sigma_0^2\boldsymbol{v} = \sigma_0^2\boldsymbol{v}$, so the orthogonal complement of that column space is exactly the eigenspace of the smallest eigenvalue σ_0^2.

By part 5 of Lemma 15, $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ almost surely; we fix from now on a realization lying in that almost sure event, so that the whole argument below is a deterministic statement about a convergent sequence of symmetric matrices.

We first make precise in what sense $\boldsymbol{P}_n \to \boldsymbol{P}_0$. Individual eigenvectors of $\boldsymbol{S}_n$ need not converge to anything, as Remark 14 recalls: they are determined only up to sign, and up to a rotation inside an eigenspace when eigenvalues collide. What converges is the projector, and it does so because it is a locally Lipschitz function of the matrix near $\boldsymbol{\Sigma}_0$, the Lipschitz constant being governed by the spectral gap δ_0. Precisely, choose N with $\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| \leqslant \frac{\delta_0}{8}$ for all $n \geqslant N$, which is possible since $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ and $\delta_0 > 0$. Part 4 of Lemma 25, applied with $\boldsymbol{S}_* = \boldsymbol{\Sigma}_0$ and $\boldsymbol{S} = \boldsymbol{S}_n$, then shows that $\boldsymbol{S}_n \in \mathcal{O}$ for every $n \geqslant N$, so that $\boldsymbol{P}_n$ is well defined, and that

$$\|\boldsymbol{P}_n - \boldsymbol{P}_0\| = \|\boldsymbol{P}_q(\boldsymbol{S}_n) - \boldsymbol{P}_q(\boldsymbol{\Sigma}_0)\| \leqslant \frac{16\rho_0}{\delta_0^2}\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| = \frac{16\,\sigma_1(\boldsymbol{W}_0)^2}{\sigma_{\min}(\boldsymbol{W}_0)^4}\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\| \longrightarrow 0. \tag{2.36}$$

The constant depends only on the true parameter, through the singular values of $\boldsymbol{W}_0$; combined with part 5 of Lemma 15 the same inequality gives the rate $\|\boldsymbol{P}_n - \boldsymbol{P}_0\| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$, which we do not need here but which is the quantitative form of the statement. For the convergence (2.36) alone, without the constant, no argument is needed at all: it is part 5 of Lemma 25, quoted there from Theorem 5.1 of Kato (1995).

By (2.15) the matrix $\boldsymbol{S}_n$ is also eventually positive definite, so Proposition 19 applies to it for all large n. Since $\operatorname{tr}\left((\boldsymbol{I}_p - \boldsymbol{P}_n)\boldsymbol{S}_n\right) = \operatorname{tr}(\boldsymbol{S}_n) - \operatorname{tr}(\boldsymbol{P}_n\boldsymbol{S}_n) = \sum_{j>q}\lambda_j(\boldsymbol{S}_n)$ by Lemma 17, formula (2.20) reads $\hat{\sigma}_n^2 = \frac{1}{p-q}\operatorname{tr}\left((\boldsymbol{I}_p - \boldsymbol{P}_n)\boldsymbol{S}_n\right)$, and the trace being a continuous bilinear function of its two arguments we get, using $\boldsymbol{P}_n \to \boldsymbol{P}_0$ and $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$,

$$\hat{\sigma}_n^2 = \frac{1}{p-q}\operatorname{tr}\left((\boldsymbol{I}_p - \boldsymbol{P}_n)\boldsymbol{S}_n\right) \longrightarrow \frac{1}{p-q}\operatorname{tr}\left((\boldsymbol{I}_p - \boldsymbol{P}_0)\boldsymbol{\Sigma}_0\right) = \frac{\sigma_0^2}{p-q}\operatorname{tr}(\boldsymbol{I}_p - \boldsymbol{P}_0) = \sigma_0^2,$$

where the third equality holds because $(\boldsymbol{I}_p - \boldsymbol{P}_0)\boldsymbol{\Sigma}_0 = \sigma_0^2(\boldsymbol{I}_p - \boldsymbol{P}_0)$, itself a consequence of $\boldsymbol{\Sigma}_0 = \boldsymbol{W}_0\boldsymbol{W}_0^T + \sigma_0^2\boldsymbol{I}_p$ and $(\boldsymbol{I}_p - \boldsymbol{P}_0)\boldsymbol{W}_0 = \boldsymbol{0}$, and the fourth because $\boldsymbol{P}_0$ is an orthogonal projector of rank q, so that $\operatorname{tr}(\boldsymbol{I}_p - \boldsymbol{P}_0) = p - \operatorname{tr}(\boldsymbol{P}_0) = p - q$. In the same way

$$\hat{\boldsymbol{W}}_n\hat{\boldsymbol{W}}_n^T = \boldsymbol{P}_n\boldsymbol{S}_n\boldsymbol{P}_n - \hat{\sigma}_n^2\boldsymbol{P}_n \longrightarrow \boldsymbol{P}_0\boldsymbol{\Sigma}_0\boldsymbol{P}_0 - \sigma_0^2\boldsymbol{P}_0 = \boldsymbol{W}_0\boldsymbol{W}_0^T,$$

using $\boldsymbol{P}_0\boldsymbol{\Sigma}_0\boldsymbol{P}_0 = \boldsymbol{W}_0\boldsymbol{W}_0^T + \sigma_0^2\boldsymbol{P}_0$.

It remains to convert this into convergence in M. We claim that if $\boldsymbol{B}_n \in \mathbb{R}^{p\times q}$ satisfy $\boldsymbol{B}_n\boldsymbol{B}_n^T \to \boldsymbol{W}_0\boldsymbol{W}_0^T$, then $\min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{B}_n\boldsymbol{Q} - \boldsymbol{W}_0\|_F \to 0$. Indeed $\|\boldsymbol{B}_n\|_F^2 = \operatorname{tr}\big(\boldsymbol{B}_n\boldsymbol{B}_n^T\big)$ is convergent, hence bounded. If the claim failed there would be $\epsilon > 0$ and a subsequence along which $\min_{\boldsymbol{Q}} \|\boldsymbol{B}_n\boldsymbol{Q} - \boldsymbol{W}_0\|_F \geqslant \epsilon$; by boundedness we may extract a further subsequence with $\boldsymbol{B}_n \to \boldsymbol{B}$, and then $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{W}_0\boldsymbol{W}_0^T$. This matrix has rank q, so $\operatorname{rank}(\boldsymbol{B}) = q$ and $(\boldsymbol{B}, \sigma_0^2) \sim (\boldsymbol{W}_0, \sigma_0^2)$; by the description of the equivalence classes recalled in Section 1.1 there is $\boldsymbol{Q}_0 \in O(q)$ with $\boldsymbol{B} = \boldsymbol{W}_0\boldsymbol{Q}_0$. Consequently $\min_{\boldsymbol{Q}} \|\boldsymbol{B}_n\boldsymbol{Q} - \boldsymbol{W}_0\|_F \leqslant \|\boldsymbol{B}_n\boldsymbol{Q}_0^T - \boldsymbol{B}\boldsymbol{Q}_0^T\|_F = \|\boldsymbol{B}_n - \boldsymbol{B}\|_F \to 0$ along that subsequence, a contradiction.

Applying the claim to $\boldsymbol{B}_n = \hat{\boldsymbol{W}}_n$ we obtain, almost surely, matrices $\boldsymbol{Q}_n' \in O(q)$ with $\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n' \to \boldsymbol{W}_0$; combined with $\hat{\sigma}_n^2 \to \sigma_0^2$ this says $(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n', \hat{\sigma}_n^2) \to (\boldsymbol{W}_0, \sigma_0^2)$ in Θ, and the continuity of π gives

$$[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2] = \pi\left((\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n', \hat{\sigma}_n^2)\right) \longrightarrow \pi\left((\boldsymbol{W}_0, \sigma_0^2)\right) = [\boldsymbol{W}_0, \sigma_0^2]$$

in M, as required. □

The proof just given passes through the spectral projector, and therefore through the perturbation theory of Lemma 25. It is worth recording that this is avoidable: once Proposition 21 is available, the whole spectral input reduces to Weyl's inequality, which we quote from Chapter 2 of Chen et al. (2021), a chapter devoted to exactly the situation at hand — there, in the notation of its equation (2.1), $\boldsymbol{M}$ is "an observed or estimated data matrix such as the sample covariance matrix" and $\boldsymbol{M}^\star$ is "the target matrix such as the population covariance matrix".

Alternative proof of Proposition 28. Fix a realization in the almost sure event $\{\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0\}$ of part 5 of Lemma 15 and set $\boldsymbol{E}_n := \boldsymbol{S}_n - \boldsymbol{\Sigma}_0$, so that $\|\boldsymbol{E}_n\| \to 0$; this is (2.1) read as the perturbation $\boldsymbol{S}_n = \boldsymbol{\Sigma}_0 + \boldsymbol{E}_n$ of Chen et al. (2021), equation (2.1). Write $\lambda_j := \lambda_j(\boldsymbol{S}_n)$.

Step 1: the eigenvalues. Lemma 2.2 of Chen et al. (2021), Weyl's inequality for eigenvalues, states that $|\lambda_j(\boldsymbol{A}) - \lambda_j(\boldsymbol{A} + \boldsymbol{E})| \leqslant \|\boldsymbol{E}\|$ for real symmetric $\boldsymbol{A}$ and $\boldsymbol{E}$ and every j; with $\boldsymbol{A} = \boldsymbol{\Sigma}_0$ and $\boldsymbol{E} = \boldsymbol{E}_n$ this gives

$$\max_{1\leqslant j\leqslant p} |\lambda_j - \lambda_j(\boldsymbol{\Sigma}_0)| \leqslant \|\boldsymbol{E}_n\| \longrightarrow 0. \tag{2.37}$$

By Lemma 16 we have $\lambda_j(\boldsymbol{\Sigma}_0) = \sigma_0^2$ for every $j > q$, $\lambda_p(\boldsymbol{\Sigma}_0) = \sigma_0^2 > 0$ and $\lambda_q(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) = \delta_0 > 0$, so (2.37) shows that for all large n the matrix $\boldsymbol{S}_n$ is positive definite and lies in $\mathcal{O}$, whence Proposition 19 applies to it, and that

$$\hat{\sigma}_n^2 = \frac{1}{p-q}\sum_{j>q}\lambda_j \longrightarrow \frac{1}{p-q}\,(p-q)\,\sigma_0^2 = \sigma_0^2, \qquad \max_{j>q}\left|\lambda_j - \hat{\sigma}_n^2\right| \leqslant 2\max_{j>q}\left|\lambda_j - \sigma_0^2\right| \leqslant 2\|\boldsymbol{E}_n\|, \tag{2.38}$$

the middle inequality because $\hat{\sigma}_n^2$ is an average of the λ_j with $j > q$, each of which is within $\|\boldsymbol{E}_n\|$ of σ_0^2.

Step 2: the loading matrix. Let $\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]) = \hat{\boldsymbol{W}}_n\hat{\boldsymbol{W}}_n^T + \hat{\sigma}_n^2\boldsymbol{I}_p$ be the fitted covariance matrix (2.22). By the second identity of (2.23) the difference $\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n$ is symmetric with eigenvalues $\hat{\sigma}_n^2 - \lambda_j$, $j > q$, so $\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| = \max_{j>q} |\hat{\sigma}_n^2 - \lambda_j| \leqslant 2\|\boldsymbol{E}_n\| \to 0$ by (2.38). Consequently $\hat{\boldsymbol{\Sigma}}_n \to \boldsymbol{\Sigma}_0$, and therefore

$$\hat{\boldsymbol{W}}_n\hat{\boldsymbol{W}}_n^T = \hat{\boldsymbol{\Sigma}}_n - \hat{\sigma}_n^2\boldsymbol{I}_p \longrightarrow \boldsymbol{\Sigma}_0 - \sigma_0^2\boldsymbol{I}_p = \boldsymbol{W}_0\boldsymbol{W}_0^T.$$

No spectral projector, and hence no continuity or analyticity statement about $\boldsymbol{P}_q$, has been used.

Step 3: convergence in M. From $\hat{\sigma}_n^2 \to \sigma_0^2$ and $\hat{\boldsymbol{W}}_n\hat{\boldsymbol{W}}_n^T \to \boldsymbol{W}_0\boldsymbol{W}_0^T$ the conclusion follows exactly as in the last part of the proof above, which is a statement about the quotient and involves no spectral theory. □

Remark 29. If the spectral projector itself is wanted, and not merely the products in which it appears, the same chapter of Chen et al. (2021) supplies it in quantitative form: the Davis–Kahan $\sin\Theta$ theorem, Theorem 2.7 there, and its Corollary 2.8, bound the distance between the eigenspaces of $\boldsymbol{\Sigma}_0$ and of $\boldsymbol{S}_n$ spanned by the q leading eigenvectors by a multiple of $\|\boldsymbol{E}_n\|$ divided by the eigengap, under a condition on $\|\boldsymbol{E}_n\|$ relative to that gap. Since $\boldsymbol{\Sigma}_0$ and, eventually, $\boldsymbol{S}_n$ are positive definite, ordering their eigenvalues by magnitude is the same as ordering them by value, so Corollary 2.8 applies to $\boldsymbol{P}_n$ and $\boldsymbol{P}_0$ as they are defined here and is an alternative to parts 4 and 5 of Lemma 25.

Remark 30. Proposition 28 recovers, for the independent and identically distributed Gaussian sampling scheme, the strong consistency result of Datta and Chakrabarty (2023) and Datta (2026), Corollary 1, which is proved there by a Wald-type argument valid under considerably weaker assumptions. The proof above is included because it is short once Lemma 25 is available, and because it makes the section self-contained.

We can now fix once and for all the normal chart in which the whole analysis will be carried out.

Lemma 31. There exists $\rho \in (0, \min(\sigma_{\min}(\boldsymbol{W}_0), \sigma_0^2))$ such that, writing $B_\rho := \{\boldsymbol{u} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M : |\boldsymbol{u}| < \rho\} \subset \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$, the map $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}$ restricted to B_ρ is a diffeomorphism onto an open subset $\mathcal{U}_0 := \mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(B_\rho) \subseteq M$, and such that $\mathcal{U}_0 \subseteq U$, where U is the neighbourhood supplied by Proposition **??** at $[\boldsymbol{W}_0, \sigma_0^2]$.

Proof. By Proposition 3 the exponential map is defined on $\mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$, and it is a local diffeomorphism at $\mathbf{0}$, so there is an open neighbourhood $\mathcal{N}$ of $\mathbf{0}$ inside $\mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$ that is mapped diffeomorphically onto an open subset of M. Since $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}$ is continuous and $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\mathbf{0}) = [\boldsymbol{W}_0, \sigma_0^2] \in U$, we may pick $\rho > 0$ so small that $B_\rho \subseteq \mathcal{N}$ and $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(B_\rho) \subseteq U$. A restriction of a diffeomorphism to an open subset is again a diffeomorphism onto its (open) image. $\square$

We fix such a ρ for the rest of the section, and define the *aligned estimation error*

$$\boldsymbol{e}_n := \left(\hat{\boldsymbol{W}}_n \boldsymbol{Q}_n - \boldsymbol{W}_0,\ \hat{\sigma}_n^2 - \sigma_0^2\right) \in \mathbb{R}^{p\times q} \times \mathbb{R}, \qquad \hat{\boldsymbol{u}}_n := \begin{cases} \mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]\right) & \text{on } \mathcal{A}_n, \\ \mathbf{0} & \text{otherwise}, \end{cases} \tag{2.39}$$

where $\boldsymbol{Q}_n$ is as in (2.32) and $\mathcal{A}_n := \left\{[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2] \in \mathcal{U}_0\right\}$.

Proposition 32. For $n \geqslant p$ we have $\mathbb{P}(\mathcal{A}_n) \to 1$; on $\mathcal{A}_n$ the vector $\hat{\boldsymbol{u}}_n$ belongs to B_ρ and its horizontal lift at $(\boldsymbol{W}_0, \sigma_0^2)$ is exactly $\boldsymbol{e}_n$, that is $\iota_{(\boldsymbol{W}_0,\sigma_0^2)}(\hat{\boldsymbol{u}}_n) = \boldsymbol{e}_n$; and $\hat{\boldsymbol{u}}_n \to \mathbf{0}$ almost surely.

Proof. By Lemma 27 the matrix $\boldsymbol{Q}_n$ is almost surely well defined for $n \geqslant p$, so $\boldsymbol{e}_n$ is. By Proposition 28 and the fact that $\mathcal{U}_0$ is an open neighbourhood of $[\boldsymbol{W}_0, \sigma_0^2]$, almost surely $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2] \in \mathcal{U}_0$ for all large n; in particular $\mathbb{P}(\mathcal{A}_n) \to 1$. On $\mathcal{A}_n$ we have $\hat{\boldsymbol{u}}_n \in B_\rho$ by Lemma 31, and since $\mathcal{U}_0 \subseteq U$ the formula of Proposition **??** applies and reads

$$\hat{\boldsymbol{u}}_n = d\pi_{(\boldsymbol{W}_0,\sigma_0^2)}\left(\left(\hat{\boldsymbol{W}}_n \boldsymbol{Q}_n - \boldsymbol{W}_0,\ \hat{\sigma}_n^2 - \sigma_0^2\right)\right) = d\pi_{(\boldsymbol{W}_0,\sigma_0^2)}(\boldsymbol{e}_n).$$

As observed in the proof of Proposition **??**, $\boldsymbol{e}_n$ is a horizontal vector at $(\boldsymbol{W}_0, \sigma_0^2)$, so $\iota_{(\boldsymbol{W}_0,\sigma_0^2)}(\hat{\boldsymbol{u}}_n) = \boldsymbol{e}_n$ by (2.2). Finally let $\epsilon \in (0, \rho)$. Then $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(B_\epsilon)$ is an open neighbourhood of $[\boldsymbol{W}_0, \sigma_0^2]$, so almost surely $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]$ lies in it for large n, and injectivity of $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}$ on B_ρ then forces $|\hat{\boldsymbol{u}}_n| < \epsilon$. $\square$

Remark 33. Assumption III.2 of Sun et al. (2026) postulates that the estimator almost surely avoids the cut locus of the true parameter, so that the logarithmic map may be applied to it, and Proposition 32 is what plays that role here. We cannot verify that assumption literally, because M is incomplete and the discussion following Proposition **??** shows that $\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}$ genuinely fails to be globally defined. What Proposition 32 provides instead is the only thing the asymptotic theory ever uses: an event of probability tending to one on which the intrinsic error is well defined, together with the explicit Procrustes description (2.39) of it. Since modifying a sequence of random vectors on a sequence of events of vanishing probability affects neither its limit in probability nor its limit in distribution, all statements below are insensitive to the convention adopted off $\mathcal{A}_n$.

Asymptotic linearity

We can now state the definition, which is Definition 3.10 of the companion draft and, in the language of Sun et al. (2026), the hypothesis of their Lemma III.3.

Definition 34. Let $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]$ be a sequence of estimators for which the intrinsic estimation error $\hat{\boldsymbol{u}}_n$ of (2.39) is defined on events of probability tending to one. The sequence is said to be *asymptotically linear at* $[\boldsymbol{W}_0, \sigma_0^2]$ *with influence operator* IF if IF : $\mathbb{R}^p \to T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ is measurable, centred and square integrable, that is

$$\mathbb{E}\left[\mathrm{IF}(\boldsymbol{X}_1)\right] = \mathbf{0} \qquad \text{and} \qquad \mathbb{E}\left[|\mathrm{IF}(\boldsymbol{X}_1)|^2\right] < \infty, \tag{2.40}$$

and if

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathrm{IF}(\boldsymbol{X}_i) + o_{\mathbb{P}}(1), \tag{2.41}$$

the remainder being $o_{\mathbb{P}}(1)$ with respect to the norm $|\cdot|$ of $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$.

Remark 35. Both sides of (2.41) live in the single fixed vector space $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$, so the expression is meaningful even though M is curved; this is exactly the device by which Sun et al. (2026) transport the classical theory to manifolds. Note also that IF is required to take values in the *tangent* space, whereas the score (1.22) is a linear functional, that is an element of the *cotangent* space; the inverse Fisher operator (2.3) is precisely what converts one into the other.

The following theorem is the main result of this section. It is the analogue of Theorem 5.3 of the companion draft, and it instantiates Corollary V.1 of Sun et al. (2026) for the probabilistic principal component analysis model.

Theorem 36. *Let $1 \leqslant q < p$ and let $[\boldsymbol{W}_0, \sigma_0^2] \in M$. Then the maximum likelihood estimator $[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]$ of Definition 20 is asymptotically linear at $[\boldsymbol{W}_0, \sigma_0^2]$ in the sense of Definition 34, with influence operator*

$$IF(\boldsymbol{x}) := G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(s(\boldsymbol{x}; [\boldsymbol{W}_0, \sigma_0^2])\right) \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M. \tag{2.42}$$

That is,

$$\sqrt{n}\, Exp^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]\right) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(s(\boldsymbol{X}_i; [\boldsymbol{W}_0, \sigma_0^2])\right) + o_{\mathbb{P}}(1), \tag{2.43}$$

and consequently

$$\sqrt{n}\, Exp^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2]\right) \rightsquigarrow N\left(\mathbf{0}, G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\right) \qquad \text{on } T_{[\boldsymbol{W}_0,\sigma_0^2]}M, \tag{2.44}$$

the Gaussian measure on $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ whose covariance operator is the inverse Fisher information operator. Equivalently, in terms of the Procrustes aligned error (2.39)*,*

$$\sqrt{n}\left(\hat{\boldsymbol{W}}_n \boldsymbol{Q}_n - \boldsymbol{W}_0,\ \hat{\sigma}_n^2 - \sigma_0^2\right) \rightsquigarrow \iota_{(\boldsymbol{W}_0,\sigma_0^2)}\left(N\left(\mathbf{0}, G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\right)\right) \tag{2.45}$$

as random elements of $\mathbb{R}^{p\times q} \times \mathbb{R}$; the limit is a centred Gaussian law carried by the d-dimensional subspace $H_{(\boldsymbol{W}_0,\sigma_0^2)}$.

Proof. Write $G := G_{[\boldsymbol{W}_0,\sigma_0^2]}$, $\iota := \iota_{(\boldsymbol{W}_0,\sigma_0^2)}$ and $s(\boldsymbol{x}) := s(\boldsymbol{x}; [\boldsymbol{W}_0, \sigma_0^2])$ throughout. The strategy is the one of Step 2 of the companion draft: in the normal chart supplied by Lemma 31 the family $\mathcal{P}$ becomes an ordinary parametric family indexed by an open subset of a Euclidean space, and $\hat{\boldsymbol{u}}_n$ becomes an ordinary M-estimator, so that Theorem 5.23 of van der Vaart (1998) applies once its four hypotheses have been verified. The essential structural point, which makes every verification a finite computation, is that by Proposition 3 the exponential map is *linear* in the horizontal lift, so no approximation of Exp is ever needed.

Step 1: the model in normal coordinates. Fix a $g_{[\boldsymbol{W}_0,\sigma_0^2]}$-orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ of $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ and identify $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ with $\mathbb{R}^d$ through it; under this identification B_ρ is the open Euclidean ball of radius ρ and $|\cdot|$ is the Euclidean norm. For $\boldsymbol{u}\in B_\rho$ put $(\boldsymbol{U}_{\boldsymbol{u}},b_{\boldsymbol{u}}):=\iota(\boldsymbol{u})$ and

$$\boldsymbol{\Sigma}_{\boldsymbol{u}}:=\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})\right)=(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})^T+(\sigma_0^2+b_{\boldsymbol{u}})\boldsymbol{I}_p, \tag{2.46}$$

using Proposition 3 and (1.6). Because ι is linear, expanding (2.46) gives the *exact* identity

$$\boldsymbol{\Sigma}_{\boldsymbol{u}}=\boldsymbol{\Sigma}_0+\left(\boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{W}_0^T+\boldsymbol{W}_0\boldsymbol{U}_{\boldsymbol{u}}^T+b_{\boldsymbol{u}}\boldsymbol{I}_p\right)+\boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{U}_{\boldsymbol{u}}^T=\boldsymbol{\Sigma}_0+D_{\boldsymbol{u}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])+\boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{U}_{\boldsymbol{u}}^T, \tag{2.47}$$

by (1.23): as a function of $\boldsymbol{u}$, the covariance matrix is a quadratic polynomial with no remainder whatsoever. Set, for $\boldsymbol{u}\in B_\rho$ and $\boldsymbol{x}\in\mathbb{R}^p$,

$$m(\boldsymbol{u};\boldsymbol{x}):=\log p\left(\boldsymbol{x};\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})\right)=-\frac{p}{2}\log(2\pi)-\frac{1}{2}\log\det\boldsymbol{\Sigma}_{\boldsymbol{u}}-\frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{x}, \tag{2.48}$$

which in the notation (1.20) is $l_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})(\boldsymbol{x})$, and write $\mathbb{M}_n(\boldsymbol{u}):=\frac{1}{n}\sum_{i=1}^n m(\boldsymbol{u};\boldsymbol{X}_i)$ and $M(\boldsymbol{u}):=\mathbb{E}\left[m(\boldsymbol{u};\boldsymbol{X}_1)\right]$. By (2.16) and Lemma 31, on the event $\mathcal{A}_n$ the vector $\hat{\boldsymbol{u}}_n$ almost surely maximizes $\mathbb{M}_n$ over B_ρ, since by Lemma 27 and Definition 20 the point $[\hat{\boldsymbol{W}}_n,\hat\sigma_n^2]$ almost surely maximizes $\mathbb{L}_n$ over all of M, and there it lies in the chart. On the remaining null event $\hat{\boldsymbol{u}}_n=\boldsymbol{0}$, because $[\hat{\boldsymbol{W}}_n,\hat\sigma_n^2]=[\boldsymbol{W}_0,\sigma_0^2]$ by Definition 20.

Step 2: uniform spectral bounds. Let $\boldsymbol{u}\in B_\rho$. Since ι is an isometry, $\|\boldsymbol{U}_{\boldsymbol{u}}\|_F\leqslant\rho$ and $|b_{\boldsymbol{u}}|\leqslant\rho$; as $\rho<\sigma_0^2$, formula (2.46) gives

$$c_1\boldsymbol{I}_p\preceq\boldsymbol{\Sigma}_{\boldsymbol{u}}\preceq c_2\boldsymbol{I}_p,\qquad c_1:=\sigma_0^2-\rho>0,\qquad c_2:=(\sigma_{\max}(\boldsymbol{W}_0)+\rho)^2+\sigma_0^2+\rho, \tag{2.49}$$

the lower bound because $(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})^T\succeq\boldsymbol{0}$, and the upper bound because $\sigma_{\max}(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})\leqslant\sigma_{\max}(\boldsymbol{W}_0)+\|\boldsymbol{U}_{\boldsymbol{u}}\|_F$. In particular $\|\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\|_2\leqslant c_1^{-1}$ and $\|\boldsymbol{\Sigma}_{\boldsymbol{u}}\|_F\leqslant\sqrt{p}\,c_2$ uniformly on B_ρ.

Step 3: smoothness and the derivative of the criterion. On the open cone of positive definite matrices the maps $\boldsymbol{\Sigma}\mapsto\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma}\mapsto\log\det\boldsymbol{\Sigma}$ are real analytic, and $\boldsymbol{u}\mapsto\boldsymbol{\Sigma}_{\boldsymbol{u}}$ is a polynomial by (2.47); hence for every fixed $\boldsymbol{x}$ the function $\boldsymbol{u}\mapsto m(\boldsymbol{u};\boldsymbol{x})$ is C^∞ on B_ρ. Its differential is computed exactly as in (1.26): for $\boldsymbol{h}\in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$,

$$\nabla m(\boldsymbol{u};\boldsymbol{x})[\boldsymbol{h}]=\frac{1}{2}\operatorname{tr}\left\{\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T-\boldsymbol{\Sigma}_{\boldsymbol{u}}\right]\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})\right)\right)\right\}, \tag{2.50}$$

where, by (1.25),

$$D_{\boldsymbol{h}}\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})\right)=\boldsymbol{U}_{\boldsymbol{h}}(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})^T+(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})\boldsymbol{U}_{\boldsymbol{h}}^T+b_{\boldsymbol{h}}\boldsymbol{I}_p, \tag{2.51}$$

which is linear in $\boldsymbol{h}$ and affine in $\boldsymbol{u}$. Taking $\boldsymbol{u}=\boldsymbol{0}$ in (2.50) and comparing with (1.22) shows

$$\nabla m(\boldsymbol{0};\boldsymbol{x})[\boldsymbol{h}]=s(\boldsymbol{x})(\boldsymbol{h}), \tag{2.52}$$

so that the gradient of the criterion at the truth is the score.

Step 4: a square integrable Lipschitz envelope. Let $\boldsymbol{u}\in B_\rho$ and let $\boldsymbol{h}$ be a unit vector. From (2.51) and $\|\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{C}^T\|_F\leqslant\|\boldsymbol{U}_{\boldsymbol{h}}\|_F\|\boldsymbol{C}\|_2$,

$$\left\|D_{\boldsymbol{h}}\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u})\right)\right\|_F\leqslant 2\left(\sigma_{\max}(\boldsymbol{W}_0)+\rho\right)+\sqrt{p}=:c_3.$$

Combining this with $|\operatorname{tr}(\boldsymbol{Z}\boldsymbol{Y})|\leqslant\|\boldsymbol{Z}\|_F\|\boldsymbol{Y}\|_F$ and the bounds of Step 2, (2.50) gives

$$\sup_{|\boldsymbol{h}|=1}|\nabla m(\boldsymbol{u};\boldsymbol{x})[\boldsymbol{h}]|\leqslant\frac{c_3}{2c_1^2}\left(\|\boldsymbol{x}\|^2+\sqrt{p}\,c_2\right)=:\dot m(\boldsymbol{x})\qquad\text{for all }\boldsymbol{u}\in B_\rho. \tag{2.53}$$

Since $\boldsymbol{X}_1$ is Gaussian, $\mathbb{E}[\|\boldsymbol{X}_1\|^4] < \infty$ and therefore $\mathbb{E}[\dot{m}(\boldsymbol{X}_1)^2] < \infty$. As B_ρ is convex and $m(\,\cdot\,;\boldsymbol{x})$ is C^1 there, the mean value theorem yields

$$|m(\boldsymbol{u}_1;\boldsymbol{x}) - m(\boldsymbol{u}_2;\boldsymbol{x})| \leqslant \dot{m}(\boldsymbol{x})|\boldsymbol{u}_1 - \boldsymbol{u}_2| \qquad \text{for all } \boldsymbol{u}_1, \boldsymbol{u}_2 \in B_\rho. \tag{2.54}$$

Step 5: the population criterion. Since $\mathbb{E}[\boldsymbol{X}_1\boldsymbol{X}_1^T] = \boldsymbol{\Sigma}_0$, (2.48) gives the closed form

$$M(\boldsymbol{u}) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det\boldsymbol{\Sigma}_{\boldsymbol{u}} - \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{\Sigma}_0\right), \tag{2.55}$$

which, by the argument of Step 3, is C^∞ on B_ρ; in particular no differentiation under the integral sign is needed anywhere below. Subtracting $M(\mathbf{0})$,

$$M(\mathbf{0}) - M(\boldsymbol{u}) = \frac{1}{2}\left[\operatorname{tr}\left(\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{\Sigma}_0\right) - p + \log\frac{\det\boldsymbol{\Sigma}_{\boldsymbol{u}}}{\det\boldsymbol{\Sigma}_0}\right] = \frac{1}{2}\sum_{j=1}^{p}\left(\mu_j - 1 - \log\mu_j\right), \tag{2.56}$$

where $\mu_1,\ldots,\mu_p > 0$ are the eigenvalues of the positive definite matrix $\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-\frac{1}{2}}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-\frac{1}{2}}$, which is similar to $\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{\Sigma}_0$. The elementary inequality $\mu - 1 - \log\mu \geqslant 0$, with equality only at $\mu = 1$, shows that $M(\mathbf{0}) - M(\boldsymbol{u}) \geqslant 0$ with equality if and only if all $\mu_j = 1$, that is if and only if $\boldsymbol{\Sigma}_{\boldsymbol{u}} = \boldsymbol{\Sigma}_0$. By the injectivity of $\boldsymbol{\Sigma}$ established in Proposition 1 this happens exactly when $\operatorname{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{u}) = [\boldsymbol{W}_0,\sigma_0^2]$, and by Lemma 31 exactly when $\boldsymbol{u} = \mathbf{0}$. Hence $\mathbf{0}$ is the unique maximizer of M over B_ρ.

We now compute the second order Taylor expansion of M at $\mathbf{0}$. Fix $\boldsymbol{h} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}\mathcal{M}$, write $(\boldsymbol{U}, b) := \iota(\boldsymbol{h})$, and let $\boldsymbol{\Sigma}(t) := \boldsymbol{\Sigma}_{t\boldsymbol{h}}$. By (2.47),

$$\boldsymbol{\Sigma}(t) = \boldsymbol{\Sigma}_0 + t\boldsymbol{\Delta} + t^2\boldsymbol{U}\boldsymbol{U}^T, \qquad \boldsymbol{\Delta} := D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2]) = \boldsymbol{U}\boldsymbol{W}_0^T + \boldsymbol{W}_0\boldsymbol{U}^T + b\boldsymbol{I}_p, \tag{2.57}$$

so $\dot{\boldsymbol{\Sigma}}(0) = \boldsymbol{\Delta}$ and $\ddot{\boldsymbol{\Sigma}}(0) = 2\boldsymbol{U}\boldsymbol{U}^T$ exactly. Using $\frac{d}{dt}\log\det\boldsymbol{\Sigma} = \operatorname{tr}\left(\boldsymbol{\Sigma}^{-1}\dot{\boldsymbol{\Sigma}}\right)$ and $\frac{d}{dt}\boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1}\dot{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1}$, the two terms of (2.55) contribute

$$\frac{d}{dt}\bigg|_{t=0}\left[-\frac{1}{2}\log\det\boldsymbol{\Sigma}(t)\right] = -\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\Delta}\right), \qquad \frac{d}{dt}\bigg|_{t=0}\left[-\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}(t)^{-1}\boldsymbol{\Sigma}_0\right)\right] = \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\Delta}\right),$$

whose sum vanishes, so $\nabla M(\mathbf{0}) = 0$, as it must since $\mathbf{0}$ is an interior maximizer. Differentiating once more,

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left[-\frac{1}{2}\log\det\boldsymbol{\Sigma}(t)\right] = -\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\ddot{\boldsymbol{\Sigma}}(0)\right) + \frac{1}{2}\operatorname{tr}\left(\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\Delta}\right)^2\right)$$

and, since $\frac{d}{dt}\left(\boldsymbol{\Sigma}^{-1}\dot{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1}\right) = \boldsymbol{\Sigma}^{-1}\ddot{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1} - 2\boldsymbol{\Sigma}^{-1}\dot{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1}\dot{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1}$,

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left[-\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}(t)^{-1}\boldsymbol{\Sigma}_0\right)\right] = \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\ddot{\boldsymbol{\Sigma}}(0)\right) - \operatorname{tr}\left(\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\Delta}\right)^2\right).$$

The terms in $\ddot{\boldsymbol{\Sigma}}(0)$ cancel identically and we are left with

$$\frac{d^2}{dt^2}\bigg|_{t=0} M(t\boldsymbol{h}) = -\frac{1}{2}\operatorname{tr}\left(\left(\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right)^2\right) = -G(\boldsymbol{h},\boldsymbol{h}) \tag{2.58}$$

by (1.24). As M is twice continuously differentiable on B_ρ, its Hessian at $\mathbf{0}$ is the symmetric bilinear form determined by the quadratic form (2.58), namely $-G$, and Taylor's theorem gives

$$M(\boldsymbol{u}) = M(\mathbf{0}) - \frac{1}{2}G(\boldsymbol{u},\boldsymbol{u}) + o\left(|\boldsymbol{u}|^2\right) \qquad \text{as } \boldsymbol{u} \to \mathbf{0}. \tag{2.59}$$

By Theorem 7 the form G is positive definite, so the second derivative matrix of M at $\mathbf{0}$ is symmetric and nonsingular. Incidentally, (2.58) is the manifold version of the classical identity between the Fisher information and the expected negative Hessian of the log-likelihood, recorded as Remark III.4 of Sun et al. (2026); here it comes out of (2.47) by a finite computation precisely because the exponential map is linear in the horizontal lift.

Step 6: application of the M-estimation theorem. We check the four hypotheses of Theorem 5.23 of van der Vaart (1998) for the criterion functions $\{m(\boldsymbol{u};\,\cdot\,)\}_{\boldsymbol{u}\in B_\rho}$ indexed by the open subset B_ρ of $\mathbb{R}^d$, at the point $\boldsymbol{u} = \mathbf{0}$.

1. For each $\boldsymbol{u}$ the map $\boldsymbol{x} \mapsto m(\boldsymbol{u};\boldsymbol{x})$ is measurable, being continuous, and for each $\boldsymbol{x}$ the map $\boldsymbol{u} \mapsto m(\boldsymbol{u};\boldsymbol{x})$ is differentiable at $\mathbf{0}$ with derivative $s(\boldsymbol{x}) \in L^2$, by Step 3 and (2.52); square integrability of the score is (1.30).

2. The Lipschitz condition with square integrable envelope is (2.54) and (2.53).

3. The map $\boldsymbol{u} \mapsto M(\boldsymbol{u})$ admits the second order Taylor expansion (2.59) at the maximizer $\mathbf{0}$, with nonsingular symmetric second derivative matrix $-G$, by Step 5.

4. By Proposition 32, $\hat{\boldsymbol{u}}_n \to \mathbf{0}$ in probability, and $\mathbb{M}_n(\hat{\boldsymbol{u}}_n) \geqslant \mathbb{M}_n(\mathbf{0})$ holds exactly: on $\mathcal{A}_n$ because $\hat{\boldsymbol{u}}_n$ maximizes $\mathbb{M}_n$ over $B_\rho \ni \mathbf{0}$ by Step 1, and off $\mathcal{A}_n$ because there $\hat{\boldsymbol{u}}_n = \mathbf{0}$. In particular $\mathbb{M}_n(\hat{\boldsymbol{u}}_n) \geqslant \mathbb{M}_n(\mathbf{0}) - o_{\mathbb{P}}(n^{-1})$.

The cited theorem therefore yields

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n = \boldsymbol{G}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\boldsymbol{s}(\boldsymbol{X}_i) + o_{\mathbb{P}}(1), \tag{2.60}$$

where, in the orthonormal coordinates of Step 1, $\boldsymbol{G}$ is the $d\times d$ matrix with entries $\boldsymbol{G}_{jk} = G(\boldsymbol{e}_j, \boldsymbol{e}_k)$ and $\boldsymbol{s}(\boldsymbol{x}) \in \mathbb{R}^d$ is the vector with entries $\boldsymbol{s}(\boldsymbol{x})_j = s(\boldsymbol{x})(\boldsymbol{e}_j)$. Now if $\mathrm{IF}(\boldsymbol{x}) = \sum_j c_j(\boldsymbol{x})\boldsymbol{e}_j$ is defined by (2.42), then (2.3) reads $\sum_k \boldsymbol{G}_{jk}c_k(\boldsymbol{x}) = s(\boldsymbol{x})(\boldsymbol{e}_j)$ for every j, that is $\boldsymbol{c}(\boldsymbol{x}) = \boldsymbol{G}^{-1}\boldsymbol{s}(\boldsymbol{x})$; so (2.60) is precisely (2.43) and (2.41).

The requirements (2.40) hold: the score has mean zero and covariance G by Theorem 7, that is $\mathbb{E}[\boldsymbol{s}(\boldsymbol{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\boldsymbol{s}(\boldsymbol{X}_1)\boldsymbol{s}(\boldsymbol{X}_1)^T] = \boldsymbol{G}$, whence $\mathbb{E}[\boldsymbol{c}(\boldsymbol{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\|\boldsymbol{c}(\boldsymbol{X}_1)\|^2] = \mathrm{tr}(\boldsymbol{G}^{-1}) < \infty$. The vectors $\boldsymbol{s}(\boldsymbol{X}_1), \boldsymbol{s}(\boldsymbol{X}_2), \ldots$ being independent, identically distributed, centred and square integrable, the multivariate central limit theorem gives $n^{-\frac{1}{2}}\sum_{i=1}^{n}\boldsymbol{s}(\boldsymbol{X}_i) \rightsquigarrow N(\mathbf{0}, \boldsymbol{G})$, so that by (2.60), Slutsky's lemma and the continuous mapping theorem,

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n \rightsquigarrow N\left(\mathbf{0}, \boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{G}^{-1}\right) = N\left(\mathbf{0}, \boldsymbol{G}^{-1}\right),$$

which is (2.44): in the $g_{[\boldsymbol{W}_0,\sigma_0^2]}$-orthonormal basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ the covariance operator $G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}$ is represented by the matrix $\boldsymbol{G}^{-1}$.

Step 7: the Procrustes form. By Proposition 32, $\boldsymbol{e}_n = \iota(\hat{\boldsymbol{u}}_n)$ on $\mathcal{A}_n$, and $\mathbb{P}(\mathcal{A}_n) \to 1$. If $\boldsymbol{\xi}_n$ and $\boldsymbol{\eta}_n$ agree on events of probability tending to one and $\boldsymbol{\xi}_n \rightsquigarrow \boldsymbol{\xi}$, then $\boldsymbol{\eta}_n \rightsquigarrow \boldsymbol{\xi}$ as well, since for every bounded Lipschitz test function ψ one has $|\mathbb{E}\psi(\boldsymbol{\xi}_n) - \mathbb{E}\psi(\boldsymbol{\eta}_n)| \leqslant 2\|\psi\|_\infty\mathbb{P}(\mathcal{A}_n^c) \to 0$ by the portmanteau theorem. Applying this and the continuous mapping theorem to the linear isometry ι turns (2.44) into (2.45); the limit is carried by $\iota(T_{[\boldsymbol{W}_0,\sigma_0^2]}M) = H_{(\boldsymbol{W}_0,\sigma_0^2)}$, which by Proposition 1 has dimension d. □

The proof just given is a proof by M-estimation. It verifies the hypotheses of a general theorem about criterion functions, and it therefore uses Theorem 7 only as a repository of formulas — the expression (1.22) for the score, its first two moments (1.30), and the positive definiteness of G — while Proposition 8 is not used at all. The substance of Theorem 7, namely that the square root likelihood is differentiable in $L^2(\mathbb{R}^p)$, never enters, and that is precisely why Step 5 must compute the Hessian of the population criterion by hand: (2.58) is the information identity, and a theorem about arbitrary criterion functions cannot be expected to know it.

The second proof below reverses that emphasis. It works throughout with the local experiments indexed by the tangent vectors $\boldsymbol{h} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$, that is with the points $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{h}/\sqrt{n})$ of M, and with their log-likelihood ratio process; its analytic engine is Proposition 8, that is Proposition III.1 of Sun et al. (2026), whose sole input is the differentiability in quadratic mean supplied by Theorem 7; and it quotes no asymptotic theorem of van der Vaart (1998). What replaces the M-estimation machinery is the geometry of (M, g). Two features of it are used, and they are used in different places. The first is that the exponential map is affine in the horizontal lift, Proposition 3, so that (2.47) is an exact polynomial identity: every expansion below terminates, and none of the curvature corrections that Lemma A.1 and Lemma A.6 of Sun et al. (2026) would supply on a general manifold is needed. The second is that the vertical and horizontal spaces meet trivially, Proposition 1, which is what makes G

positive definite in Theorem 7 and hence what produces the quadratic separation (2.75) of the Kullback–Leibler divergence; that separation, and not a maximal inequality, is what delivers the $\sqrt{n}$ rate.

Throughout the proof G, ι and $s(\boldsymbol{x})$ keep the meaning fixed at the beginning of the proof of Theorem 36; the orthonormal basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_d$, the identification $T_{[\boldsymbol{W}_0,\sigma_0^2]}M \cong \mathbb{R}^d$, the ball B_ρ, the matrix $\boldsymbol{G}$ and the coordinate vector $\boldsymbol{s}(\boldsymbol{x})$ are those of its Steps 1 and 6; the functions m, $\mathbb{M}_n$ and M are those of (2.48) and (2.55); and c_1, c_2, c_3 are the constants of its Steps 2 and 4. We write $\lambda_{\min}(\boldsymbol{G}) > 0$ for the smallest eigenvalue of $\boldsymbol{G}$, positive by Theorem 7, and we use freely that all norms on $\mathrm{Sym}(p)$ are equivalent.

Alternative proof of Theorem 36. **Step 1: the local experiment, and the score as a linear functional of $\boldsymbol{S}_n$.** Abbreviate

$$\boldsymbol{Z}_n := \sqrt{n}\,(\boldsymbol{S}_n - \Sigma_0) \in \mathrm{Sym}(p), \qquad \boldsymbol{\Delta}_n := \frac{1}{\sqrt{n}}\sum_{i=1}^n \boldsymbol{s}(\boldsymbol{X}_i) \in \mathbb{R}^d, \tag{2.61}$$

and, for $\boldsymbol{h} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ with $|\boldsymbol{h}| < \rho\sqrt{n}$, let

$$\Lambda_n(\boldsymbol{h}) := \sum_{i=1}^n \log \frac{p\left(\boldsymbol{X}_i; \mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i; [\boldsymbol{W}_0,\sigma_0^2])} = n\left[\mathbb{M}_n\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) - \mathbb{M}_n(\boldsymbol{0})\right] \tag{2.62}$$

be the log-likelihood ratio process of the local experiments, which is the object appearing in Proposition 8. Since $\boldsymbol{x}\boldsymbol{x}^T$ enters (1.22) linearly and $\boldsymbol{Z}_n = n^{-\frac{1}{2}}\sum_{i=1}^n \left(\boldsymbol{X}_i\boldsymbol{X}_i^T - \Sigma_0\right)$, formula (1.22) gives the exact identity

$$\begin{gathered}\boldsymbol{\Delta}_n^T\boldsymbol{h} = \frac{1}{\sqrt{n}}\sum_{i=1}^n s(\boldsymbol{X}_i)(\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\left\{\Sigma_0^{-1}\boldsymbol{Z}_n\Sigma_0^{-1}D_{\boldsymbol{h}}\Sigma([\boldsymbol{W}_0,\sigma_0^2])\right\} = \frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{B}_{\boldsymbol{h}}\boldsymbol{Z}_n\right), \\ \boldsymbol{B}_{\boldsymbol{h}} := \Sigma_0^{-1}D_{\boldsymbol{h}}\Sigma([\boldsymbol{W}_0,\sigma_0^2])\Sigma_0^{-1} \in \mathrm{Sym}(p).\end{gathered} \tag{2.63}$$

The score is thus an exact linear image of the single random matrix $\boldsymbol{Z}_n$; this is the sufficiency of $\boldsymbol{S}_n$, part 4 of Lemma 15, made quantitative, and it is what allows the entire stochastic side of the argument to rest on the one weak limit (2.12). Two consequences are recorded at once. First, by (2.10) applied with $\boldsymbol{B} = \boldsymbol{B}_{\boldsymbol{h}_1}$ and $\boldsymbol{C} = \boldsymbol{B}_{\boldsymbol{h}_2}$,

$$\mathbb{E}\left[\boldsymbol{\Delta}_n\right] = \boldsymbol{0}, \qquad \mathrm{Cov}\left(\boldsymbol{\Delta}_n^T\boldsymbol{h}_1, \boldsymbol{\Delta}_n^T\boldsymbol{h}_2\right) = \frac{n}{4}\cdot\frac{2}{n}\,\mathrm{tr}\left(\boldsymbol{B}_{\boldsymbol{h}_1}\Sigma_0\boldsymbol{B}_{\boldsymbol{h}_2}\Sigma_0\right) = G(\boldsymbol{h}_1,\boldsymbol{h}_2) \tag{2.64}$$

for every n, the last equality being (1.24). Second, applying the continuous mapping theorem to the linear map $\boldsymbol{Z} \mapsto \frac{1}{2}\left(\mathrm{tr}\left(\boldsymbol{B}_{\boldsymbol{e}_j}\boldsymbol{Z}\right)\right)_{j=1}^d$ and to (2.12),

$$\boldsymbol{\Delta}_n \rightsquigarrow N(\boldsymbol{0},\boldsymbol{G}), \qquad \text{and in particular} \qquad \boldsymbol{\Delta}_n = O_{\mathbb{P}}(1), \quad \|\boldsymbol{Z}_n\|_F = O_{\mathbb{P}}(1). \tag{2.65}$$

No separate central limit theorem for the score is needed.

Step 2: the criterion is exactly affine in $\boldsymbol{S}_n$. By (2.48) and $\boldsymbol{x}^T\Sigma_{\boldsymbol{u}}^{-1}\boldsymbol{x} = \mathrm{tr}\left(\Sigma_{\boldsymbol{u}}^{-1}\boldsymbol{x}\boldsymbol{x}^T\right)$,

$$\mathbb{M}_n(\boldsymbol{u}) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma_{\boldsymbol{u}} - \frac{1}{2}\,\mathrm{tr}\left(\Sigma_{\boldsymbol{u}}^{-1}\boldsymbol{S}_n\right), \qquad \boldsymbol{u} \in B_\rho,$$

which differs from (2.55) only in that $\boldsymbol{S}_n$ replaces Σ_0; hence $\mathbb{M}_n(\boldsymbol{u}) - M(\boldsymbol{u}) = -\frac{1}{2}\,\mathrm{tr}\left(\Sigma_{\boldsymbol{u}}^{-1}(\boldsymbol{S}_n - \Sigma_0)\right)$ and, writing $K(\boldsymbol{u}) := M(\boldsymbol{0}) - M(\boldsymbol{u})$ for the Kullback–Leibler divergence evaluated in (2.56),

$$\mathbb{M}_n(\boldsymbol{0}) - \mathbb{M}_n(\boldsymbol{u}) = K(\boldsymbol{u}) + \frac{1}{2}\,\mathrm{tr}\left[\left(\Sigma_{\boldsymbol{u}}^{-1} - \Sigma_0^{-1}\right)(\boldsymbol{S}_n - \Sigma_0)\right] \qquad \text{for every } \boldsymbol{u} \in B_\rho \text{ and every } n. \tag{2.66}$$

This identity is exact: no remainder, no expectation, no limit. Equivalently, in terms of (2.62),

$$\Lambda_n(\boldsymbol{h}) = -\,nK\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) - \frac{\sqrt{n}}{2}\,\mathrm{tr}\left[\left(\Sigma_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1}\right)\boldsymbol{Z}_n\right], \qquad |\boldsymbol{h}| < \rho\sqrt{n}. \tag{2.67}$$

The log-likelihood ratio process is therefore a deterministic smooth function of $\boldsymbol{h}$ plus a linear functional of $\boldsymbol{Z}_n$ whose coefficient is again deterministic and smooth in $\boldsymbol{h}$. This is the structural reason why no empirical process, no envelope function and no maximal inequality will be required below.

Step 3: elementary bounds on the chart. Let $\boldsymbol{u} \in B_\rho$. Since ι is an isometry, $\|\boldsymbol{U}_{\boldsymbol{u}}\|_F \leqslant |\boldsymbol{u}|$, and $\|D_{\boldsymbol{u}}\boldsymbol{\Sigma}([\boldsymbol{W}_0, \sigma_0^2])\|_F \leqslant c_3|\boldsymbol{u}|$ by the bound of Step 4 of the previous proof read at $\boldsymbol{u} = \boldsymbol{0}$ together with linearity in the direction, so that (2.47) gives

$$\|\boldsymbol{\Sigma}_{\boldsymbol{u}} - \boldsymbol{\Sigma}_0\|_F \leqslant c_3|\boldsymbol{u}| + |\boldsymbol{u}|^2 \leqslant (c_3 + \rho)|\boldsymbol{u}|. \tag{2.68}$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1} - \boldsymbol{\Sigma}_0^{-1} = \boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}(\boldsymbol{\Sigma}_0 - \boldsymbol{\Sigma}_{\boldsymbol{u}})\boldsymbol{\Sigma}_0^{-1}$ and $\|\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\|_2 \leqslant c_1^{-1}$, $\|\boldsymbol{\Sigma}_0^{-1}\|_2 \leqslant c_1^{-1}$ by (2.49),

$$\|\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1} - \boldsymbol{\Sigma}_0^{-1}\|_F \leqslant c_1^{-2}\|\boldsymbol{\Sigma}_{\boldsymbol{u}} - \boldsymbol{\Sigma}_0\|_F \leqslant c_4|\boldsymbol{u}|, \qquad c_4 := \frac{c_3 + \rho}{c_1^2}. \tag{2.69}$$

Consequently, for every $K > 0$, every $\boldsymbol{h}$ with $|\boldsymbol{h}| \leqslant K$ and every $n > K^2/\rho^2$,

$$\left\|\sqrt{n}\left(\boldsymbol{\Sigma}_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1} - \boldsymbol{\Sigma}_0^{-1}\right) + \boldsymbol{B}_{\boldsymbol{h}}\right\|_F \leqslant \frac{c_5(K)}{\sqrt{n}}, \qquad c_5(K) := \left(c_4 c_3 c_1^{-1} + c_1^{-2}\right) K^2. \tag{2.70}$$

Indeed, taking $\boldsymbol{u} = \boldsymbol{h}/\sqrt{n}$ in the factorization above and using (2.47),

$$\sqrt{n}\left(\boldsymbol{\Sigma}_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1} - \boldsymbol{\Sigma}_0^{-1}\right) + \boldsymbol{B}_{\boldsymbol{h}} = -\left(\boldsymbol{\Sigma}_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1} - \boldsymbol{\Sigma}_0^{-1}\right) D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0, \sigma_0^2])\boldsymbol{\Sigma}_0^{-1} - \frac{1}{\sqrt{n}}\boldsymbol{\Sigma}_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1}\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}_{\boldsymbol{h}}^T\boldsymbol{\Sigma}_0^{-1},$$

and (2.69) bounds the first term in Frobenius norm by $c_4 c_3 c_1^{-1} K^2 n^{-\frac{1}{2}}$ and the second by $c_1^{-2}K^2 n^{-\frac{1}{2}}$. Combining (2.70) with (2.63) and $|\mathrm{tr}(\boldsymbol{Y}\boldsymbol{Z})| \leqslant \|\boldsymbol{Y}\|_F\|\boldsymbol{Z}\|_F$,

$$\sup_{|\boldsymbol{h}| \leqslant K}\left|-\frac{\sqrt{n}}{2}\,\mathrm{tr}\left[\left(\boldsymbol{\Sigma}_{\frac{\boldsymbol{h}}{\sqrt{n}}}^{-1} - \boldsymbol{\Sigma}_0^{-1}\right)\boldsymbol{Z}_n\right] - \boldsymbol{\Delta}_n^T\boldsymbol{h}\right| \leqslant \frac{c_5(K)}{2\sqrt{n}}\|\boldsymbol{Z}_n\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right). \tag{2.71}$$

Step 4: the Hessian of the Kullback–Leibler divergence, read off from local asymptotic normality. By (2.47) the map $\boldsymbol{u} \mapsto \boldsymbol{\Sigma}_{\boldsymbol{u}}$ is polynomial and, by (2.49), takes its values in the open cone of positive definite matrices, on which $\log\det$ and inversion are real analytic; hence K is C^∞ on B_ρ. By (2.56) we have $K \geqslant 0$ and $K(\boldsymbol{0}) = 0$, so $\boldsymbol{0}$ is an interior minimizer of K and therefore

$$K(\boldsymbol{0}) = 0, \qquad \nabla K(\boldsymbol{0}) = \boldsymbol{0}; \tag{2.72}$$

this requires no computation, only the nonnegativity of the Kullback–Leibler divergence. Now fix $\boldsymbol{h} \in T_{[\boldsymbol{W}_0, \sigma_0^2]}M$. Proposition 8, applied at the true parameter $[\boldsymbol{W}_0, \sigma_0^2]$ and written in the coordinates of Step 1, states that $\Lambda_n(\boldsymbol{h}) = \boldsymbol{\Delta}_n^T\boldsymbol{h} - \frac{1}{2}G(\boldsymbol{h}, \boldsymbol{h}) + o_{\mathbb{P}}(1)$. Substituting this and (2.71) into (2.67),

$$nK\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) = -\Lambda_n(\boldsymbol{h}) + \boldsymbol{\Delta}_n^T\boldsymbol{h} + o_{\mathbb{P}}(1) = \frac{1}{2}G(\boldsymbol{h}, \boldsymbol{h}) + o_{\mathbb{P}}(1).$$

The left hand side is a deterministic sequence of real numbers, and a deterministic sequence which equals a constant up to $o_{\mathbb{P}}(1)$ converges to that constant; hence $nK(\boldsymbol{h}/\sqrt{n}) \to \frac{1}{2}G(\boldsymbol{h}, \boldsymbol{h})$ for every $\boldsymbol{h}$. On the other hand (2.72) and Taylor's theorem give $nK(\boldsymbol{h}/\sqrt{n}) = \frac{1}{2}\boldsymbol{h}^T\nabla^2 K(\boldsymbol{\xi}_{n,\boldsymbol{h}})\boldsymbol{h}$ for some $\boldsymbol{\xi}_{n,\boldsymbol{h}}$ on the segment joining $\boldsymbol{0}$ to $\boldsymbol{h}/\sqrt{n}$, and $\nabla^2 K$ is continuous, so $nK(\boldsymbol{h}/\sqrt{n}) \to \frac{1}{2}\boldsymbol{h}^T\nabla^2 K(\boldsymbol{0})\boldsymbol{h}$. Comparing the two limits and polarizing,

$$\nabla^2 K(\boldsymbol{0}) = \boldsymbol{G}, \qquad \text{equivalently} \qquad -\nabla^2 M(\boldsymbol{0}) = \boldsymbol{G}. \tag{2.73}$$

This is (2.58) again, the manifold form of the identity between the Fisher information and the Hessian of the Kullback–Leibler divergence recorded as Remark III.4 of Sun et al. (2026). It has not been computed here: it has been read off the local asymptotic normality expansion, which is itself only a restatement of the differentiability in quadratic mean of Theorem 7. Two corollaries follow. First, since $\nabla^2 K$ is continuous, the Taylor argument above is uniform on compacta,

$$\sup_{|\boldsymbol{h}|\leqslant K}\left|nK\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)-\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\right|\leqslant\frac{K^2}{2}\sup_{|\boldsymbol{\xi}|\leqslant\frac{K}{\sqrt{n}}}\left\|\nabla^2K(\boldsymbol{\xi})-\nabla^2K(\mathbf{0})\right\|\longrightarrow 0,$$

so that (2.67) and (2.71) upgrade Proposition 8 to its locally uniform form: for every $K>0$,

$$\sup_{|\boldsymbol{h}|\leqslant K}\left|\Lambda_n(\boldsymbol{h})-\boldsymbol{\Delta}_n^T\boldsymbol{h}+\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\right|\longrightarrow 0\qquad\text{in probability.}\tag{2.74}$$

The upgrade is free precisely because, by (2.67), all the dependence of Λ_n on $\boldsymbol{h}$ is deterministic and smooth, the only randomness being the single matrix $\boldsymbol{Z}_n$. Second, (2.72), (2.73) and Taylor's theorem give $K(\boldsymbol{u})=\frac{1}{2}\boldsymbol{u}^T\boldsymbol{G}\boldsymbol{u}+o(|\boldsymbol{u}|^2)$, so there is $\rho_1\in(0,\rho]$ with

$$K(\boldsymbol{u})\geqslant\frac{\lambda_{\min}(\boldsymbol{G})}{4}|\boldsymbol{u}|^2\qquad\text{whenever }|\boldsymbol{u}|\leqslant\rho_1.\tag{2.75}$$

This quadratic separation is where the positive definiteness of the Fisher operator, that is, by the proof of Theorem 7, the transversality $V_{(\boldsymbol{W}_0,\sigma_0^2)}\cap H_{(\boldsymbol{W}_0,\sigma_0^2)}=\{\mathbf{0}\}$ of Proposition 1, is used quantitatively.

Step 5: the $\sqrt{n}$ rate, by the basic inequality. On $\mathcal{A}_n$ the vector $\hat{\boldsymbol{u}}_n$ maximizes $\mathbb{M}_n$ over B_ρ, by Step 1 of the previous proof, and $\mathbf{0}\in B_\rho$; hence $\mathbb{M}_n(\mathbf{0})-\mathbb{M}_n(\hat{\boldsymbol{u}}_n)\leqslant 0$, and (2.66) together with (2.69) gives

$$K(\hat{\boldsymbol{u}}_n)\leqslant\frac{1}{2}\left|\operatorname{tr}\left[\left(\boldsymbol{\Sigma}_{\hat{\boldsymbol{u}}_n}^{-1}-\boldsymbol{\Sigma}_0^{-1}\right)(\boldsymbol{S}_n-\boldsymbol{\Sigma}_0)\right]\right|\leqslant\frac{c_4}{2}|\hat{\boldsymbol{u}}_n|\,\|\boldsymbol{S}_n-\boldsymbol{\Sigma}_0\|_F\qquad\text{on }\mathcal{A}_n.\tag{2.76}$$

By Proposition 32, $\hat{\boldsymbol{u}}_n\to\mathbf{0}$ almost surely, so for every $\varepsilon>0$ there is n_ε with $\mathbb{P}\left(|\hat{\boldsymbol{u}}_n|\leqslant\rho_1\text{ for all }n\geqslant n_\varepsilon\right)\geqslant 1-\varepsilon$. On the intersection of that event with $\mathcal{A}_n$ the separation (2.75) applies to $\hat{\boldsymbol{u}}_n$ and, combined with (2.76), yields $\frac{\lambda_{\min}(\boldsymbol{G})}{4}|\hat{\boldsymbol{u}}_n|^2\leqslant\frac{c_4}{2}|\hat{\boldsymbol{u}}_n|\|\boldsymbol{S}_n-\boldsymbol{\Sigma}_0\|_F$, that is

$$|\hat{\boldsymbol{u}}_n|\leqslant\frac{2c_4}{\lambda_{\min}(\boldsymbol{G})}\|\boldsymbol{S}_n-\boldsymbol{\Sigma}_0\|_F.$$

Since $\mathbb{P}(\mathcal{A}_n)\to 1$, since ε was arbitrary, and since $\|\boldsymbol{S}_n-\boldsymbol{\Sigma}_0\|_F=O_{\mathbb{P}}(n^{-\frac{1}{2}})$ by part 5 of Lemma 15, we conclude that

$$\hat{\boldsymbol{h}}_n:=\sqrt{n}\,\hat{\boldsymbol{u}}_n=O_{\mathbb{P}}(1).\tag{2.77}$$

This is the classical basic inequality argument in its simplest possible form. It is available here because (2.66) is exact and affine in $\boldsymbol{S}_n$, so that the stochastic part of the criterion is controlled by the single Lipschitz bound (2.69), and because its deterministic part is quadratically separated by (2.75). Beyond Proposition 28, only the geometry of the chart has been used.

Step 6: identification of the local maximizer. By (2.62) and Step 1 of the previous proof, on $\mathcal{A}_n$ the vector $\hat{\boldsymbol{h}}_n$ maximizes Λ_n over $\{|\boldsymbol{h}|<\rho\sqrt{n}\}$. Since $\boldsymbol{G}$ is positive definite by Theorem 7, the quadratic

$$Q_n(\boldsymbol{h}):=\boldsymbol{\Delta}_n^T\boldsymbol{h}-\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}=\frac{1}{2}\boldsymbol{\Delta}_n^T\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n-\frac{1}{2}\left(\boldsymbol{h}-\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right)^T\boldsymbol{G}\left(\boldsymbol{h}-\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right)\tag{2.78}$$

has the unique maximizer $\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n$, which is $O_{\mathbb{P}}(1)$ by (2.65). Let $\varepsilon>0$. By (2.77), (2.65) and $\mathbb{P}(\mathcal{A}_n)\to 1$ there are $K>0$ and $n_0>K^2/\rho^2$ such that the event

$$\mathcal{E}_n:=\mathcal{A}_n\cap\left\{\left|\hat{\boldsymbol{h}}_n\right|\leqslant K\right\}\cap\left\{\left|\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right|\leqslant K\right\}$$

has $\mathbb{P}(\mathcal{E}_n)\geqslant 1-\varepsilon$ for all $n\geqslant n_0$; on $\mathcal{E}_n$ both $\hat{\boldsymbol{h}}_n$ and $\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n$ then lie in the ball $\{|\boldsymbol{h}|\leqslant K\}$, on which Λ_n is defined. Put $r_n:=\sup_{|\boldsymbol{h}|\leqslant K}|\Lambda_n(\boldsymbol{h})-Q_n(\boldsymbol{h})|$, which is $o_{\mathbb{P}}(1)$ by (2.74). On $\mathcal{E}_n$, using the maximality of $\hat{\boldsymbol{h}}_n$ for Λ_n in the middle step,

$$Q_n(\hat{\boldsymbol{h}}_n)\geqslant\Lambda_n(\hat{\boldsymbol{h}}_n)-r_n\geqslant\Lambda_n\left(\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right)-r_n\geqslant Q_n\left(\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right)-2r_n,$$

which, by the completed square in (2.78), is exactly

$$\frac{1}{2}\left(\hat{\boldsymbol{h}}_n - \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right)^T \boldsymbol{G}\left(\hat{\boldsymbol{h}}_n - \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right) \leqslant 2r_n, \qquad \text{whence} \qquad \left|\hat{\boldsymbol{h}}_n - \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right|^2 \leqslant \frac{4r_n}{\lambda_{\min}(\boldsymbol{G})}.$$

Therefore $\limsup_n \mathbb{P}\left(\left|\hat{\boldsymbol{h}}_n - \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n\right| > \delta\right) \leqslant \varepsilon$ for every $\delta > 0$, and letting $\varepsilon \downarrow 0$,

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n = \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n + o_{\mathbb{P}}(1), \tag{2.79}$$

which is (2.60).

Step 7: conclusion. The passage from (2.79) to (2.43), the verification of (2.40) — for which (2.64) may be quoted in place of Theorem 7 — and the Procrustes form (2.45) are word for word Steps 6 and 7 of the previous proof and are not repeated. Finally (2.79), (2.65) and Slutsky's lemma give $\sqrt{n}\hat{\boldsymbol{u}}_n \rightsquigarrow N\left(\mathbf{0}, \boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{G}^{-1}\right) = N\left(\mathbf{0}, \boldsymbol{G}^{-1}\right)$, which is (2.44). □

Remark 37. Theorem 36 has the shape of Corollary V.1 of Sun et al. (2026), but neither proof given above is an application of it. That corollary is deduced there from their Proposition V.1, whose standing hypothesis is that the loss $\boldsymbol{x} \mapsto -\log p(\boldsymbol{x};\,\cdot\,)$ be almost surely *geodesically convex* on the parameter manifold, in the sense of their Definition V.1. That hypothesis is not available here. A function f convex along a geodesic γ with $\gamma(0) = \theta$ and $\nabla f(\theta) = \mathbf{0}$ satisfies $f(\gamma(t)) \geqslant f(\gamma(0))$ for all $t \in [0,1]$, so a geodesically convex function has no stationary point that fails to minimize it among the points that a geodesic joins it to. But the stationarity equations (7) and (8) of Tipping and Bishop (1999), recalled in the proof of Proposition 19, exhibit a stationary point of (2.16) for every selection of q eigenvectors of $\boldsymbol{S}$ whose eigenvalues exceed the resulting $\hat{\sigma}^2$, and only the selection of the q largest ones gives the maximizer. As soon as $p - q \geqslant 2$ a second admissible selection exists for an open set of $\boldsymbol{S} \in \mathcal{O}$ — take $\lambda_1, \ldots, \lambda_{q+1}$ nearly equal and much larger than $\lambda_{q+2}, \ldots, \lambda_p$, and retain $\lambda_2, \ldots, \lambda_{q+1}$ — and by part 2 of Lemma 15 the law of $\boldsymbol{S}_n$ charges every such open set once $n \geqslant p$. So $-\mathbb{L}_n$ has, with positive probability, stationary points on M that are not global minimizers, and the geodesic convexity of Definition V.1 of Sun et al. (2026) could hold only in the vacuous way permitted by the incompleteness of M, that is if no geodesic of (M, g) joined such a point to the maximizer. Neither proof of Theorem 36 needs it. What the second proof does use of Sun et al. (2026) is their Definition III.1 and Proposition III.1, verified for $\mathcal{P}$ in Theorem 7 and Proposition 8; their Remark III.4, obtained here as (2.73) rather than assumed; and their Assumption III.2, whose role is played by Proposition 32.

Corollary 38. The influence operator (2.42) satisfies the influence relation of Sun et al. (2026), Lemma III.3, namely

$$\mathbb{E}\left[\mathrm{IF}(\boldsymbol{X}_1) \otimes s(\boldsymbol{X}_1; [\boldsymbol{W}_0, \sigma_0^2])\right] = \mathrm{id}_{T_{[\boldsymbol{W}_0,\sigma_0^2]}M}. \tag{2.80}$$

Moreover it is the only influence operator with values in the span of the score for which (2.41) can hold.

Proof. In the coordinates of Step 6 of the previous proof, (2.80) reads

$$\mathbb{E}\left[\boldsymbol{c}(\boldsymbol{X}_1)\boldsymbol{s}(\boldsymbol{X}_1)^T\right] = \boldsymbol{G}^{-1}\mathbb{E}\left[\boldsymbol{s}(\boldsymbol{X}_1)\boldsymbol{s}(\boldsymbol{X}_1)^T\right] = \boldsymbol{G}^{-1}\boldsymbol{G} = \boldsymbol{I}_d.$$

For the uniqueness, an influence operator of the stated form is $\mathrm{IF}'(\boldsymbol{x}) = \boldsymbol{C}\boldsymbol{s}(\boldsymbol{x})$ for some $d \times d$ matrix $\boldsymbol{C}$, and (2.80), which Sun et al. (2026) show to be forced for regular asymptotically linear sequences, gives $\boldsymbol{C}\boldsymbol{G} = \boldsymbol{I}_d$, that is $\boldsymbol{C} = \boldsymbol{G}^{-1}$. □

Remark 39. Three comments on the content of Theorem 36.

First, (2.45) is the well posed replacement of the false naive statement discussed at the beginning of this section: it is not the raw error $\hat{\boldsymbol{W}}_n - \boldsymbol{W}_0$ that is asymptotically Gaussian, but the error measured after the optimal rotational alignment $\hat{\boldsymbol{W}}_n \mapsto \hat{\boldsymbol{W}}_n\boldsymbol{Q}_n$, and the limit lives on the horizontal space $H_{(\boldsymbol{W}_0,\sigma_0^2)}$ only. Viewed inside $\mathbb{R}^{p\times q} \times \mathbb{R} \cong \mathbb{R}^{pq+1}$ the limiting Gaussian law is degenerate, carrying no mass in the $\frac{q(q-1)}{2}$ vertical directions of Proposition 1. This degeneracy is a feature and not a defect: those are exactly the directions along which the likelihood (2.16) is constant, so no data can ever be informative about them.

Second, (2.44) contains as a by-product the rate of convergence, since it gives $\left|\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}([\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2])\right| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and hence, on $\mathcal{A}_n$, that the Riemannian distance between $[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2]$ and $[\boldsymbol{W}_0,\sigma_0^2]$ is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$. This is the quantitative refinement of Proposition 28.

Third, no assumption whatsoever has been made on the multiplicities of the singular values of $\boldsymbol{W}_0$. This is worth emphasizing, because the classical route to such a theorem, namely perturbation of the eigenvectors of $\boldsymbol{S}_n$ followed by the delta method, requires the leading q eigenvalues of $\boldsymbol{\Sigma}_0$ to be simple. When they are not, individual principal directions are genuinely not asymptotically normal, and yet the point $[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2]$ of M still is: the quotient level limit simply does not see the multiplicities. The reason is Theorem 7, where positive definiteness of $G_{[\boldsymbol{W}_0,\sigma_0^2]}$ was obtained from $V_{(\boldsymbol{W}_0,\sigma_0^2)} \cap H_{(\boldsymbol{W}_0,\sigma_0^2)} = \{\boldsymbol{0}\}$ alone.

Remark 40. Theorem 36 is the input required by the optimality theory of Sun et al. (2026). Together with the local asymptotic normality expansion of Proposition 8 it yields, through Le Cam's third lemma, that the maximum likelihood estimator is locally regular in the sense of their Definition III.5, and then their convolution theorem, Theorem III.1, and their local asymptotic minimax theorem, Theorem III.2, identify $G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}$ as the efficiency bound over all regular estimator sequences; compare their Theorem V.1. Carrying this out requires the parallel transport of (M,g) and a comparison between the transported and the coordinate errors, which are Proposition 41 and Lemma 44 of Section 2.1; the conclusions are Theorem 49, Theorem 51 and Theorem 52 there.

2.1 Parallel Transport, Local Regularity and Optimality

Theorem 36 is a statement made entirely under the single law $\mathbb{P}_{[\boldsymbol{W}_0,\sigma_0^2]}$, and it concerns a single estimator. The present section adds the two things it does not contain: a statement about the behaviour of the maximum likelihood estimator under *neighbouring* parameters, which is local regularity, and a statement about *all other* estimators, which is optimality. Both require comparing tangent vectors based at two different points of M, so both require the parallel transport of (M,g), which we compute first.

Throughout this section we abbreviate

$$\theta_0 := [\boldsymbol{W}_0,\sigma_0^2] \in M, \qquad \hat{\theta}_n := [\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2] \in M, \qquad G := G_{\theta_0}, \qquad \iota := \iota_{(\boldsymbol{W}_0,\sigma_0^2)}. \tag{2.81}$$

The letter θ, which in Section 1.1 denoted a point of Θ, was retired at the beginning of Section 1.4 when the coordinates on Θ were renamed $(\boldsymbol{W},\sigma^2)$, so no confusion can arise: from here on θ always denotes a point of M. We keep the orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ of $T_{\theta_0}M$ and the identification $T_{\theta_0}M \cong \mathbb{R}^d$ of Step 1 of the proof of Theorem 36, and we write $\boldsymbol{G}$ for the matrix of G and $\boldsymbol{s}(\boldsymbol{x}) \in \mathbb{R}^d$ for the coordinate vector of the score in that basis, exactly as in (2.60). Finally we abbreviate the normalized score

$$\boldsymbol{\Delta}_n := \frac{1}{\sqrt{n}}\sum_{i=1}^n \boldsymbol{s}(\boldsymbol{X}_i) \in \mathbb{R}^d, \tag{2.82}$$

so that Theorem 36 reads $\sqrt{n}\hat{\boldsymbol{u}}_n = \boldsymbol{G}^{-1}\boldsymbol{\Delta}_n + o_{\mathbb{P}}(1)$ and Proposition 8 reads, for $\boldsymbol{h} \in T_{\theta_0}M$ with coordinate vector still written $\boldsymbol{h}$,

$$\Lambda_n(\boldsymbol{h}) := \sum_{i=1}^n \log\frac{p\left(\boldsymbol{X}_i;\mathrm{Exp}_{\theta_0}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i;\theta_0)} = \boldsymbol{\Delta}_n^T\boldsymbol{h} - \frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h} + o_{\mathbb{P}}(1), \tag{2.83}$$

because $s(\boldsymbol{x})(\boldsymbol{h}) = \boldsymbol{s}(\boldsymbol{x})^T\boldsymbol{h}$ by bilinearity and $\sum_i s(\boldsymbol{X}_i)(\boldsymbol{h}/\sqrt{n}) = \boldsymbol{\Delta}_n^T\boldsymbol{h}$.

Parallel transport along the geodesics of M

The metric g was constructed in Section 1.1 so that π is a Riemannian submersion from the *flat* manifold $\Theta \subseteq \mathbb{R}^{pq+1}$, and Proposition 3 exploited this to make the exponential map affine in the horizontal lift. The same structure makes the parallel transport an explicit linear ordinary differential equation.

Proposition 41. Let $[\boldsymbol{A}, s] \in M$, let $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ with horizontal lift $(\boldsymbol{V}, c) := \iota_{(\boldsymbol{A},s)}(\boldsymbol{v})$, and let

$$\gamma(t) := \mathrm{Exp}_{[\boldsymbol{A},s]}(t\boldsymbol{v}) = [\boldsymbol{A}_t, s+tc], \qquad \boldsymbol{A}_t := \boldsymbol{A} + t\boldsymbol{V}, \qquad t \in [0,1], \tag{2.84}$$

be the geodesic of Proposition 3.

1. $\mathrm{rank}(\boldsymbol{A}_t) = q$ for every $t \in [0,1]$, and for every t and every skew-symmetric $\boldsymbol{K} \in \mathbb{R}^{q\times q}$ the Lyapunov equation

$$\left(\boldsymbol{A}_t^T\boldsymbol{A}_t\right)\boldsymbol{\Omega} + \boldsymbol{\Omega}\left(\boldsymbol{A}_t^T\boldsymbol{A}_t\right) = \boldsymbol{K} \tag{2.85}$$

   has a unique solution $\boldsymbol{\Omega} =: \mathcal{L}_t(\boldsymbol{K})$; it is skew-symmetric, $\mathcal{L}_t$ is linear in $\boldsymbol{K}$, and $(t, \boldsymbol{K}) \mapsto \mathcal{L}_t(\boldsymbol{K})$ is smooth.

2. Let $\boldsymbol{x}(\,\cdot\,)$ be a vector field along γ, that is $\boldsymbol{x}(t) \in T_{\gamma(t)}M$, and let $(\boldsymbol{X}(t), \beta(t)) := \iota_{(\boldsymbol{A}_t, s+tc)}(\boldsymbol{x}(t)) \in H_{(\boldsymbol{A}_t,s+tc)}$ be its horizontal lift. Then $\boldsymbol{x}$ is parallel along γ if and only if

$$\dot{\boldsymbol{X}}(t) = \boldsymbol{A}_t\,\mathcal{L}_t\left(\boldsymbol{X}(t)^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}(t)\right), \qquad \dot\beta(t) = 0. \tag{2.86}$$

3. Equation (2.86) is a linear ordinary differential equation on $\mathbb{R}^{p\times q} \times \mathbb{R}$ whose coefficients are smooth in $(t, \boldsymbol{A}, \boldsymbol{V}, c)$; its flow preserves the horizontal subspaces, that is $(\boldsymbol{X}(0), \beta(0)) \in H_{(\boldsymbol{A},s)}$ implies $(\boldsymbol{X}(t), \beta(t)) \in H_{(\boldsymbol{A}_t,s+tc)}$ for all t, and it preserves the Euclidean norm. Consequently the parallel transport along γ exists, is a linear isometry, and depends smoothly on all the data.

Proof. **1.** Since $\iota_{(\boldsymbol{A},s)}$ is an isometry, $\|\boldsymbol{V}\| \leqslant \|\boldsymbol{V}\|_F \leqslant |\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$, so Weyl's perturbation inequality, as quoted in the proof of Proposition 3, gives $\sigma_{\min}(\boldsymbol{A}_t) \geqslant \sigma_{\min}(\boldsymbol{A}) - t\|\boldsymbol{V}\| > 0$ for $t \in [0,1]$; hence $\boldsymbol{A}_t$ has rank q and $\boldsymbol{P}_t := \boldsymbol{A}_t^T\boldsymbol{A}_t$ is positive definite. The linear operator $\boldsymbol{\Omega} \mapsto \boldsymbol{P}_t\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{P}_t$ on $\mathbb{R}^{q\times q}$ is invertible: diagonalizing $\boldsymbol{P}_t = \boldsymbol{O}\mathrm{diag}(p_1, \dots, p_q)\boldsymbol{O}^T$ with $\boldsymbol{O} \in O(q)$ and $p_i > 0$ and conjugating by $\boldsymbol{O}$ turns it into the entrywise multiplication by $p_i + p_j > 0$. This proves existence, uniqueness and linearity, and smoothness in $(t, \boldsymbol{K})$ follows since the inverse of an invertible matrix depends smoothly on it and $t \mapsto \boldsymbol{P}_t$ is polynomial. For the skew-symmetry, transposing (2.85) gives $\boldsymbol{P}_t(-\boldsymbol{\Omega}^T) + (-\boldsymbol{\Omega}^T)\boldsymbol{P}_t = -\boldsymbol{K}^T = \boldsymbol{K}$, so $-\boldsymbol{\Omega}^T$ solves the same equation and uniqueness forces $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$.

2. The curve $c(t) := (\boldsymbol{A}_t, s+tc)$ is a straight line in Θ with horizontal initial velocity, hence, as recalled in the proof of Proposition 3, a horizontal geodesic of Θ projecting to γ. Because π is a Riemannian submersion, the horizontal lift of the covariant derivative $D_t\boldsymbol{x}$ along γ is the horizontal part of the covariant derivative of the lift along c (O'Neill, 1967, Section 2; Valencia, 2021, Section 3). The statement there is for basic vector fields, and it extends to fields along a curve because both sides of the identity are $\mathbb{R}$-linear, satisfy the Leibniz rule in the scalar factor, and agree on fields that extend to basic fields, which are the three properties characterizing the covariant derivative along a curve. Since the metric of Θ is the Euclidean one (1.4), its covariant derivative along c is the ordinary derivative $(\dot{\boldsymbol{X}}, \dot\beta)$. Therefore $\boldsymbol{x}$ is parallel if and only if $(\dot{\boldsymbol{X}}(t), \dot\beta(t)) \in V_{(\boldsymbol{A}_t,s+tc)}$, which by Proposition 1 means

$$\dot{\boldsymbol{X}}(t) = \boldsymbol{A}_t\boldsymbol{\Omega}(t) \text{ for some skew-symmetric } \boldsymbol{\Omega}(t), \qquad \dot\beta(t) = 0. \tag{2.87}$$

It remains to identify $\boldsymbol{\Omega}(t)$. By Proposition 1 again, horizontality of the lift at every t says that $\boldsymbol{S}(t) := \boldsymbol{A}_t^T\boldsymbol{X}(t) - \boldsymbol{X}(t)^T\boldsymbol{A}_t$ vanishes identically, so $\dot{\boldsymbol{S}}(t) = \boldsymbol{0}$; substituting (2.87) and $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$,

$$\mathbf{0} = \dot{\boldsymbol{S}}(t) = \boldsymbol{V}^T\boldsymbol{X} + \boldsymbol{A}_t^T\boldsymbol{A}_t\boldsymbol{\Omega} - \boldsymbol{\Omega}^T\boldsymbol{A}_t^T\boldsymbol{A}_t - \boldsymbol{X}^T\boldsymbol{V} = \left(\boldsymbol{A}_t^T\boldsymbol{A}_t\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{A}_t^T\boldsymbol{A}_t\right) - \left(\boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}\right), \tag{2.88}$$

which is (2.85) with $\boldsymbol{K} = \boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}$, a skew-symmetric matrix. By part 1 this determines $\boldsymbol{\Omega}(t)$ uniquely and gives (2.86). Conversely, (2.86) is of the form (2.87) with a skew-symmetric $\boldsymbol{\Omega}(t)$, so it implies parallelism provided the lift stays horizontal, which is part 3.

3. The right hand side of (2.86) is linear in $\boldsymbol{X}(t)$, with coefficients smooth in t and in $(\boldsymbol{A}, \boldsymbol{V}, c)$ by part 1; solutions therefore exist on all of $[0,1]$ and depend smoothly on the initial condition and on the parameters, by the standard theory of linear systems and smooth dependence (Lee, 2013, Theorem D.1 and Theorem D.6). For the preservation of horizontality, let $\boldsymbol{X}$ solve (2.86) and put again $\boldsymbol{S}(t) := \boldsymbol{A}_t^T\boldsymbol{X}(t) - \boldsymbol{X}(t)^T\boldsymbol{A}_t$. The computation (2.88), read in the reverse direction, gives $\dot{\boldsymbol{S}}(t) = \mathbf{0}$ identically, and $\boldsymbol{S}(0) = \mathbf{0}$ because $(\boldsymbol{X}(0), \beta(0))$ is horizontal; hence $\boldsymbol{S} \equiv \mathbf{0}$. For the preservation of the norm, write $\boldsymbol{\Omega}(t)$ for the skew-symmetric matrix appearing in (2.86) and note that $\boldsymbol{X}^T\boldsymbol{A}_t = (\boldsymbol{A}_t^T\boldsymbol{X})^T = \boldsymbol{A}_t^T\boldsymbol{X}$ is symmetric by the horizontality just proved, so

$$\frac{d}{dt}\left[\operatorname{tr}\left(\boldsymbol{X}\boldsymbol{X}^T\right) + \beta^2\right] = 2\operatorname{tr}\left(\boldsymbol{X}^T\dot{\boldsymbol{X}}\right) + 0 = 2\operatorname{tr}\left(\boldsymbol{X}^T\boldsymbol{A}_t\boldsymbol{\Omega}\right) = 2\operatorname{tr}\left(\left(\boldsymbol{A}_t^T\boldsymbol{X}\right)\boldsymbol{\Omega}\right) = 0,$$

the last equality because the trace of the product of a symmetric and a skew-symmetric matrix vanishes. Since $\iota_{(\boldsymbol{A}_t, s+tc)}$ is an isometry by (2.2), this is exactly the statement that $|\boldsymbol{x}(t)|$ is constant. □

Definition 42. For $\boldsymbol{u} \in B_\rho$ we write

$$\Gamma_{\boldsymbol{u}} : T_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}M \longrightarrow T_{\theta_0}M \tag{2.89}$$

for the parallel transport from $t = 1$ to $t = 0$ along the radial geodesic $t \mapsto \mathrm{Exp}_{\theta_0}(t\boldsymbol{u})$ of the normal chart of Lemma 31, that is the inverse of the linear isometry supplied by Proposition 41 applied with $[\boldsymbol{A}, s] = \theta_0$ and $\boldsymbol{v} = \boldsymbol{u}$. This is the transport along the minimizing geodesic joining the two points, which is the map denoted Γ in Definition III.5 of Sun et al. (2026). By Proposition 41, $\Gamma_{\boldsymbol{u}}$ is a linear isometry, $\Gamma_{\mathbf{0}} = \mathrm{id}$, and, after both tangent spaces are identified with subspaces of $\mathbb{R}^{p\times q} \times \mathbb{R}$ through the horizontal lifts $\iota_{(\boldsymbol{W}_0, \sigma_0^2)}$ and $\iota_{(\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}}, \sigma_0^2 + b_{\boldsymbol{u}})}$, the map $(\boldsymbol{u}, \boldsymbol{z}) \mapsto \Gamma_{\boldsymbol{u}}(\boldsymbol{z})$ is smooth on $B_\rho \times \mathbb{R}^{p\times q} \times \mathbb{R}$.

The transport comparison

Definition III.5 of Sun et al. (2026) compares $\mathrm{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}(\hat{\theta}_n)$, transported back to $T_{\theta_0}M$, with $\mathrm{Exp}^{-1}_{\theta_0}(\hat{\theta}_n)$. On a flat space the two would differ by exactly the vector joining the two base points; on M they differ by that vector plus a curvature correction, and the whole content of this subsection is that the correction is quadratically small. We first record that the logarithm may be taken at a moving base point.

Lemma 43. There exists $r \in \left(0, \frac{\rho}{2}\right)$ such that, writing $\mathcal{V} := \mathrm{Exp}_{\theta_0}(B_r)$, the following hold.

1. For every $x \in \mathcal{V}$ the map Exp_x^{-1} is defined and smooth on $\mathcal{V}$, and $\mathrm{Exp}_x^{-1}(x) = \mathbf{0}$.
2. The map $(\boldsymbol{u}, \boldsymbol{w}) \mapsto \mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}\left(\mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right)$ is smooth on $B_r \times B_r$, as a map into $\mathbb{R}^{p\times q} \times \mathbb{R}$ after the horizontal lift identification of Definition 42.

Proof. Let $\mathcal{E} \subseteq TM$ be the set of pairs $(x, \boldsymbol{v})$ with $\boldsymbol{v} \in \mathcal{E}_x$, the ball of Proposition 3. Its radius at $x = [\boldsymbol{A}, s]$ is $\min(\sigma_{\min}(\boldsymbol{A}), s)$, which is well defined on M because the singular values of $\boldsymbol{A}$ are unchanged by $\boldsymbol{A} \mapsto \boldsymbol{A}\boldsymbol{Q}$ with $\boldsymbol{Q} \in O(q)$, and is a continuous function of x; hence $\mathcal{E}$ is open in TM and contains the zero section, and $\mathrm{Exp} : \mathcal{E} \to M$ is smooth by Proposition 3, being affine in the horizontal lift. Consider

$$E : \mathcal{E} \longrightarrow M \times M, \qquad E(x, \boldsymbol{v}) := (x, \mathrm{Exp}_x(\boldsymbol{v})).$$

Its differential at $(\theta_0, \mathbf{0})$ is invertible, because in the splitting $T_{(\theta_0,\mathbf{0})}TM \cong T_{\theta_0}M \oplus T_{\theta_0}M$ it is $(\boldsymbol{a}, \boldsymbol{b}) \mapsto (\boldsymbol{a}, \boldsymbol{a}+\boldsymbol{b})$, using $d(\mathrm{Exp}_{\theta_0})_{\mathbf{0}} = \mathrm{id}$. The inverse function theorem therefore gives open sets $\mathcal{N} \ni (\theta_0, \mathbf{0})$ in $\mathcal{E}$ and $\mathcal{D} \ni (\theta_0, \theta_0)$

in $M \times M$ such that $E : \mathcal{N} \to \mathcal{D}$ is a diffeomorphism. This is the standard construction of uniformly normal neighbourhoods (Lee, 2018, Theorem 5.14 and its proof); note that completeness of M, which fails here, is nowhere used, only the openness of the domain $\mathcal{E}$ of Exp. Writing $E^{-1}(x,y) = (x, \Xi(x,y))$, the map Ξ is smooth on $\mathcal{D}$ and satisfies $\mathrm{Exp}_x(\Xi(x,y)) = y$ and $\Xi(x,x) = \mathbf{0}$; and $\Xi(x,\cdot\,)$ is a smooth inverse of Exp_x on a neighbourhood of x, so it deserves the name Exp_x^{-1}. Since Exp_{θ_0} is continuous and $(\theta_0,\theta_0) \in \mathcal{D}$, we may choose $r \in (0, \rho/2)$ so small that $\mathcal{V} \times \mathcal{V} \subseteq \mathcal{D}$ with $\mathcal{V} = \mathrm{Exp}_{\theta_0}(B_r)$, which is part 1. Part 2 follows by composing Ξ with the smooth map $(\boldsymbol{u},\boldsymbol{w}) \mapsto (\mathrm{Exp}_{\theta_0}(\boldsymbol{u}), \mathrm{Exp}_{\theta_0}(\boldsymbol{w}))$ and with the horizontal lift, itself smooth because by Proposition 1 the horizontal space at $(\boldsymbol{A}, s)$ is the kernel of a matrix depending polynomially on $\boldsymbol{A}$, of constant rank. $\square$

We may and do shrink r further so that $\mathrm{Exp}_{\theta_0}^{-1}$ agrees on $\mathcal{V}$ with the map of Lemma 31, which is automatic since both invert Exp_{θ_0} on $B_r \subseteq B_\rho$ and the latter is injective there. Define

$$F : B_r \times B_r \longrightarrow T_{\theta_0}M, \qquad F(\boldsymbol{u},\boldsymbol{w}) := \Gamma_{\boldsymbol{u}}\left(\mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}\left(\mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right)\right). \tag{2.90}$$

Lemma 44 (Transport comparison)**.** After a further shrinking of r there is a constant $C < \infty$, depending only on $[\boldsymbol{W}_0, \sigma_0^2]$ and r, such that

$$|F(\boldsymbol{u},\boldsymbol{w}) - (\boldsymbol{w}-\boldsymbol{u})| \leqslant C\,|\boldsymbol{u}|\,|\boldsymbol{w}-\boldsymbol{u}| \qquad \text{for all } \boldsymbol{u},\boldsymbol{w} \in B_r. \tag{2.91}$$

Proof. By Lemma 43 and Definition 42, F is smooth on $B_r \times B_r$; put

$$R(\boldsymbol{u},\boldsymbol{w}) := F(\boldsymbol{u},\boldsymbol{w}) - (\boldsymbol{w}-\boldsymbol{u}),$$

a smooth $T_{\theta_0}M$-valued function on $B_r \times B_r$. It vanishes on two slices:

- $R(\boldsymbol{u},\boldsymbol{u}) = \mathbf{0}$ for every $\boldsymbol{u} \in B_r$, because $\mathrm{Exp}_x^{-1}(x) = \mathbf{0}$ and $\Gamma_{\boldsymbol{u}}$ is linear, so $F(\boldsymbol{u},\boldsymbol{u}) = \mathbf{0} = \boldsymbol{u}-\boldsymbol{u}$;
- $R(\mathbf{0},\boldsymbol{w}) = \mathbf{0}$ for every $\boldsymbol{w} \in B_r$, because $\Gamma_{\mathbf{0}} = \mathrm{id}$ and $\mathrm{Exp}_{\theta_0}^{-1}(\mathrm{Exp}_{\theta_0}(\boldsymbol{w})) = \boldsymbol{w}$, so $F(\mathbf{0},\boldsymbol{w}) = \boldsymbol{w} = \boldsymbol{w}-\mathbf{0}$.

Replace r by $r/2$, so that the closed balls $\overline{B_r} \times \overline{B_r}$ are compact subsets of the open set on which R is smooth, and set

$$C := \sup_{(\boldsymbol{u},\boldsymbol{w}) \in \overline{B_r}\times\overline{B_r}} \|\partial_1\partial_2 R(\boldsymbol{u},\boldsymbol{w})\| < \infty,$$

the norm being the one induced on bilinear maps $T_{\theta_0}M \times T_{\theta_0}M \to T_{\theta_0}M$ by $|\cdot|$; the supremum is finite because $\partial_1\partial_2 R$ is continuous on a compact set. Let $\boldsymbol{u},\boldsymbol{w} \in B_r$. The ball B_r being convex, the segment from $\boldsymbol{u}$ to $\boldsymbol{w}$ lies in it, and the fundamental theorem of calculus in the second variable, together with $R(\boldsymbol{u},\boldsymbol{u}) = \mathbf{0}$, gives

$$R(\boldsymbol{u},\boldsymbol{w}) = \int_0^1 \partial_2 R\left(\boldsymbol{u}, \boldsymbol{u} + \tau(\boldsymbol{w}-\boldsymbol{u})\right)[\boldsymbol{w}-\boldsymbol{u}]\,\mathrm{d}\tau. \tag{2.92}$$

Fix $\boldsymbol{z} \in B_r$. The segment from $\mathbf{0}$ to $\boldsymbol{u}$ also lies in B_r, and $\partial_2 R(\mathbf{0},\boldsymbol{z}) = \mathbf{0}$ because $R(\mathbf{0},\cdot\,)$ vanishes identically on B_r; so the same argument in the first variable gives

$$\partial_2 R(\boldsymbol{u},\boldsymbol{z}) = \int_0^1 \partial_1\partial_2 R(\varsigma\boldsymbol{u},\boldsymbol{z})[\boldsymbol{u}]\,\mathrm{d}\varsigma, \qquad \text{whence} \qquad \|\partial_2 R(\boldsymbol{u},\boldsymbol{z})\| \leqslant C|\boldsymbol{u}|. \tag{2.93}$$

Substituting (2.93) into (2.92) yields (2.91). $\square$

Remark 45. Lemma 44 is the only place where the curvature of (M,g) enters the asymptotic theory, and (2.91) says exactly how much it costs. In the applications below $\boldsymbol{u}$ is a local perturbation of size $n^{-\frac{1}{2}}$ and $\boldsymbol{w}-\boldsymbol{u}$ is an estimation error of size $O_{\mathbb{P}}(n^{-\frac{1}{2}})$, so the right hand side is $O_{\mathbb{P}}(n^{-1})$; after the multiplication by $\sqrt{n}$ that all these statements carry, it is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and disappears. The two vanishing slices used in the proof are the two trivial cases of the comparison — no displacement of the base point, and no error to compare — and it is their combination, not either alone, that produces the product $|\boldsymbol{u}||\boldsymbol{w}-\boldsymbol{u}|$.

Local regularity

We can now state the definition. For $\boldsymbol{h} \in T_{\theta_0}M$ and n large enough that $\frac{\boldsymbol{h}}{\sqrt{n}} \in B_r$ we write

$$\theta_{n,\boldsymbol{h}} := \mathrm{Exp}_{\theta_0}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) \in \mathcal{V}, \tag{2.94}$$

the *geodesic local alternative* in the direction $\boldsymbol{h}$, and $\mathbb{P}^n_\theta$ for the law of $(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)$ when the true parameter is θ, that is the n-fold product of $p(\,\cdot\,;\theta)\,\mathrm{d}\boldsymbol{x}$. Given an M-valued estimator sequence T_n we put, in the notation of Lemma 43,

$$\hat{\boldsymbol{u}}_n[T_n] := \begin{cases} \mathrm{Exp}^{-1}_{\theta_0}(T_n) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \qquad \hat{\boldsymbol{u}}^{\boldsymbol{h}}_n[T_n] := \begin{cases} \Gamma_{\frac{\boldsymbol{h}}{\sqrt{n}}}\left(\mathrm{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}(T_n)\right) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \tag{2.95}$$

both taking values in the single vector space $T_{\theta_0}M$; the second is defined as soon as $\frac{\boldsymbol{h}}{\sqrt{n}} \in B_r$, by part 1 of Lemma 43. For the maximum likelihood estimator, $\hat{\boldsymbol{u}}_n[\hat{\theta}_n]$ coincides with the vector $\hat{\boldsymbol{u}}_n$ of (2.39) on the event $\left\{\hat{\theta}_n \in \mathcal{V}\right\}$, which by the last part of the proof of Proposition 32, applied with $\epsilon = r$, has probability tending to one; we therefore keep the notation $\hat{\boldsymbol{u}}_n$ for it, the two conventions differing only on events of vanishing probability, which by the portmanteau argument of Step 7 of the proof of Theorem 36 changes nothing below.

Definition 46. An M-valued estimator sequence T_n is *locally regular at* θ_0 *with limit law* L, where L is a Borel probability measure on $T_{\theta_0}M$, if for every $\boldsymbol{h} \in T_{\theta_0}M$

$$\sqrt{n}\,\hat{\boldsymbol{u}}^{\boldsymbol{h}}_n[T_n] \rightsquigarrow L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}, \tag{2.96}$$

the limit L being the same for every $\boldsymbol{h}$. This is Definition III.5 of Sun et al. (2026) with $\psi = \mathrm{id}$ and $\Psi = M$.

Lemma 47 (Contiguity)**.** For every $\boldsymbol{h} \in T_{\theta_0}M$ the sequences $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ and $\mathbb{P}^n_{\theta_0}$ are mutually contiguous.

Proof. By (1.34) of Proposition 8 the log-likelihood ratio $\Lambda_n(\boldsymbol{h})$ of (2.83) converges in distribution under $\mathbb{P}^n_{\theta_0}$ to $N(-\frac{1}{2}\varsigma^2, \varsigma^2)$ with $\varsigma^2 := G(\boldsymbol{h}, \boldsymbol{h})$. If $Z \sim N(-\frac{1}{2}\varsigma^2, \varsigma^2)$ then $\mathbb{E}[e^Z] = e^{-\frac{1}{2}\varsigma^2 + \frac{1}{2}\varsigma^2} = 1$, so Le Cam's first lemma in the form of Example 6.5 of van der Vaart (1998) gives mutual contiguity. ☐

The next lemma is the bridge: it says that on this manifold, regularity in the intrinsic sense of Definition 46 is the same thing as regularity of the ordinary Euclidean kind read in the normal chart of Lemma 31. It is what allows the classical theory to be applied verbatim below.

Lemma 48 (Chart form of regularity)**.** Let T_n be an M-valued estimator sequence such that $\mathbb{P}^n_{\theta_0}(T_n \in \mathcal{V}) \to 1$ and $\sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$. Then, for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\sqrt{n}\,\hat{\boldsymbol{u}}^{\boldsymbol{h}}_n[T_n] = \sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} + o_{\mathbb{P}}(1) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}. \tag{2.97}$$

Consequently T_n is locally regular at θ_0 with limit law L if and only if

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} \rightsquigarrow L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}} \text{ for every } \boldsymbol{h} \in T_{\theta_0}M. \tag{2.98}$$

Moreover the two standing hypotheses are automatically satisfied by any locally regular sequence.

Proof. Fix $\boldsymbol{h}$ and abbreviate $\hat{\boldsymbol{u}}_n := \hat{\boldsymbol{u}}_n[T_n]$, $\boldsymbol{u}_n := \frac{\boldsymbol{h}}{\sqrt{n}}$ and $\mathcal{B}_n := \{T_n \in \mathcal{V}\}$. By hypothesis $\mathbb{P}^n_{\theta_0}(\mathcal{B}_n) \to 1$, hence $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{B}_n) \to 1$ by Lemma 47; and $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well. The latter is the only property of contiguity used beyond the transfer of $o_{\mathbb{P}}(1)$ statements, and it follows from the definition: were it false, there would be $\varepsilon > 0$, a subsequence n_k and constants $M_k \to \infty$ with $\mathbb{P}^{n_k}_{\theta_{n_k,\boldsymbol{h}}}\left(\sqrt{n_k}|\hat{\boldsymbol{u}}_{n_k}| > M_k\right) > \varepsilon$ for all k, whereas $\mathbb{P}^{n_k}_{\theta_0}\left(\sqrt{n_k}|\hat{\boldsymbol{u}}_{n_k}|>M_k\right)\to 0$ by boundedness in probability under $\mathbb{P}^n_{\theta_0}$, and these two are incompatible with contiguity. On $\mathcal{B}_n$, and for n large enough that $\boldsymbol{u}_n\in B_r$, we have $\hat{\boldsymbol{u}}_n\in B_r$ and, by (2.90) and (2.95),

$$\hat{\boldsymbol{u}}^{\boldsymbol{h}}_n[T_n]=\Gamma_{\boldsymbol{u}_n}\left(\mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u}_n)}\left(\mathrm{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)\right)\right)=F(\boldsymbol{u}_n,\hat{\boldsymbol{u}}_n),$$

because $T_n=\mathrm{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)$ there. Lemma 44 therefore gives, on $\mathcal{B}_n$,

$$\left|\sqrt{n}\,\hat{\boldsymbol{u}}^{\boldsymbol{h}}_n[T_n]-\left(\sqrt{n}\hat{\boldsymbol{u}}_n-\boldsymbol{h}\right)\right|=\sqrt{n}\,|F(\boldsymbol{u}_n,\hat{\boldsymbol{u}}_n)-(\hat{\boldsymbol{u}}_n-\boldsymbol{u}_n)|\leqslant\sqrt{n}\,C\,\frac{|\boldsymbol{h}|}{\sqrt{n}}\left|\hat{\boldsymbol{u}}_n-\frac{\boldsymbol{h}}{\sqrt{n}}\right|=\frac{C|\boldsymbol{h}|}{\sqrt{n}}\left|\sqrt{n}\hat{\boldsymbol{u}}_n-\boldsymbol{h}\right|. \tag{2.99}$$

The right hand side is $O_{\mathbb{P}}(n^{-\frac{1}{2}})=o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, and $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{B}^c_n)\to 0$, which proves (2.97). The equivalence of (2.96) and (2.98) is then Slutsky's lemma.

For the last assertion, suppose T_n is locally regular with limit law L and take $\boldsymbol{h}=\boldsymbol{0}$ in (2.96): since $\theta_{n,\boldsymbol{0}}=\theta_0$ and $\Gamma_{\boldsymbol{0}}=\mathrm{id}$, this reads $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n]\rightsquigarrow L$ under $\mathbb{P}^n_{\theta_0}$, which is boundedness in probability by Prohorov's theorem; and (2.95) is only meaningful under the convention that $T_n\in\mathcal{V}$ with probability tending to one, which we have built into the definition. □

Theorem 49 (Local regularity of the maximum likelihood estimator)**.** *Let $1\leqslant q<p$ and let $\theta_0=[\boldsymbol{W}_0,\sigma_0^2]\in M$. Then the maximum likelihood estimator $\hat{\theta}_n$ of Definition 20 is locally regular at θ_0 in the sense of Definition 46, with limit law*

$$L=N\left(\boldsymbol{0},G^{-1}_{\theta_0}\right)\qquad \textit{on } T_{\theta_0}M, \tag{2.100}$$

which does not depend on $\boldsymbol{h}$. Explicitly, for every $\boldsymbol{h}\in T_{\theta_0}M$,

$$\Gamma_{\frac{\boldsymbol{h}}{\sqrt{n}}}\left(\sqrt{n}\,\mathit{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}\left(\hat{\theta}_n\right)\right)\rightsquigarrow N\left(\boldsymbol{0},G^{-1}_{\theta_0}\right)\qquad \textit{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}. \tag{2.101}$$

Proof. We work in the coordinates of (2.81). Fix $\boldsymbol{h}\in T_{\theta_0}M$.

Step 1: a single linear statistic drives everything. By Theorem 36, in the form (2.60),

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n=\boldsymbol{G}^{-1}\boldsymbol{\Delta}_n+o_{\mathbb{P}}(1)\qquad\text{under } \mathbb{P}^n_{\theta_0}, \tag{2.102}$$

and by (2.83), $\Lambda_n(\boldsymbol{h})=\boldsymbol{\Delta}_n^T\boldsymbol{h}-\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}+o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$. Both expansions are linear in the *same* vector $\boldsymbol{\Delta}_n$ of (2.82), so no separate joint central limit theorem is required: the vectors $\boldsymbol{s}(\boldsymbol{X}_1),\boldsymbol{s}(\boldsymbol{X}_2),\ldots$ are independent, identically distributed, centred and square integrable with covariance matrix $\boldsymbol{G}$ by Theorem 7, so the multivariate central limit theorem gives $\boldsymbol{\Delta}_n\rightsquigarrow N(\boldsymbol{0},\boldsymbol{G})$ under $\mathbb{P}^n_{\theta_0}$, and the continuous mapping theorem applied to the linear map $\boldsymbol{\delta}\mapsto(\boldsymbol{G}^{-1}\boldsymbol{\delta},\ \boldsymbol{\delta}^T\boldsymbol{h}-\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h})$ together with Slutsky's lemma yields

$$\left(\sqrt{n}\hat{\boldsymbol{u}}_n,\ \Lambda_n(\boldsymbol{h})\right)\rightsquigarrow N\left(\begin{pmatrix}\boldsymbol{0}\\ -\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix},\begin{pmatrix}\boldsymbol{G}^{-1} & \boldsymbol{h}\\ \boldsymbol{h}^T & \boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix}\right)\qquad\text{under } \mathbb{P}^n_{\theta_0}, \tag{2.103}$$

the off-diagonal block being $\mathrm{Cov}(\boldsymbol{G}^{-1}\boldsymbol{\Delta},\boldsymbol{\Delta}^T\boldsymbol{h})=\boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{h}=\boldsymbol{h}$ for $\boldsymbol{\Delta}\sim N(\boldsymbol{0},\boldsymbol{G})$.

Step 2: Le Cam's third lemma. The limit in (2.103) has the form required by Example 6.7 of van der Vaart (1998), the Gaussian case of Le Cam's third lemma (Theorem 6.6 there): the last coordinate is the log-likelihood ratio and is $N(-\frac{1}{2}\varsigma^2,\varsigma^2)$ with $\varsigma^2=\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}=G(\boldsymbol{h},\boldsymbol{h})$. The cited result therefore gives

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n\rightsquigarrow N\left(\boldsymbol{h},\boldsymbol{G}^{-1}\right)\qquad\text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}: \tag{2.104}$$

the limit under the contiguous alternative is the limit under the null shifted by the covariance $\boldsymbol{h}$ between the statistic and the log-likelihood ratio. Equivalently $\sqrt{n}\hat{\boldsymbol{u}}_n-\boldsymbol{h}\rightsquigarrow N(\boldsymbol{0},\boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, a law free of $\boldsymbol{h}$.

Step 3: from the chart to the transported error. The hypotheses of Lemma 48 hold for $T_n=\hat{\theta}_n$: the event $\left\{\hat{\theta}_n\in\mathcal{V}\right\}$ has probability tending to one, as recorded after (2.95), and $\sqrt{n}\hat{\boldsymbol{u}}_n=O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$ by

(2.44). The equivalence (2.98) of that lemma turns (2.104) into (2.101), with limit law $N(\mathbf{0}, \boldsymbol{G}^{-1})$, which in the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ represents $N(\mathbf{0}, G_{\theta_0}^{-1})$ as in Step 6 of the proof of Theorem 36. □

Remark 50. It is worth being explicit about what (2.101) excludes, since Theorem 36 alone excludes nothing of the kind. Fix a point $\theta_* \in M$, known to the statistician, and let $\hat{\theta}_n^{\mathrm{H}}$ be the Hodges-type modification of $\hat{\theta}_n$ that replaces it by θ_* whenever $\hat{\theta}_n \in \mathrm{Exp}_{\theta_*}(B)$ with B the ball of radius $n^{-\frac{1}{4}}$ in $T_{\theta_*}M$, and leaves it unchanged otherwise. For any fixed true parameter $\theta_0 \neq \theta_*$ the modification occurs on an event of probability tending to zero, by Proposition 28, so $\hat{\theta}_n^{\mathrm{H}}$ satisfies the exact analogue of (2.44) there. For $\theta_0 = \theta_*$ it does better: since $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}}) \ll n^{-\frac{1}{4}}$ by (2.44), we have $\hat{\theta}_n^{\mathrm{H}} = \theta_*$ with probability tending to one, so that $\sqrt{n}\hat{\boldsymbol{u}}_n[\hat{\theta}_n^{\mathrm{H}}] \rightsquigarrow \delta_{\mathbf{0}}$, a limit strictly more concentrated than $N(\mathbf{0}, G_{\theta_0}^{-1})$; this is the superefficiency phenomenon, and it shows that no bound whatsoever can be extracted from pointwise statements of the type of (2.44) alone. And yet, at $\theta_0 = \theta_*$ and $\boldsymbol{h} \neq \mathbf{0}$, the same computation gives $\sqrt{n}\hat{\boldsymbol{u}}_n[\hat{\theta}_n^{\mathrm{H}}] \rightsquigarrow \delta_{\mathbf{0}}$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well — the estimator is pinned to θ_* on an event of probability tending to one, since $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}}) \ll n^{-\frac{1}{4}}$ under the alternative too, by Lemma 47 — so by (2.97) its transported error converges to $\delta_{-\boldsymbol{h}}$, which does depend on $\boldsymbol{h}$. Hence $\hat{\theta}_n^{\mathrm{H}}$ is superefficient at θ_* and is not locally regular there. Local regularity is precisely the statement that the asymptotics of Theorem 36 do not degrade as the truth is moved at the scale $n^{-\frac{1}{2}}$, and it is a strictly stronger property.

Optimality

Theorem 49 concerns one estimator. We now turn to the statements that concern *all* estimators, and which are therefore lower bounds rather than limit theorems. The key point is Lemma 48: because on this manifold intrinsic regularity is equivalent to ordinary regularity read in the normal chart of Lemma 31, and because by Proposition 8 the local experiments in that chart are asymptotically normal in the sense of Chapter 7 of van der Vaart (1998), with nonsingular information matrix $\boldsymbol{G}$ by Theorem 7, the classical bounds apply verbatim. No curvature enters: it has already been absorbed by Lemma 44.

Throughout this subsection the *local experiments* are

$$\mathcal{E}_n := \left(\mathbb{P}^n_{\mathrm{Exp}_{\theta_0}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)} \;:\; \boldsymbol{h} \in T_{\theta_0}M \cong \mathbb{R}^d \right), \tag{2.105}$$

and (2.83) says exactly that $\mathcal{E}_n$ is asymptotically normal at $\boldsymbol{h} = \mathbf{0}$ with score $\boldsymbol{\Delta}_n$ and information $\boldsymbol{G}$.

Theorem 51 (Convolution bound)**.** *Let T_n be any estimator sequence that is locally regular at θ_0 in the sense of Definition 46, with limit law L. Then there exists a Borel probability measure Λ on $T_{\theta_0}M$ such that*

$$L = N\left(\mathbf{0}, G_{\theta_0}^{-1}\right) * \Lambda, \tag{2.106}$$

the convolution of the Gaussian law with covariance operator the inverse Fisher information and an independent factor. In particular:

1. *if L has a finite second moment then so does Λ and the covariance operators satisfy $\mathrm{cov}(L) = G_{\theta_0}^{-1} + \mathrm{cov}(\Lambda) \succeq G_{\theta_0}^{-1}$;*
2. *the maximum likelihood estimator attains $\Lambda = \delta_{\mathbf{0}}$, by Theorem 49, and is therefore best regular: no locally regular estimator sequence for $[\boldsymbol{W}, \sigma^2]$ has a limit law less dispersed than that of $\hat{\theta}_n$.*

Proof. By the last assertion of Lemma 48 the sequence T_n satisfies the two standing hypotheses of that lemma, and by (2.98)

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} \rightsquigarrow L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}} \text{ for every } \boldsymbol{h} \in \mathbb{R}^d.$$

In the chart, $\hat{\boldsymbol{u}}_n[T_n]$ is an estimator of the local parameter $\boldsymbol{u}$, the true value is $\boldsymbol{u} = \mathbf{0}$, the local alternative $\theta_{n,\boldsymbol{h}}$ is the parameter value $\boldsymbol{u} = \boldsymbol{h}/\sqrt{n}$, and the display reads $\sqrt{n}\left(\hat{\boldsymbol{u}}_n[T_n] - \frac{\boldsymbol{h}}{\sqrt{n}}\right) \rightsquigarrow L$ under $\boldsymbol{h}$; this is precisely the definition of a regular estimator sequence of Section 8.5 of van der Vaart (1998), with ψ the identity map. The experiments $\mathcal{E}_n$ being asymptotically normal with nonsingular $\boldsymbol{G}$, the convolution theorem, Theorem 8.8 of van der Vaart (1998), gives a probability measure Λ on $\mathbb{R}^d$ with $L = N(\mathbf{0}, \boldsymbol{G}^{-1}) * \Lambda$, which is (2.106) once $\mathbb{R}^d$ is identified with $T_{\theta_0}M$ through the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_d$, in which $\boldsymbol{G}^{-1}$ represents $G_{\theta_0}^{-1}$.

Part 1 is the elementary fact that if Z and Y are independent with $Z \sim N(\mathbf{0}, \boldsymbol{G}^{-1})$ and $Z + Y \sim L$ square integrable then Y is square integrable, its law being the image of L under a measurable map determined by the deconvolution, and covariances add on independent summands; concretely, $\operatorname{cov}(L) = \operatorname{cov}(Z) + \operatorname{cov}(Y) = \boldsymbol{G}^{-1} + \operatorname{cov}(\Lambda)$. Part 2 is Theorem 49, which gives $L = N(\mathbf{0}, G_{\theta_0}^{-1})$, whence $\Lambda = \delta_{\mathbf{0}}$ by uniqueness of the deconvolution of a Gaussian, or directly by comparing characteristic functions: $\hat{L}(\xi) = \hat{N}(\xi)\hat{\Lambda}(\xi)$ with $\hat{N}$ nowhere vanishing forces $\hat{\Lambda} \equiv 1$. □

The convolution theorem restricts attention to regular estimators. The local asymptotic minimax theorem removes that restriction, at the price of replacing the limit law by a maximal risk over a shrinking neighbourhood. Recall that a function $\ell : T_{\theta_0}M \to [0, \infty)$ is called *bowl-shaped* if its sublevel sets $\{\ell \leqslant c\}$ are convex and symmetric about the origin for every $c \geqslant 0$; the squared norm $\ell(\boldsymbol{z}) = |\boldsymbol{z}|^2$ and its truncations $|\boldsymbol{z}|^2 \wedge m$ are the examples to keep in mind.

Theorem 52 (Local asymptotic minimax bound and its attainment)**.** *Let ℓ be bowl-shaped.*

1. (Lower bound.) *For every sequence $\hat{\boldsymbol{v}}_n$ of measurable maps $(\mathbb{R}^p)^n \to T_{\theta_0}M$ — in particular for $\hat{\boldsymbol{v}}_n = \hat{\boldsymbol{u}}_n[T_n]$ arising from an arbitrary M-valued estimator sequence T_n, regular or not —*

$$\sup_I \liminf_{n\to\infty} \sup_{\boldsymbol{h}\in I} \mathbb{E}_{\theta_{n,\boldsymbol{h}}} \left[\ell\left(\sqrt{n}\,\hat{\boldsymbol{v}}_n - \boldsymbol{h}\right)\right] \geqslant \int \ell \,\mathrm{d}N\left(\mathbf{0}, G_{\theta_0}^{-1}\right), \tag{2.107}$$

the first supremum being over all finite subsets $I \subset T_{\theta_0}M$.

2. (Attainment.) *If ℓ is in addition bounded and continuous, then the maximum likelihood estimator turns* (2.107) *into an equality:*

$$\sup_I \liminf_{n\to\infty} \sup_{\boldsymbol{h}\in I} \mathbb{E}_{\theta_{n,\boldsymbol{h}}} \left[\ell\left(\sqrt{n}\,\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right)\right] = \int \ell \,\mathrm{d}N\left(\mathbf{0}, G_{\theta_0}^{-1}\right), \tag{2.108}$$

and the same holds with $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}$ replaced by the transported intrinsic error $\sqrt{n}\,\hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[\hat{\theta}_n]$ of (2.95).

Proof. **1.** As in the proof of Theorem 51, in the normal chart the family $\mathcal{E}_n$ is a sequence of asymptotically normal experiments indexed by $\boldsymbol{h} \in \mathbb{R}^d$ with nonsingular information $\boldsymbol{G}$, and $\hat{\boldsymbol{v}}_n$ is an arbitrary estimator sequence for the local parameter. Inequality (2.107) is then Theorem 8.11 of van der Vaart (1998), the local asymptotic minimax theorem, whose only hypotheses are the asymptotic normality of the experiments and the bowl-shapedness of ℓ; no property of the estimator sequence is required, which is why the bound is estimator-agnostic and closes the superefficiency loophole illustrated in Remark 50.

2. Fix a finite $I \subset T_{\theta_0}M$. By Step 2 of the proof of Theorem 49, $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\mathbf{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ for each $\boldsymbol{h}$. Since ℓ is bounded and continuous, the portmanteau theorem gives $\mathbb{E}_{\theta_{n,\boldsymbol{h}}}[\ell(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h})] \to \int \ell \,\mathrm{d}N(\mathbf{0}, \boldsymbol{G}^{-1})$ for each $\boldsymbol{h} \in I$, a limit not depending on $\boldsymbol{h}$. As I is finite, the maximum over $\boldsymbol{h} \in I$ of finitely many sequences with a common limit converges to that limit, so

$$\liminf_{n\to\infty} \sup_{\boldsymbol{h}\in I} \mathbb{E}_{\theta_{n,\boldsymbol{h}}} \left[\ell\left(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right)\right] = \int \ell \,\mathrm{d}N(\mathbf{0}, \boldsymbol{G}^{-1})$$

for every finite I, and taking the supremum over I leaves the value unchanged. This is (2.108), and it matches the lower bound of part 1 exactly. For the last claim, (2.97) gives $\sqrt{n}\hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[\hat{\theta}_n] = \sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} + o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, so the two have the same limit in distribution and, ℓ being bounded and continuous, the same limiting risk. □

Remark 53. Part 2 is stated for bounded ℓ because weak convergence alone controls the risk only for bounded losses; for $\ell(\boldsymbol{z}) = |\boldsymbol{z}|^2$ the identity (2.108) requires in addition the uniform integrability of $|\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}|^2$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, which we have not established and which does not follow from anything proved above. The lower bound of part 1 is unaffected: it holds for every bowl-shaped ℓ, bounded or not. Applying part 2 to the truncations $\ell \wedge m$, which are again bounded, continuous and bowl-shaped, and letting $m \to \infty$ recovers the bound but not the equality, monotone convergence acting on the wrong side of the inequality.

Consequences

The first consequence closes the small gap left in Corollary 38, whose uniqueness clause was proved by invoking the influence relation, which Sun et al. (2026) establish in their Lemma III.3 for sequences that are *both* asymptotically linear *and* regular. Theorem 49 supplies the missing half of that hypothesis. It is worth recording that the uniqueness in fact needs neither.

Proposition 54. Let T_n be an estimator sequence that is asymptotically linear at θ_0 in the sense of Definition 34 with influence operator IF, and suppose IF$'$ is another influence operator for which (2.41) holds. Then $\mathrm{IF}'(\boldsymbol{X}_1) = \mathrm{IF}(\boldsymbol{X}_1)$ almost surely. In particular the influence operator (2.42) of the maximum likelihood estimator is the unique one, with no restriction to the span of the score, and $\hat{\theta}_n$ is a regular asymptotically linear sequence, so that the influence relation (2.80) is now available with all the hypotheses of Sun et al. (2026), Lemma III.3, verified.

Proof. Subtracting the two expansions (2.41) gives $n^{-\frac{1}{2}} \sum_{i=1}^n \boldsymbol{D}(\boldsymbol{X}_i) = o_{\mathbb{P}}(1)$ with $\boldsymbol{D} := \mathrm{IF} - \mathrm{IF}'$, which is centred and square integrable by (2.40). The multivariate central limit theorem gives $n^{-\frac{1}{2}} \sum_i \boldsymbol{D}(\boldsymbol{X}_i) \rightsquigarrow N(\boldsymbol{0}, \mathrm{cov}(\boldsymbol{D}(\boldsymbol{X}_1)))$, and a sequence converging in distribution to a limit and also to $\boldsymbol{0}$ in probability forces that limit to be $\delta_{\boldsymbol{0}}$, so $\mathrm{cov}(\boldsymbol{D}(\boldsymbol{X}_1)) = \boldsymbol{0}$; being centred, $\boldsymbol{D}(\boldsymbol{X}_1) = \boldsymbol{0}$ almost surely. The remaining assertions are Theorem 36 and Theorem 49. $\square$

The second consequence is the inferential payoff, and it is the one that distinguishes local regularity from Theorem 36 in practice. We first record that the information operator may be evaluated at the estimator.

Lemma 55. For $\boldsymbol{u} \in B_r$ let $\boldsymbol{G}(\boldsymbol{u})$ be the $d \times d$ matrix

$$\boldsymbol{G}(\boldsymbol{u})_{jk} := G_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})} \left(\Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_j), \Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_k) \right), \qquad j, k = 1, \ldots, d, \tag{2.109}$$

the matrix of the Fisher information operator at $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ in the frame obtained by transporting $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ along the radial geodesic. Then $\boldsymbol{u} \mapsto \boldsymbol{G}(\boldsymbol{u})$ is smooth on B_r, with $\boldsymbol{G}(\boldsymbol{0}) = \boldsymbol{G}$, and it is positive definite everywhere.

Proof. Write $(\boldsymbol{W}_{\boldsymbol{u}}, \sigma^2_{\boldsymbol{u}}) := (\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2 + b_{\boldsymbol{u}})$ for the representative of $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ used in Definition 42, which is affine in $\boldsymbol{u}$, and let $(\tilde{\boldsymbol{U}}_j(\boldsymbol{u}), \tilde{b}_j(\boldsymbol{u})) := \iota_{(\boldsymbol{W}_{\boldsymbol{u}}, \sigma^2_{\boldsymbol{u}})}(\Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_j))$ be the horizontal lift there. By (1.24) and (1.23) applied at the point $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$,

$$\boldsymbol{G}(\boldsymbol{u})_{jk} = \frac{1}{2} \operatorname{tr}\left\{ \boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1} \boldsymbol{\Delta}_j(\boldsymbol{u}) \boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1} \boldsymbol{\Delta}_k(\boldsymbol{u}) \right\}, \qquad \boldsymbol{\Delta}_j(\boldsymbol{u}) := \tilde{\boldsymbol{U}}_j(\boldsymbol{u}) \boldsymbol{W}_{\boldsymbol{u}}^T + \boldsymbol{W}_{\boldsymbol{u}} \tilde{\boldsymbol{U}}_j(\boldsymbol{u})^T + \tilde{b}_j(\boldsymbol{u}) \boldsymbol{I}_p.$$

Each ingredient is smooth in $\boldsymbol{u}$: the matrix $\boldsymbol{\Sigma}_{\boldsymbol{u}}$ is a polynomial by (2.47) and is bounded below by $c_1 \boldsymbol{I}_p$ by (2.49), so $\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}$ is smooth; $\Gamma_{\boldsymbol{u}}^{-1}$ is smooth by Proposition 41 and Definition 42; and $\iota_{(\boldsymbol{W}_{\boldsymbol{u}}, \sigma^2_{\boldsymbol{u}})}$ is smooth in $\boldsymbol{u}$, as noted at the end of the proof of Lemma 43. That $\boldsymbol{G}(\boldsymbol{0}) = \boldsymbol{G}$ is $\Gamma_{\boldsymbol{0}} = \mathrm{id}$, and positive definiteness is Theorem 7 at the point $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$, the change of frame being invertible. $\square$

Corollary 56 (Locally uniform Wald confidence sets)**.** Let $\alpha \in (0,1)$, let $\chi^2_{d,1-\alpha}$ be the $(1-\alpha)$ quantile of the chi-squared law with d degrees of freedom, and define the random subset of M

$$\mathcal{C}_{n,1-\alpha} := \mathrm{Exp}_{\hat{\theta}_n}\left(\left\{ \boldsymbol{v} \in T_{\hat{\theta}_n} M \ :\ n\, G_{\hat{\theta}_n}(\boldsymbol{v}, \boldsymbol{v}) \leqslant \chi^2_{d,1-\alpha} \right\} \right), \tag{2.110}$$

which is well defined on an event of probability tending to one. Then, for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}\left(\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}\right) \longrightarrow 1-\alpha. \tag{2.111}$$

In particular the coverage is asymptotically correct not only at θ_0, which is the case $\boldsymbol{h} = \mathbf{0}$ and already follows from Theorem 36, but along every sequence of local alternatives at the estimation scale $n^{-\frac{1}{2}}$.

Proof. Fix $\boldsymbol{h}$ and work under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, which by Lemma 47 is contiguous to $\mathbb{P}^n_{\theta_0}$; all the $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ statements of Theorem 36 and Proposition 32 therefore remain valid. In particular $\hat{\boldsymbol{u}}_n \to \mathbf{0}$ in probability and $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$.

By Lemma 55 and the continuous mapping theorem, $\boldsymbol{G}(\hat{\boldsymbol{u}}_n) \to \boldsymbol{G}$ in probability; in particular $\lambda_{\min}(\boldsymbol{G}(\hat{\boldsymbol{u}}_n))$ is bounded away from zero in probability, so the ellipsoid in (2.110) is contained in a ball of radius $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and hence, with probability tending to one, in the domain $\mathcal{E}_{\hat{\theta}_n}$ of $\operatorname{Exp}_{\hat{\theta}_n}$ supplied by Proposition 3; this is the well definedness.

Since $\operatorname{Exp}_{\hat{\theta}_n}$ is injective on that ball for large n, membership $\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}$ is equivalent, on an event of probability tending to one, to

$$n\, G_{\hat{\theta}_n}(\boldsymbol{v}_n, \boldsymbol{v}_n) \leqslant \chi^2_{d,1-\alpha}, \qquad \boldsymbol{v}_n := \operatorname{Exp}^{-1}_{\hat{\theta}_n}(\theta_{n,\boldsymbol{h}}). \tag{2.112}$$

Put $\boldsymbol{z}_n := \Gamma_{\hat{\boldsymbol{u}}_n}(\boldsymbol{v}_n) \in T_{\theta_0}M$. Because $\Gamma^{-1}_{\hat{\boldsymbol{u}}_n}\boldsymbol{e}_1, \ldots, \Gamma^{-1}_{\hat{\boldsymbol{u}}_n}\boldsymbol{e}_d$ is a basis of $T_{\hat{\theta}_n}M$ whose coordinates for $\boldsymbol{v}_n$ are exactly the coordinates of $\boldsymbol{z}_n$ in $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$, the definition (2.109) gives

$$G_{\hat{\theta}_n}(\boldsymbol{v}_n, \boldsymbol{v}_n) = \boldsymbol{z}_n^T\, \boldsymbol{G}(\hat{\boldsymbol{u}}_n)\, \boldsymbol{z}_n. \tag{2.113}$$

Now apply Lemma 44 with $\boldsymbol{u} = \hat{\boldsymbol{u}}_n$ and $\boldsymbol{w} = \boldsymbol{h}/\sqrt{n}$, both in B_r with probability tending to one, noting that $F(\hat{\boldsymbol{u}}_n, \boldsymbol{h}/\sqrt{n}) = \boldsymbol{z}_n$ by (2.90):

$$\left|\sqrt{n}\boldsymbol{z}_n - \left(\boldsymbol{h} - \sqrt{n}\hat{\boldsymbol{u}}_n\right)\right| \leqslant \sqrt{n}\, C\, |\hat{\boldsymbol{u}}_n| \left|\frac{\boldsymbol{h}}{\sqrt{n}} - \hat{\boldsymbol{u}}_n\right| = \frac{C}{\sqrt{n}} \left|\sqrt{n}\hat{\boldsymbol{u}}_n\right| \left|\boldsymbol{h} - \sqrt{n}\hat{\boldsymbol{u}}_n\right| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

so that $\sqrt{n}\boldsymbol{z}_n = -(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}) + o_{\mathbb{P}}(1)$. By Step 2 of the proof of Theorem 49, $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\mathbf{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, and the Gaussian law is symmetric, so $\sqrt{n}\boldsymbol{z}_n \rightsquigarrow N(\mathbf{0}, \boldsymbol{G}^{-1})$. Combining this with $\boldsymbol{G}(\hat{\boldsymbol{u}}_n) \to \boldsymbol{G}$ in probability, Slutsky's lemma and the continuous mapping theorem applied to $(\boldsymbol{z}, \boldsymbol{A}) \mapsto \boldsymbol{z}^T\boldsymbol{A}\boldsymbol{z}$ give, through (2.113),

$$n\, G_{\hat{\theta}_n}(\boldsymbol{v}_n, \boldsymbol{v}_n) = \left(\sqrt{n}\boldsymbol{z}_n\right)^T \boldsymbol{G}(\hat{\boldsymbol{u}}_n)\left(\sqrt{n}\boldsymbol{z}_n\right) \rightsquigarrow \boldsymbol{Z}^T\boldsymbol{G}\boldsymbol{Z}, \qquad \boldsymbol{Z} \sim N(\mathbf{0}, \boldsymbol{G}^{-1}),$$

and $\boldsymbol{Z}^T\boldsymbol{G}\boldsymbol{Z} = \|\boldsymbol{G}^{\frac{1}{2}}\boldsymbol{Z}\|^2 \sim \chi^2_d$ because $\boldsymbol{G}^{\frac{1}{2}}\boldsymbol{Z} \sim N(\mathbf{0}, \boldsymbol{I}_d)$. The distribution function of χ^2_d being continuous, the probability of (2.112) converges to $\mathbb{P}(\chi^2_d \leqslant \chi^2_{d,1-\alpha}) = 1-\alpha$, which is (2.111). □

Remark 57. Corollary 56 is stated along each fixed sequence of local alternatives, which is the exact strength of Definition 46 and is already enough to rule out the failure exhibited in Remark 50: for the Hodges-type estimator the analogous set is a single point with probability tending to one, and its coverage of $\theta_{n,\boldsymbol{h}}$ tends to 0 for every $\boldsymbol{h} \neq \mathbf{0}$. The genuinely uniform statement, $\sup_{|\boldsymbol{h}|\leqslant c}\left|\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}) - (1-\alpha)\right| \to 0$, requires the expansion (2.83) along arbitrary bounded sequences $\boldsymbol{h}_n$ rather than along fixed $\boldsymbol{h}$, which Proposition 8 as stated does not provide; we do not need it anywhere.

The last consequence records what all of this says about the object the model is really about, namely the fitted covariance matrix. It is the analogue of Corollary 5.5 of the companion draft. Recall from (1.23) that

$$\mathcal{D}_0 : T_{\theta_0}M \longrightarrow \operatorname{Sym}(p), \qquad \mathcal{D}_0(\boldsymbol{h}) := D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0) = \boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{W}_0^T + \boldsymbol{W}_0\boldsymbol{U}_{\boldsymbol{h}}^T + b_{\boldsymbol{h}}\boldsymbol{I}_p \tag{2.114}$$

is linear; it is the differential $d\boldsymbol{\Sigma}_{\theta_0}$ of the map $\boldsymbol{\Sigma}$ of (1.6), read through the exponential chart. We write $\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}(\hat{\theta}_n) = \hat{\boldsymbol{W}}_n\hat{\boldsymbol{W}}_n^T + \hat{\sigma}_n^2\boldsymbol{I}_p$ for the fitted covariance matrix, which unlike $\hat{\theta}_n$ itself is an honest element of the fixed vector space $\operatorname{Sym}(p)$.

Corollary 58 (Covariance level central limit theorem)**.** Let $1 \leqslant q < p$ and $\theta_0 \in M$.

1. Under $\mathbb{P}^n_{\theta_0}$ the fitted covariance matrix is asymptotically linear in $\mathrm{Sym}(p)$ with influence function $\mathcal{D}_0 \circ \mathrm{IF}$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \mathcal{D}_0\left(\mathrm{IF}(\boldsymbol{X}_i)\right) + o_{\mathbb{P}}(1), \tag{2.115}$$

   and consequently

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) \rightsquigarrow \mathcal{D}_0(\boldsymbol{Z}), \qquad \boldsymbol{Z} \sim N\left(\boldsymbol{0}, G^{-1}_{\theta_0}\right) \text{ on } T_{\theta_0}M. \tag{2.116}$$

   The limit is a centred Gaussian law on $\mathrm{Sym}(p)$ carried by the d-dimensional subspace $\mathcal{D}_0(T_{\theta_0}M)$, the tangent space at $\boldsymbol{\Sigma}_0$ to the immersed submanifold $\boldsymbol{\Sigma}(M)$ of Proposition 1; it is degenerate in $\mathrm{Sym}(p)$ exactly when $p - q \geqslant 2$, the codimension being

$$\frac{p(p+1)}{2} - d = \frac{(p-q+2)(p-q-1)}{2}. \tag{2.117}$$

2. The convergence (2.116) is locally regular: for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}\left(\theta_{n,\boldsymbol{h}}\right)\right) \rightsquigarrow \mathcal{D}_0(\boldsymbol{Z}) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}, \tag{2.118}$$

   with the same limit law, free of $\boldsymbol{h}$.

Explicitly, $\hat{\boldsymbol{\Sigma}}_n$ retains the q leading eigenpairs of $\boldsymbol{S}_n$ and replaces the trailing $p-q$ eigenvalues by their average, by (2.23), and the blocks of $\mathcal{D}_0(\boldsymbol{Z})$ in the singular basis of $\boldsymbol{W}_0$ are computed in Theorem 65 and Remark 68.

Proof. **1.** On $\mathcal{A}_n$ we have $\hat{\theta}_n = \mathrm{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)$, so (2.47) applied at $\boldsymbol{u} = \hat{\boldsymbol{u}}_n$ is the *exact* identity

$$\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0 = \mathcal{D}_0(\hat{\boldsymbol{u}}_n) + \boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}^T_{\hat{\boldsymbol{u}}_n} \qquad \text{on } \mathcal{A}_n, \tag{2.119}$$

with no remainder, because Exp_{θ_0} is affine in the horizontal lift by Proposition 3. Since ι is a linear isometry, $\|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}^T_{\hat{\boldsymbol{u}}_n}\|_F \leqslant \|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\|_F^2 \leqslant |\hat{\boldsymbol{u}}_n|^2$, and $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ by (2.44); hence $\sqrt{n}\|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}^T_{\hat{\boldsymbol{u}}_n}\|_F = O_{\mathbb{P}}(n^{-\frac{1}{2}}) = o_{\mathbb{P}}(1)$. Multiplying (2.119) by $\sqrt{n}$ therefore gives

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) = \mathcal{D}_0\left(\sqrt{n}\,\hat{\boldsymbol{u}}_n\right) + o_{\mathbb{P}}(1) \qquad \text{on } \mathcal{A}_n, \tag{2.120}$$

and the restriction to $\mathcal{A}_n$ is harmless: the two sides of (2.120) differ only on $\mathcal{A}_n^c$, whose probability tends to zero by Proposition 32, so their difference is $o_{\mathbb{P}}(1)$ and they have the same weak limits, exactly as in Step 7 of the proof of Theorem 36. Now (2.115) follows from (2.43) and the linearity of $\mathcal{D}_0$, and (2.116) from (2.44), Slutsky's lemma and the continuous mapping theorem applied to the linear, hence continuous, map $\mathcal{D}_0$.

The map $\mathcal{D}_0$ is injective: if $\mathcal{D}_0(\boldsymbol{h}) = \boldsymbol{0}$ then $G_{\theta_0}(\boldsymbol{h},\boldsymbol{h}) = 0$ by (1.24), whence $\boldsymbol{h} = \boldsymbol{0}$ by Theorem 7. Its image is therefore a d-dimensional subspace of $\mathrm{Sym}(p)$, and it is the image of $d\boldsymbol{\Sigma}_{\theta_0}$ because $\mathcal{D}_0(\boldsymbol{h}) = \frac{d}{dt}\big|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{\theta_0}(t\boldsymbol{h}))$ by (1.23) and $t \mapsto \mathrm{Exp}_{\theta_0}(t\boldsymbol{h})$ has velocity $\boldsymbol{h}$ at $t = 0$. Finally, by (2.1),

$$\frac{p(p+1)}{2} - d = \frac{p(p+1)}{2} - pq - 1 + \frac{q(q-1)}{2} = \frac{(p-q)^2 + (p-q) - 2}{2} = \frac{(p-q+2)(p-q-1)}{2},$$

which vanishes if and only if $p - q = 1$, since $p - q + 2 \geqslant 3$.

2. Fix $\boldsymbol{h}$ and work under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, which is contiguous to $\mathbb{P}^n_{\theta_0}$ by Lemma 47; in particular $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{A}_n) \to 1$ and the $O_{\mathbb{P}}$ statements above persist, as in the proof of Corollary 56. Applying (2.47) at $\boldsymbol{u} = \boldsymbol{h}/\sqrt{n}$ gives the second exact identity

$$\boldsymbol{\Sigma}\left(\theta_{n,\boldsymbol{h}}\right) - \boldsymbol{\Sigma}_0 = \frac{1}{\sqrt{n}}\mathcal{D}_0(\boldsymbol{h}) + \frac{1}{n}\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}^T_{\boldsymbol{h}},$$

so subtracting it from (2.119) and multiplying by $\sqrt{n}$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}(\theta_{n,\boldsymbol{h}})\right) = \mathcal{D}_0\left(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right) + \sqrt{n}\,\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}^T - \frac{1}{\sqrt{n}}\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}_{\boldsymbol{h}}^T \qquad \text{on } \mathcal{A}_n.$$

The last term is deterministic and $O(n^{-\frac{1}{2}})$. By (2.104) in the proof of Theorem 49, $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$; in particular $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ there, so the middle term is again $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ by the bound of part 1. The continuous mapping theorem applied to $\mathcal{D}_0$ now gives (2.118), the limit being the image of $N(\boldsymbol{0}, \boldsymbol{G}^{-1})$, that is $\mathcal{D}_0(\boldsymbol{Z})$, and free of $\boldsymbol{h}$. □

Remark 59 (What part 1 does and does not use)**.** Part 1 of Corollary 58 uses nothing from this section. Its proof invokes Theorem 36, the exact expansion (2.47), Proposition 32 and the continuous mapping theorem, and no more; neither the parallel transport of Proposition 41, nor the transport comparison of Lemma 44, nor the contiguity of Lemma 47, nor the local regularity of Theorem 49 appears anywhere in it. It could have been stated at the end of Section 1.4; it is placed here only because part 2 belongs here, and because it is the covariance level reading of the whole chapter.

The reason no transport is needed is worth isolating, because it is the converse of the phenomenon that made this section necessary in the first place. The intrinsic error $\operatorname{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}(\hat{\theta}_n)$ of Definition 46 lives in $T_{\theta_{n,\boldsymbol{h}}}M$, a vector space that *moves with* n, which is exactly why $\Gamma_{\boldsymbol{h}/\sqrt{n}}$ has to be inserted before any limit can be taken. The covariance level error $\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}(\theta)$, by contrast, is a difference of two elements of the single fixed vector space $\operatorname{Sym}(p)$, for every θ at once; there is nothing to transport, and the delta method applies verbatim. Curvature is invisible at the covariance level because $\boldsymbol{\Sigma}(M) \subseteq \operatorname{Sym}(p)$ is being viewed extrinsically.

What part 2 costs is correspondingly small: given (2.104), which is the substance of Theorem 49 and does use Lemma 47 and Le Cam's third lemma, the passage to (2.118) is three lines of exact algebra. One should resist reading more into part 2 than it says: it asserts that the *maximum likelihood* fitted covariance is locally regular as an estimator of $\boldsymbol{\Sigma}(\theta)$, and nothing about other estimators. The corresponding lower bound would require the convolution theorem for the differentiable functional $\theta \mapsto \boldsymbol{\Sigma}(\theta)$, which is Theorem IV.1 of Sun et al. (2026) rather than Theorem 51, and which we do not develop.

Remark 60. It may be useful to summarize what Section 2.1 adds to Theorem 36, since the three groups of results answer three different questions.

Theorem 49 is about *other parameters*: it says that the conclusion of Theorem 36 is stable under perturbations of the truth at the scale $n^{-\frac{1}{2}}$. Nothing in (2.43)–(2.45) implies this, as Remark 50 shows by exhibiting an estimator that satisfies the analogue of (2.44) at every point of M and is not regular anywhere.

Theorems 51 and 52 are about *other estimators*: they are lower bounds over a class of procedures, and are therefore statements of a different kind from anything in Section 1.4, which concerns one estimator only. The two are complementary. The convolution bound is sharp and distributional but restricted to regular sequences; the local asymptotic minimax bound holds for every estimator sequence whatsoever but only in the weaker, maximal-risk form. It is worth noting where each hypothesis is used: the lower bound (2.107) needs nothing beyond Theorem 7 and Proposition 8, so it was already within reach at the end of Section 1.4; what needs Theorem 49 is the *attainment* (2.108), because the supremum there is taken over a neighbourhood that shrinks at the estimation scale and therefore probes exactly the local alternatives.

Finally, Proposition 41 and Lemmas 43–44 are geometric results about (M, g) in their own right, independent of any statistics; Lemma 47 is a probabilistic one. They are what the two-base-point comparison costs, and Theorem 36, which lives entirely inside the single vector space $T_{\theta_0}M$, needed none of them.

In the terminology of Sun et al. (2026), Theorem 49 verifies their Definition III.5 for the probabilistic principal component analysis model, Theorem 51 is the instance of their Theorem III.1, Theorem 52 of their Theorem III.2, and the two together are their Theorem V.1 for this model. We have proved them here through the chart reduction of Lemma 48 and the classical Euclidean theorems of van der Vaart (1998), rather than by citing the manifold versions, because Lemma 44 makes the reduction exact and because it isolates precisely where the curvature of M is felt: in the $O(|\boldsymbol{u}||\boldsymbol{w} - \boldsymbol{u}|)$ term of (2.91), and nowhere else.

2.2 The Fisher Information Operator in Closed Form

Theorem 36 identifies the limiting covariance of the maximum likelihood estimator as $G_{\theta_0}^{-1}$, and Theorems 51 and 52 identify the same operator as the efficiency bound; but nothing so far says what $G_{\theta_0}^{-1}$ *is*. This section computes it. The only inputs are (1.23), (1.22) and (1.24) of Section 1.3, the spectral decomposition (2.13) of Lemma 16, and the description (2.2) of the horizontal lift; no asymptotics are used until Theorem 65, which merely reads Theorem 36 in the coordinates constructed here. The computation follows Section 4 of the companion draft.

Two features of the answer deserve to be announced. First, the whole calculation is a 2×2 block computation in the singular basis of $\boldsymbol{W}_0$, and it produces a set of coordinates in which G_{θ_0} is block diagonal with blocks of size 1, 1 and $q+1$; the only genuinely coupled block is the one that mixes the q leading directions with the noise variance, and it is inverted by a one-line Schur complement. Second, no assumption whatsoever is made on the multiplicities of the singular values of $\boldsymbol{W}_0$: the formulas below are valid when $\sigma_1(\boldsymbol{W}_0) = \cdots = \sigma_q(\boldsymbol{W}_0)$, a case in which individual eigenvectors of $\boldsymbol{S}_n$ are not asymptotically normal at all. This is the quantitative form of the third comment of Remark 39.

Block coordinates at the true parameter

Throughout this section we keep the notation (2.81), we fix the singular value decomposition $\boldsymbol{W}_0 = \boldsymbol{U}_0 \boldsymbol{D}_0 \boldsymbol{V}_0^T$ of Lemma 16, and we complete $\boldsymbol{U}_0$ to an orthogonal matrix: we let $\boldsymbol{U}_0^{\perp} \in \mathbb{R}^{p \times (p-q)}$ have orthonormal columns spanning the orthogonal complement of the column space of $\boldsymbol{W}_0$, so that

$$\boldsymbol{J}_0 := \left[\boldsymbol{U}_0 \;\; \boldsymbol{U}_0^{\perp}\right] \in O(p), \qquad \boldsymbol{I}_p - \boldsymbol{P}_0 = \boldsymbol{U}_0^{\perp} \left(\boldsymbol{U}_0^{\perp}\right)^T. \tag{2.121}$$

We abbreviate the eigenvalues of $\boldsymbol{\Sigma}_0$ supplied by Lemma 16 as

$$\tau_j := \lambda_j(\boldsymbol{\Sigma}_0) = \sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2 > \sigma_0^2 \qquad (j = 1, \ldots, q), \qquad \boldsymbol{T}_0 := \boldsymbol{D}_0^2 + \sigma_0^2 \boldsymbol{I}_q = \operatorname{diag}(\tau_1, \ldots, \tau_q). \tag{2.122}$$

The letters $\boldsymbol{B}$, $\boldsymbol{C}$, $\boldsymbol{\Psi}$, $\boldsymbol{y}$, $\boldsymbol{z}$, $\boldsymbol{J}_0$, $\boldsymbol{T}_0$, τ_j, κ and $\mathcal{G}$ are local to this section.

Lemma 61 (Block form of the differential)**.** Let $\boldsymbol{h} \in T_{\theta_0} M$ have horizontal lift $\iota(\boldsymbol{h}) = (\boldsymbol{U}_{\boldsymbol{h}}, b_{\boldsymbol{h}})$ as in (2.2), and set

$$\boldsymbol{B} := \boldsymbol{U}_0^T \boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{V}_0 \in \mathbb{R}^{q \times q}, \qquad \boldsymbol{C} := \left(\boldsymbol{U}_0^{\perp}\right)^T \boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{V}_0 \in \mathbb{R}^{(p-q) \times q}, \qquad \boldsymbol{\Psi} := \frac{1}{2}\left(\boldsymbol{B}\boldsymbol{D}_0 + \boldsymbol{D}_0 \boldsymbol{B}^T\right) \in \operatorname{Sym}(q). \tag{2.123}$$

Then:

1. $\boldsymbol{J}_0^T \boldsymbol{\Sigma}_0 \boldsymbol{J}_0 = \begin{pmatrix} \boldsymbol{T}_0 & \boldsymbol{0} \\ \boldsymbol{0} & \sigma_0^2 \boldsymbol{I}_{p-q} \end{pmatrix}$ and $\boldsymbol{J}_0^T \boldsymbol{\Sigma}_0^{-1} \boldsymbol{J}_0 = \begin{pmatrix} \boldsymbol{T}_0^{-1} & \boldsymbol{0} \\ \boldsymbol{0} & \sigma_0^{-2} \boldsymbol{I}_{p-q} \end{pmatrix}$.
2. $\boldsymbol{U}_{\boldsymbol{h}} = \boldsymbol{U}_0 \boldsymbol{B} \boldsymbol{V}_0^T + \boldsymbol{U}_0^{\perp} \boldsymbol{C} \boldsymbol{V}_0^T$ and $|\boldsymbol{h}|^2 = \|\boldsymbol{B}\|_F^2 + \|\boldsymbol{C}\|_F^2 + b_{\boldsymbol{h}}^2$.
3. $\boldsymbol{D}_0 \boldsymbol{B} \in \operatorname{Sym}(q)$; conversely every triple $(\boldsymbol{B}, \boldsymbol{C}, b)$ with $\boldsymbol{D}_0 \boldsymbol{B}$ symmetric arises from exactly one $\boldsymbol{h} \in T_{\theta_0} M$.
4. In the basis (2.121),
$$\boldsymbol{J}_0^T \, D_{\boldsymbol{h}} \boldsymbol{\Sigma}(\theta_0) \, \boldsymbol{J}_0 = \begin{pmatrix} 2\boldsymbol{\Psi} + b_{\boldsymbol{h}} \boldsymbol{I}_q & \boldsymbol{D}_0 \boldsymbol{C}^T \\ \boldsymbol{C} \boldsymbol{D}_0 & b_{\boldsymbol{h}} \boldsymbol{I}_{p-q} \end{pmatrix}. \tag{2.124}$$

Proof. 1. Substitute $\boldsymbol{P}_0 = \boldsymbol{U}_0 \boldsymbol{U}_0^T$ and $\boldsymbol{I}_p - \boldsymbol{P}_0 = \boldsymbol{U}_0^{\perp} (\boldsymbol{U}_0^{\perp})^T$ into (2.13) and use $\boldsymbol{U}_0^T \boldsymbol{U}_0 = \boldsymbol{I}_q$, $(\boldsymbol{U}_0^{\perp})^T \boldsymbol{U}_0^{\perp} = \boldsymbol{I}_{p-q}$, $\boldsymbol{U}_0^T \boldsymbol{U}_0^{\perp} = \boldsymbol{0}$.

2. Since $\boldsymbol{J}_0 \in O(p)$ and $\boldsymbol{V}_0 \in O(q)$, the matrix $\boldsymbol{J}_0^T \boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{V}_0$ has blocks $\boldsymbol{B}$ and $\boldsymbol{C}$ stacked, which is the first identity; the Frobenius norm is invariant under multiplication by orthogonal matrices on either side, so $\|\boldsymbol{U}_{\boldsymbol{h}}\|_F^2 = \|\boldsymbol{B}\|_F^2 + \|\boldsymbol{C}\|_F^2$, and the second identity is (2.2).

3. By part 3 of Proposition 1 the pair $(\boldsymbol{U}_{\boldsymbol{h}}, b_{\boldsymbol{h}})$ is horizontal at $(\boldsymbol{W}_0, \sigma_0^2)$ if and only if $\boldsymbol{U}_{\boldsymbol{h}}^T \boldsymbol{W}_0 = \boldsymbol{W}_0^T \boldsymbol{U}_{\boldsymbol{h}}$. Using part 2 and $\boldsymbol{W}_0 = \boldsymbol{U}_0 \boldsymbol{D}_0 \boldsymbol{V}_0^T$,

$$\boldsymbol{W}_0^T \boldsymbol{U}_{\boldsymbol{h}} = \boldsymbol{V}_0 \boldsymbol{D}_0 \boldsymbol{U}_0^T \left(\boldsymbol{U}_0 \boldsymbol{B} \boldsymbol{V}_0^T + \boldsymbol{U}_0^\perp \boldsymbol{C} \boldsymbol{V}_0^T \right) = \boldsymbol{V}_0 \left(\boldsymbol{D}_0 \boldsymbol{B} \right) \boldsymbol{V}_0^T,$$

which is symmetric if and only if $\boldsymbol{D}_0 \boldsymbol{B}$ is. Conversely, given such a triple, part 2 defines $\boldsymbol{U}_{\boldsymbol{h}}$, the displayed identity makes $(\boldsymbol{U}_{\boldsymbol{h}}, b)$ horizontal, and $\boldsymbol{h} := d\pi_{(\boldsymbol{W}_0, \sigma_0^2)}(\boldsymbol{U}_{\boldsymbol{h}}, b)$ is the unique preimage because ι is a bijection onto $H_{(\boldsymbol{W}_0, \sigma_0^2)}$.

4. By (1.23), $D_{\boldsymbol{h}} \boldsymbol{\Sigma}(\theta_0) = \boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{W}_0^T + \boldsymbol{W}_0 \boldsymbol{U}_{\boldsymbol{h}}^T + b_{\boldsymbol{h}} \boldsymbol{I}_p$. By part 2,

$$\boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{W}_0^T = \left(\boldsymbol{U}_0 \boldsymbol{B} \boldsymbol{V}_0^T + \boldsymbol{U}_0^\perp \boldsymbol{C} \boldsymbol{V}_0^T \right) \boldsymbol{V}_0 \boldsymbol{D}_0 \boldsymbol{U}_0^T = \boldsymbol{U}_0 \boldsymbol{B} \boldsymbol{D}_0 \boldsymbol{U}_0^T + \boldsymbol{U}_0^\perp \boldsymbol{C} \boldsymbol{D}_0 \boldsymbol{U}_0^T,$$

so that $\boldsymbol{J}_0^T \boldsymbol{U}_{\boldsymbol{h}} \boldsymbol{W}_0^T \boldsymbol{J}_0 = \begin{pmatrix} \boldsymbol{B}\boldsymbol{D}_0 & \boldsymbol{0} \\ \boldsymbol{C}\boldsymbol{D}_0 & \boldsymbol{0} \end{pmatrix}$. Adding the transpose of this matrix and $b_{\boldsymbol{h}} \boldsymbol{I}_p$, and using $\boldsymbol{B}\boldsymbol{D}_0 + (\boldsymbol{B}\boldsymbol{D}_0)^T = 2\boldsymbol{\Psi}$, gives (2.124). □

The three blocks have direct meanings, and it is worth recording them before any computation. By (2.124) the matrix $2\boldsymbol{\Psi} + b_{\boldsymbol{h}} \boldsymbol{I}_q$ is the first order change of $\boldsymbol{\Sigma}$ inside the signal block, the matrix $\boldsymbol{C}\boldsymbol{D}_0$ is its change in the signal–noise block, and $b_{\boldsymbol{h}} \boldsymbol{I}_{p-q}$ is its change in the noise block — where the model forces the change to be a *multiple of the identity*, which is the whole source of the efficiency gain computed below. Note also that $\boldsymbol{\Psi}$, not $\boldsymbol{B}$, is the natural unknown: the map $\boldsymbol{B} \mapsto \boldsymbol{\Psi}$ is exactly the operation that quotients out the $\frac{q(q-1)}{2}$ directions along which $\boldsymbol{\Sigma}$ does not move.

Lemma 62 (Linear coordinates on $T_{\theta_0} M$)**.** The map

$$\kappa : T_{\theta_0} M \longrightarrow \mathsf{Sym}(q) \times \mathbb{R}^{(p-q)\times q} \times \mathbb{R}, \qquad \kappa(\boldsymbol{h}) := (\boldsymbol{\Psi}, \boldsymbol{C}, b_{\boldsymbol{h}}) \tag{2.125}$$

of (2.123) is a linear isomorphism.

Proof. Linearity is clear: ι is linear by (2.2), and $\boldsymbol{U}_{\boldsymbol{h}} \mapsto (\boldsymbol{B}, \boldsymbol{C}) \mapsto \boldsymbol{\Psi}$ is a composition of linear maps.

For injectivity, suppose $\kappa(\boldsymbol{h}) = \boldsymbol{0}$. Then $b_{\boldsymbol{h}} = 0$ and $\boldsymbol{C} = \boldsymbol{0}$, and $\boldsymbol{\Psi} = \boldsymbol{0}$ reads $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0 \boldsymbol{B}^T$. Since $\operatorname{rank}(\boldsymbol{W}_0) = q$, all $\sigma_j(\boldsymbol{W}_0)$ are strictly positive and $\boldsymbol{D}_0$ is invertible, so part 3 of Lemma 61, that is $\boldsymbol{D}_0 \boldsymbol{B} = \boldsymbol{B}^T \boldsymbol{D}_0$, gives $\boldsymbol{B}^T = \boldsymbol{D}_0 \boldsymbol{B} \boldsymbol{D}_0^{-1}$. Substituting this into $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0 \boldsymbol{B}^T$ yields $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0^2 \boldsymbol{B} \boldsymbol{D}_0^{-1}$, and multiplying on the right by $\boldsymbol{D}_0$,

$$\boldsymbol{B}\boldsymbol{D}_0^2 + \boldsymbol{D}_0^2 \boldsymbol{B} = \boldsymbol{0}.$$

Both $\boldsymbol{D}_0^2$ factors are diagonal, so the (i,j) entry of the left hand side is $(\sigma_i(\boldsymbol{W}_0)^2 + \sigma_j(\boldsymbol{W}_0)^2)\, \boldsymbol{B}_{ij}$, and the bracket is strictly positive; hence $\boldsymbol{B} = \boldsymbol{0}$. By part 2 of Lemma 61 this forces $\boldsymbol{U}_{\boldsymbol{h}} = \boldsymbol{0}$, so $\iota(\boldsymbol{h}) = (\boldsymbol{0}, 0)$ and $\boldsymbol{h} = \boldsymbol{0}$.

For surjectivity it suffices to compare dimensions, since κ is injective and linear:

$$\dim \left(\mathsf{Sym}(q) \times \mathbb{R}^{(p-q)\times q} \times \mathbb{R} \right) = \frac{q(q+1)}{2} + (p-q)q + 1 = pq + 1 - \frac{q(q-1)}{2} = d$$

by (2.1). □

Lemma 62 lets us treat $(\boldsymbol{\Psi}, \boldsymbol{C}, b)$ as coordinates on $T_{\theta_0} M$. We use throughout the *scalar* coordinates

$$(\boldsymbol{\Psi}_{ij})_{1 \leqslant i \leqslant j \leqslant q}, \qquad (\boldsymbol{C}_{ki})_{1 \leqslant k \leqslant p-q,\ 1 \leqslant i \leqslant q}, \qquad b, \tag{2.126}$$

which identify the target of κ with $\mathbb{R}^d$, and we let $\mathcal{G}$ denote the $d \times d$ symmetric matrix representing the Fisher form in them, that is

$$G_{\theta_0}(\boldsymbol{h}_1, \boldsymbol{h}_2) = \kappa(\boldsymbol{h}_1)^T \, \mathcal{G} \, \kappa(\boldsymbol{h}_2) \qquad \text{for all } \boldsymbol{h}_1, \boldsymbol{h}_2 \in T_{\theta_0} M. \tag{2.127}$$

Note that κ is *not* a g_{θ_0}-isometry, so $\mathcal{G}$ is not the matrix $\boldsymbol{G}$ of Step 1 of the proof of Theorem 36; the relation between the two is recorded in the proof of Theorem 65 and is the only place where it is needed.

The Fisher form and its inverse

Proposition 63 (Explicit Fisher form)**.** For every $\boldsymbol{h} \in T_{\theta_0}M$, in the notation (2.123),

$$G_{\theta_0}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{T}_0^{-1}\left(2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q\right)\right)^2\right] + \frac{1}{\sigma_0^2}\operatorname{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) + \frac{(p-q)\,b_{\boldsymbol{h}}^2}{2\sigma_0^4}, \tag{2.128}$$

and, in the scalar coordinates (2.126),

$$G_{\theta_0}(\boldsymbol{h},\boldsymbol{h}) = \underbrace{\sum_{i=1}^{q}\frac{\left(2\boldsymbol{\Psi}_{ii} + b_{\boldsymbol{h}}\right)^2}{2\tau_i^2} + \frac{(p-q)b_{\boldsymbol{h}}^2}{2\sigma_0^4}}_{\text{coupled block}} + \underbrace{\sum_{1\leqslant i<j\leqslant q}\frac{4\boldsymbol{\Psi}_{ij}^2}{\tau_i\tau_j}}_{\boldsymbol{\Psi}\text{ off-diagonal}} + \underbrace{\sum_{i=1}^{q}\left(\frac{1}{\sigma_0^2} - \frac{1}{\tau_i}\right)\|\boldsymbol{C}\boldsymbol{e}_i\|^2}_{\boldsymbol{C}\text{-block}}. \tag{2.129}$$

In particular $\mathcal{G}$ is block diagonal with respect to the partition of (2.126) into the three groups $\{\boldsymbol{\Psi}_{ij}\}_{i<j}$, $\{\boldsymbol{C}_{ki}\}$ and $(\boldsymbol{\Psi}_{11},\ldots,\boldsymbol{\Psi}_{qq},b)$, with

$$\mathcal{G}_{\boldsymbol{\Psi}_{ij}\boldsymbol{\Psi}_{ij}} = \frac{4}{\tau_i\tau_j}\quad(i<j),\qquad \mathcal{G}_{\boldsymbol{C}_{ki}\boldsymbol{C}_{ki}} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} = \frac{\sigma_i(\boldsymbol{W}_0)^2}{\sigma_0^2\tau_i},\qquad \mathcal{G}_{\mathrm{c}} = \begin{pmatrix}\operatorname{diag}\left(2\tau_i^{-2}\right) & \boldsymbol{v}\\ \boldsymbol{v}^T & \gamma\end{pmatrix}, \tag{2.130}$$

the last being the coupled block in the ordering $(\boldsymbol{\Psi}_{11},\ldots,\boldsymbol{\Psi}_{qq},b)$, where

$$\boldsymbol{v} := \left(\tau_1^{-2},\ldots,\tau_q^{-2}\right)^T \in \mathbb{R}^q,\qquad \gamma := \frac{1}{2}\sum_{i=1}^{q}\frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}. \tag{2.131}$$

Proof. Write $\boldsymbol{M} := 2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q$ and $b := b_{\boldsymbol{h}}$. Conjugating by $\boldsymbol{J}_0$ does not change traces, so by (1.24) with $\boldsymbol{h}_1 = \boldsymbol{h}_2 = \boldsymbol{h}$ it suffices to compute the trace of the square of $\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0)\boldsymbol{J}_0$. Multiplying the two matrices of Lemma 61,

$$\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0)\boldsymbol{J}_0 = \begin{pmatrix}\boldsymbol{T}_0^{-1}\boldsymbol{M} & \boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T\\ \sigma_0^{-2}\boldsymbol{C}\boldsymbol{D}_0 & \sigma_0^{-2}b\,\boldsymbol{I}_{p-q}\end{pmatrix}. \tag{2.132}$$

The two diagonal blocks of the square of (2.132) are

$$\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2 + \sigma_0^{-2}\boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{D}_0 \qquad\text{and}\qquad \sigma_0^{-2}\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T + \sigma_0^{-4}b^2\boldsymbol{I}_{p-q},$$

so, using the cyclic property of the trace and the fact that the diagonal matrices $\boldsymbol{T}_0$ and $\boldsymbol{D}_0$ commute,

$$\operatorname{tr}\left[\left(\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0)\right)^2\right] = \operatorname{tr}\left[\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2\right] + \frac{2}{\sigma_0^2}\operatorname{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) + \frac{(p-q)b^2}{\sigma_0^4}.$$

Halving gives (2.128).

For (2.129), both $\boldsymbol{T}_0^{-1}$ and $\boldsymbol{D}_0^2$ are diagonal. Hence $\operatorname{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) = \sum_{i=1}^{q}\frac{\sigma_i(\boldsymbol{W}_0)^2}{\tau_i}\left(\boldsymbol{C}^T\boldsymbol{C}\right)_{ii} = \sum_{i=1}^{q}\frac{\sigma_i(\boldsymbol{W}_0)^2}{\tau_i}\|\boldsymbol{C}\boldsymbol{e}_i\|^2$, and $\frac{\sigma_i(\boldsymbol{W}_0)^2}{\sigma_0^2\tau_i} = \frac{\tau_i - \sigma_0^2}{\sigma_0^2\tau_i} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i}$ by (2.122). For the first term, $\boldsymbol{M}$ is symmetric, so

$$\operatorname{tr}\left[\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2\right] = \sum_{i,j=1}^{q}\frac{\boldsymbol{M}_{ij}\boldsymbol{M}_{ji}}{\tau_i\tau_j} = \sum_{i=1}^{q}\frac{\boldsymbol{M}_{ii}^2}{\tau_i^2} + 2\sum_{i<j}\frac{\boldsymbol{M}_{ij}^2}{\tau_i\tau_j},$$

and $\boldsymbol{M}_{ii} = 2\boldsymbol{\Psi}_{ii} + b$, $\boldsymbol{M}_{ij} = 2\boldsymbol{\Psi}_{ij}$ for $i \neq j$; halving gives the first two groups of terms in (2.129).

Finally, (2.129) is a sum of three quadratic forms in *disjoint* sets of the scalar coordinates (2.126), which is exactly the assertion of block diagonality; reading off the coefficients gives the first two entries of (2.130). For the third, expanding the coupled part of (2.129),

$$\sum_i\frac{\left(2\boldsymbol{\Psi}_{ii} + b\right)^2}{2\tau_i^2} + \frac{(p-q)b^2}{2\sigma_0^4} = \sum_i\frac{2\boldsymbol{\Psi}_{ii}^2}{\tau_i^2} + 2b\sum_i\frac{\boldsymbol{\Psi}_{ii}}{\tau_i^2} + b^2\left(\frac{1}{2}\sum_i\frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}\right),$$

whose matrix in the ordering $(\boldsymbol{\Psi}_{11},\dots,\boldsymbol{\Psi}_{qq},b)$ is $\mathcal{G}_{\mathrm{c}}$ of (2.130): the coefficient of $\boldsymbol{\Psi}_{ii}^2$ is $2\tau_i^{-2}$, that of $b\,\boldsymbol{\Psi}_{ii}$ is $2\tau_i^{-2}=2\boldsymbol{v}_i$, and that of b^2 is γ. □

Proposition 64 (Explicit inverse)**.** The matrix $\mathcal{G}$ of (2.127) is invertible, and $\mathcal{G}^{-1}$ is block diagonal with respect to the same partition, with

$$\left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ij}\boldsymbol{\Psi}_{ij}}=\frac{\tau_i\tau_j}{4}\quad(i<j),\qquad\left(\mathcal{G}^{-1}\right)_{\boldsymbol{C}_{ki}\boldsymbol{C}_{ki}}=\frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2}=\left(\frac{1}{\sigma_0^2}-\frac{1}{\tau_i}\right)^{-1},\tag{2.133}$$

these two groups being diagonal, and, on the coupled block,

$$\left(\mathcal{G}^{-1}\right)_{bb}=\frac{2\sigma_0^4}{p-q},\qquad\left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ii}b}=-\frac{\sigma_0^4}{p-q},\qquad\left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ii}\boldsymbol{\Psi}_{jj}}=\delta_{ij}\frac{\tau_i^2}{2}+\frac{\sigma_0^4}{2(p-q)}.\tag{2.134}$$

Proof. $\mathcal{G}$ is positive definite because G_{θ_0} is, by Theorem 7, and κ is a bijection; so it is invertible, and the inverse of a block diagonal matrix is block diagonal with the inverted blocks. The first two groups are diagonal by (2.130), and their inverses are the reciprocals, which is (2.133).

For the coupled block write $\mathcal{G}_{\mathrm{c}}=\begin{pmatrix}\boldsymbol{P}&\boldsymbol{v}\\\boldsymbol{v}^T&\gamma\end{pmatrix}$ with $\boldsymbol{P}:=\operatorname{diag}(2\tau_i^{-2})$ as in (2.130). Then $\boldsymbol{P}^{-1}=\operatorname{diag}(\tau_i^2/2)$ and

$$\boldsymbol{P}^{-1}\boldsymbol{v}=\left(\frac{\tau_i^2}{2}\cdot\frac{1}{\tau_i^2}\right)_{i=1}^{q}=\frac{1}{2}\mathbf{1}_q,\tag{2.135}$$

so the Schur complement of $\boldsymbol{P}$ in $\mathcal{G}_{\mathrm{c}}$ is

$$\varsigma:=\gamma-\boldsymbol{v}^T\boldsymbol{P}^{-1}\boldsymbol{v}=\gamma-\frac{1}{2}\sum_{i=1}^{q}\frac{1}{\tau_i^2}=\frac{p-q}{2\sigma_0^4}\tag{2.136}$$

by (2.131); it is strictly positive because $q<p$, which is where the standing assumption $q<p$ enters the inversion. The standard block inversion formula

$$\begin{pmatrix}\boldsymbol{P}&\boldsymbol{v}\\\boldsymbol{v}^T&\gamma\end{pmatrix}^{-1}=\begin{pmatrix}\boldsymbol{P}^{-1}+\varsigma^{-1}\boldsymbol{P}^{-1}\boldsymbol{v}\boldsymbol{v}^T\boldsymbol{P}^{-1}&-\varsigma^{-1}\boldsymbol{P}^{-1}\boldsymbol{v}\\-\varsigma^{-1}\boldsymbol{v}^T\boldsymbol{P}^{-1}&\varsigma^{-1}\end{pmatrix}$$

together with (2.135) and (2.136) gives $\varsigma^{-1}=\frac{2\sigma_0^4}{p-q}$, then $-\varsigma^{-1}\left(\boldsymbol{P}^{-1}\boldsymbol{v}\right)_i=-\frac{2\sigma_0^4}{p-q}\cdot\frac{1}{2}=-\frac{\sigma_0^4}{p-q}$, and finally

$$\left(\boldsymbol{P}^{-1}\right)_{ij}+\varsigma^{-1}\left(\boldsymbol{P}^{-1}\boldsymbol{v}\right)_i\left(\boldsymbol{P}^{-1}\boldsymbol{v}\right)_j=\delta_{ij}\frac{\tau_i^2}{2}+\frac{2\sigma_0^4}{p-q}\cdot\frac{1}{2}\cdot\frac{1}{2}=\delta_{ij}\frac{\tau_i^2}{2}+\frac{\sigma_0^4}{2(p-q)},$$

which is (2.134). □

Theorem 65 (Theorem 36 in block coordinates)**.** *Let $1\leqslant q<p$ and $\theta_0=[\boldsymbol{W}_0,\sigma_0^2]\in M$. Define, on the event $\mathcal{A}_n$ of (2.39), the aligned error blocks*

$$\begin{gathered}\boldsymbol{B}_n:=\boldsymbol{U}_0^T\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n-\boldsymbol{W}_0\right)\boldsymbol{V}_0,\qquad\boldsymbol{C}_n:=\left(\boldsymbol{U}_0^{\perp}\right)^T\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n-\boldsymbol{W}_0\right)\boldsymbol{V}_0,\\\boldsymbol{\Psi}_n:=\frac{1}{2}\left(\boldsymbol{B}_n\boldsymbol{D}_0+\boldsymbol{D}_0\boldsymbol{B}_n^T\right),\qquad b_n:=\hat{\sigma}_n^2-\sigma_0^2,\end{gathered}\tag{2.137}$$

and $(\boldsymbol{\Psi}_n,\boldsymbol{C}_n,b_n):=(\mathbf{0},\mathbf{0},0)$ off $\mathcal{A}_n$. Then

$$\sqrt{n}\left(\boldsymbol{\Psi}_n,\boldsymbol{C}_n,b_n\right)\rightsquigarrow N\left(\mathbf{0},\mathcal{G}^{-1}\right)\qquad\text{in } Sym(q)\times\mathbb{R}^{(p-q)\times q}\times\mathbb{R},\tag{2.138}$$

with $\mathcal{G}^{-1}$ given by Proposition 64. Explicitly, writing $\boldsymbol{Z}=(\boldsymbol{Z}_{\boldsymbol{\Psi}},\boldsymbol{Z}_{\boldsymbol{C}},Z_b)$ for the limit:

1. *the $(p-q)q$ entries of $\boldsymbol{Z}_{\boldsymbol{C}}$ are mutually independent, independent of $(\boldsymbol{Z}_{\boldsymbol{\Psi}},Z_b)$, and Var $((\boldsymbol{Z}_{\boldsymbol{C}})_{ki})=\frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2}$;*

2. *the $\frac{q(q-1)}{2}$ strictly upper triangular entries of $\boldsymbol{Z}_{\boldsymbol{\Psi}}$ are mutually independent, independent of everything else, and Var $((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ij}) = \frac{\tau_i\tau_j}{4}$ for $i < j$;*

3. *the remaining $q+1$ coordinates $((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{11}, \ldots, (\boldsymbol{Z}_{\boldsymbol{\Psi}})_{qq}, Z_b)$ are jointly Gaussian with the covariances (2.134); in particular*

$$\sqrt{n}\left(\hat{\sigma}_n^2 - \sigma_0^2\right) \rightsquigarrow N\left(0, \frac{2\sigma_0^4}{p-q}\right). \tag{2.139}$$

Proof. By Proposition 32 we have $\iota(\hat{\boldsymbol{u}}_n) = \boldsymbol{e}_n = (\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n - \boldsymbol{W}_0,\ \hat{\sigma}_n^2 - \sigma_0^2)$ on $\mathcal{A}_n$, so by (2.123) the triple $(\boldsymbol{\Psi}_n, \boldsymbol{C}_n, b_n)$ of (2.137) is exactly $\kappa(\hat{\boldsymbol{u}}_n)$ there, and both vanish off $\mathcal{A}_n$ because $\hat{\boldsymbol{u}}_n = \boldsymbol{0}$ there by (2.39). Hence $(\boldsymbol{\Psi}_n, \boldsymbol{C}_n, b_n) = \kappa(\hat{\boldsymbol{u}}_n)$ identically.

Let $\boldsymbol{K}$ denote the matrix of κ with respect to the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ of Step 1 of the proof of Theorem 36 on the source and the scalar coordinates (2.126) on the target. Comparing (2.127) with $\boldsymbol{G}_{jk} = G_{\theta_0}(\boldsymbol{e}_j, \boldsymbol{e}_k)$ gives $\boldsymbol{G} = \boldsymbol{K}^T\mathcal{G}\boldsymbol{K}$, so

$$\boldsymbol{K}\boldsymbol{G}^{-1}\boldsymbol{K}^T = \boldsymbol{K}\left(\boldsymbol{K}^T\mathcal{G}\boldsymbol{K}\right)^{-1}\boldsymbol{K}^T = \boldsymbol{K}\boldsymbol{K}^{-1}\mathcal{G}^{-1}\boldsymbol{K}^{-T}\boldsymbol{K}^T = \mathcal{G}^{-1}. \tag{2.140}$$

By (2.44) we have $\sqrt{n}\hat{\boldsymbol{u}}_n \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}^{-1})$ in the coordinates of that basis, so the continuous mapping theorem applied to the linear map $\boldsymbol{K}$ and (2.140) give (2.138).

The three assertions are then read off Proposition 64: a Gaussian vector whose covariance matrix is block diagonal has independent blocks, and within the first two groups the blocks are diagonal, so the coordinates are mutually independent. Statement (2.139) is the (b, b) entry of (2.134) together with $b_n = \hat{\sigma}_n^2 - \sigma_0^2$. □

The score and the influence function in coordinates

The same block decomposition makes the score (1.22) and the influence operator (2.42) completely explicit. Throughout, for $\boldsymbol{x} \in \mathbb{R}^p$ we write

$$\boldsymbol{y} := \boldsymbol{U}_0^T\boldsymbol{x} \in \mathbb{R}^q, \qquad \boldsymbol{z} := \left(\boldsymbol{U}_0^\perp\right)^T\boldsymbol{x} \in \mathbb{R}^{p-q}, \tag{2.141}$$

so that, by part 1 of Lemma 61, when $\boldsymbol{X} \sim N_p(\boldsymbol{0}, \boldsymbol{\Sigma}_0)$ the vectors $\boldsymbol{y} = \boldsymbol{U}_0^T\boldsymbol{X}$ and $\boldsymbol{z} = (\boldsymbol{U}_0^\perp)^T\boldsymbol{X}$ are independent, $\boldsymbol{y} \sim N_q(\boldsymbol{0}, \boldsymbol{T}_0)$ and $\boldsymbol{z} \sim N_{p-q}(\boldsymbol{0}, \sigma_0^2\boldsymbol{I}_{p-q})$.

Proposition 66 (Score and influence function in coordinates)**.** Let $\boldsymbol{s}(\boldsymbol{x}) \in \mathbb{R}^d$ be the coordinate vector of the score, defined by $s(\boldsymbol{x}; \theta_0)(\boldsymbol{h}) = \boldsymbol{s}(\boldsymbol{x})^T\kappa(\boldsymbol{h})$ for all $\boldsymbol{h} \in T_{\theta_0}M$. Then, in the scalar coordinates (2.126),

$$\boldsymbol{s}_{\boldsymbol{\Psi}_{ii}} = \frac{\boldsymbol{y}_i^2 - \tau_i}{\tau_i^2}, \qquad \boldsymbol{s}_{\boldsymbol{\Psi}_{ij}} = \frac{2\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j} \quad (i < j), \qquad \boldsymbol{s}_{\boldsymbol{C}_{ki}} = \frac{\sigma_i(\boldsymbol{W}_0)\,\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_0^2\tau_i}, \tag{2.142}$$

$$\boldsymbol{s}_b = \frac{1}{2}\left[\sum_{i=1}^q \frac{\boldsymbol{y}_i^2 - \tau_i}{\tau_i^2} + \frac{\|\boldsymbol{z}\|^2 - (p-q)\sigma_0^2}{\sigma_0^4}\right]. \tag{2.143}$$

Moreover $\mathbb{E}[\boldsymbol{s}(\boldsymbol{X})] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{s}(\boldsymbol{X})\boldsymbol{s}(\boldsymbol{X})^T] = \mathcal{G}$, the influence operator (2.42) has coordinate vector $\kappa(\mathrm{IF}(\boldsymbol{x})) = \mathcal{G}^{-1}\boldsymbol{s}(\boldsymbol{x})$, and explicitly

$$\mathrm{IF}_b(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right)\boldsymbol{x} - (p-q)\sigma_0^2}{p-q}, \qquad \mathrm{IF}_{\boldsymbol{C}_{ki}}(\boldsymbol{x}) = \frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)}, \tag{2.144}$$

$$\mathrm{IF}_{\boldsymbol{\Psi}_{ij}}(\boldsymbol{x}) = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{2} \quad (i < j), \qquad \mathrm{IF}_{\boldsymbol{\Psi}_{ii}}(\boldsymbol{x}) = \frac{1}{2}\left[\boldsymbol{y}_i^2 - \tau_i - \mathrm{IF}_b(\boldsymbol{x})\right]. \tag{2.145}$$

Proof. Abbreviate $\boldsymbol{E} := \boldsymbol{\Sigma}_0^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}_0\right]\boldsymbol{\Sigma}_0^{-1}$, so that (1.22) reads $s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \frac{1}{2}\operatorname{tr}\{\boldsymbol{E}D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0)\}$. Since $\boldsymbol{J}_0^T\boldsymbol{x} = (\boldsymbol{y}^T, \boldsymbol{z}^T)^T$ by (2.141), part 1 of Lemma 61 gives

$$\boldsymbol{J}_0^T\boldsymbol{E}\boldsymbol{J}_0 = \begin{pmatrix} \boldsymbol{T}_0^{-1}\left(\boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{T}_0\right)\boldsymbol{T}_0^{-1} & \sigma_0^{-2}\boldsymbol{T}_0^{-1}\boldsymbol{y}\boldsymbol{z}^T \\ \sigma_0^{-2}\boldsymbol{z}\boldsymbol{y}^T\boldsymbol{T}_0^{-1} & \sigma_0^{-4}\left(\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}\right)\end{pmatrix} =: \begin{pmatrix} \boldsymbol{E}_{11} & \boldsymbol{E}_{12} \\ \boldsymbol{E}_{12}^T & \boldsymbol{E}_{22}\end{pmatrix}. \tag{2.146}$$

Pairing (2.146) with (2.124), the two off-diagonal contributions to the trace are equal, because $\operatorname{tr}\left(\boldsymbol{E}_{12}^T\boldsymbol{D}_0\boldsymbol{C}^T\right) = \operatorname{tr}\left(\left(\boldsymbol{E}_{12}^T\boldsymbol{D}_0\boldsymbol{C}^T\right)^T\right) = \operatorname{tr}\left(\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{E}_{12}\right) = \operatorname{tr}\left(\boldsymbol{E}_{12}\boldsymbol{C}\boldsymbol{D}_0\right)$, and therefore

$$2\,s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \operatorname{tr}\left(\boldsymbol{E}_{11}\left[2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q\right]\right) + \frac{2}{\sigma_0^2}\,\boldsymbol{z}^T\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{T}_0^{-1}\boldsymbol{y} + b_{\boldsymbol{h}}\operatorname{tr}\boldsymbol{E}_{22}. \tag{2.147}$$

Now read off the coefficient of each scalar coordinate. The matrix $\boldsymbol{\Psi}$ is symmetric, so $\operatorname{tr}(\boldsymbol{E}_{11}\boldsymbol{\Psi}) = \sum_i(\boldsymbol{E}_{11})_{ii}\boldsymbol{\Psi}_{ii} + 2\sum_{i<j}(\boldsymbol{E}_{11})_{ij}\boldsymbol{\Psi}_{ij}$, while $(\boldsymbol{E}_{11})_{ii} = \frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2}$ and $(\boldsymbol{E}_{11})_{ij} = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j}$ for $i \neq j$; dividing the resulting $2\operatorname{tr}(\boldsymbol{E}_{11}\boldsymbol{\Psi})$ by the factor 2 in (2.147) gives the first two formulas of (2.142). The second term of (2.147) equals $\frac{2}{\sigma_0^2}\sum_{k,i}\boldsymbol{C}_{ki}\boldsymbol{z}_k\frac{\sigma_i(\boldsymbol{W}_0)\boldsymbol{y}_i}{\tau_i}$, which after division by 2 gives the third formula of (2.142). The coefficient of $b_{\boldsymbol{h}}$ is $\frac{1}{2}\left[\operatorname{tr}\boldsymbol{E}_{11} + \operatorname{tr}\boldsymbol{E}_{22}\right]$, which is (2.143).

The identities $\mathbb{E}[\boldsymbol{s}] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{s}\boldsymbol{s}^T] = \mathcal{G}$ are Theorem 7 read through (2.127), since for all $\boldsymbol{h}_1, \boldsymbol{h}_2 \in T_{\theta_0}M$

$$\mathbb{E}\left[s(\boldsymbol{X};\theta_0)(\boldsymbol{h}_1)\,s(\boldsymbol{X};\theta_0)(\boldsymbol{h}_2)\right] = G_{\theta_0}(\boldsymbol{h}_1, \boldsymbol{h}_2) = \kappa(\boldsymbol{h}_1)^T\mathcal{G}\kappa(\boldsymbol{h}_2)$$

and κ is onto. (They can also be checked directly from (2.142): with $\boldsymbol{y}$ and $\boldsymbol{z}$ independent centred Gaussians, $\mathbb{E}[\boldsymbol{s}_{\boldsymbol{\Psi}_{ii}}^2] = \operatorname{Var}(\boldsymbol{y}_i^2)/\tau_i^4 = 2/\tau_i^2$, $\mathbb{E}[\boldsymbol{s}_{\boldsymbol{\Psi}_{ij}}^2] = 4/(\tau_i\tau_j)$, $\mathbb{E}[\boldsymbol{s}_{\boldsymbol{C}_{ki}}^2] = \sigma_i(\boldsymbol{W}_0)^2\tau_i\sigma_0^2/(\sigma_0^4\tau_i^2) = \sigma_i(\boldsymbol{W}_0)^2/(\sigma_0^2\tau_i)$, $\mathbb{E}[\boldsymbol{s}_{\boldsymbol{\Psi}_{ii}}\boldsymbol{s}_b] = 1/\tau_i^2$ and $\mathbb{E}[\boldsymbol{s}_b^2] = \gamma$, every product containing an odd power of a coordinate of $\boldsymbol{z}$ having mean zero; these are exactly the entries (2.130).)

By (2.3) the vector $\operatorname{IF}(\boldsymbol{x})$ is characterized by $G_{\theta_0}(\operatorname{IF}(\boldsymbol{x}), \boldsymbol{h}) = s(\boldsymbol{x};\theta_0)(\boldsymbol{h})$ for all $\boldsymbol{h}$, that is, by (2.127), $\kappa(\operatorname{IF}(\boldsymbol{x}))^T\mathcal{G}\kappa(\boldsymbol{h}) = \boldsymbol{s}(\boldsymbol{x})^T\kappa(\boldsymbol{h})$ for all $\boldsymbol{h}$; since κ is onto, $\mathcal{G}\kappa(\operatorname{IF}(\boldsymbol{x})) = \boldsymbol{s}(\boldsymbol{x})$.

It remains to compute $\mathcal{G}^{-1}\boldsymbol{s}$ block by block. For $i < j$, by (2.133), $\operatorname{IF}_{\boldsymbol{\Psi}_{ij}} = \frac{\tau_i\tau_j}{4}\cdot\frac{2\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j} = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{2}$, and $\operatorname{IF}_{\boldsymbol{C}_{ki}} = \frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2}\cdot\frac{\sigma_i(\boldsymbol{W}_0)\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_0^2\tau_i} = \frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)}$, which are the second formula of (2.144) and the first of (2.145). On the coupled block, (2.134) and (2.143) give

$$\begin{aligned}\operatorname{IF}_b &= \frac{2\sigma_0^4}{p-q}\,\boldsymbol{s}_b - \frac{\sigma_0^4}{p-q}\sum_{i=1}^q\boldsymbol{s}_{\boldsymbol{\Psi}_{ii}} \\ &= \frac{\sigma_0^4}{p-q}\sum_{i=1}^q\frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2} + \frac{\|\boldsymbol{z}\|^2 - (p-q)\sigma_0^2}{p-q} - \frac{\sigma_0^4}{p-q}\sum_{i=1}^q\frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2} = \frac{\|\boldsymbol{z}\|^2 - (p-q)\sigma_0^2}{p-q},\end{aligned}$$

the $\boldsymbol{y}$ terms cancelling identically; since $\|\boldsymbol{z}\|^2 = \boldsymbol{x}^T\boldsymbol{U}_0^\perp(\boldsymbol{U}_0^\perp)^T\boldsymbol{x} = \boldsymbol{x}^T(\boldsymbol{I}_p - \boldsymbol{P}_0)\boldsymbol{x}$ by (2.121), this is the first formula of (2.144). Similarly

$$\begin{aligned}\operatorname{IF}_{\boldsymbol{\Psi}_{ii}} &= \sum_{j=1}^q\left[\delta_{ij}\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}\right]\boldsymbol{s}_{\boldsymbol{\Psi}_{jj}} - \frac{\sigma_0^4}{p-q}\,\boldsymbol{s}_b \\ &= \frac{\boldsymbol{y}_i^2-\tau_i}{2} + \frac{\sigma_0^4}{2(p-q)}\sum_{j=1}^q\frac{\boldsymbol{y}_j^2-\tau_j}{\tau_j^2} - \frac{\sigma_0^4}{2(p-q)}\sum_{j=1}^q\frac{\boldsymbol{y}_j^2-\tau_j}{\tau_j^2} - \frac{\|\boldsymbol{z}\|^2 - (p-q)\sigma_0^2}{2(p-q)},\end{aligned}$$

which is the second formula of (2.145). □

Corollary 67 (The noise variance)**.** The maximum likelihood estimator of the noise variance satisfies (2.139), and it is asymptotically linear with influence function

$$\mathrm{IF}_{\sigma^2}(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right)\boldsymbol{x} - (p-q)\sigma_0^2}{p-q}, \qquad \mathbb{E}\left[\mathrm{IF}_{\sigma^2}(\boldsymbol{X})\right] = 0, \qquad \mathbb{E}\left[\mathrm{IF}_{\sigma^2}(\boldsymbol{X})^2\right] = \frac{2\sigma_0^4}{p-q}. \tag{2.148}$$

Consequently $\frac{2\sigma_0^4}{p-q}$ cannot be improved: by Theorem 51 the limit law of any locally regular estimator sequence of θ_0 is $N\left(\mathbf{0}, G_{\theta_0}^{-1}\right) * \Lambda$, so the noise variance component of such a sequence has limit law $N\left(0, \frac{2\sigma_0^4}{p-q}\right) * \Lambda_b$ for some probability measure Λ_b on $\mathbb{R}$.

Proof. The weak limit is (2.139). The influence function is the first formula of (2.144), because $b_n = \hat{\sigma}_n^2 - \sigma_0^2$ is the b coordinate of $\kappa(\hat{\boldsymbol{u}}_n)$ by (2.137), so that the b coordinate of (2.43) reads $\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) = n^{-\frac{1}{2}}\sum_{i=1}^n \mathrm{IF}_b(\boldsymbol{X}_i) + o_{\mathbb{P}}(1)$. For the moments, $\boldsymbol{z} \sim N_{p-q}(\mathbf{0}, \sigma_0^2\boldsymbol{I}_{p-q})$ gives $\mathbb{E}\|\boldsymbol{z}\|^2 = (p-q)\sigma_0^2$ and $\mathrm{Var}(\|\boldsymbol{z}\|^2) = 2(p-q)\sigma_0^4$, whence $\mathbb{E}[\mathrm{IF}_{\sigma^2}^2] = \frac{2(p-q)\sigma_0^4}{(p-q)^2} = \frac{2\sigma_0^4}{p-q}$, consistent with (2.139) as it must be, since the asymptotic covariance of an asymptotically linear sequence is the covariance of its influence function. For the last assertion, a linear map carries a convolution of two laws to the convolution of their images, so it suffices to push $L = N(\mathbf{0}, G_{\theta_0}^{-1}) * \Lambda$ forward by the linear map $\boldsymbol{h} \mapsto b_{\boldsymbol{h}}$, which by (2.140) sends $N(\mathbf{0}, G_{\theta_0}^{-1})$ to $N\left(0, (\mathcal{G}^{-1})_{bb}\right) = N\left(0, \frac{2\sigma_0^4}{p-q}\right)$ by (2.134). $\square$

Remark 68 (Where the model helps, and where it does not)**.** Formulas (2.144)–(2.145) answer a question the asymptotic theory alone does not: in which directions is the constrained estimator $\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}([\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2])$ actually better than the sample covariance matrix $\boldsymbol{S}_n$? The answer is: in exactly one block, and the reason is Proposition 21.

By (2.47) the quadratic term of $\boldsymbol{\Sigma}_{\boldsymbol{u}} - \boldsymbol{\Sigma}_0$ is $O_{\mathbb{P}}(n^{-1})$ under Theorem 36, so $\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) = \sqrt{n}D_{\hat{\boldsymbol{u}}_n}\boldsymbol{\Sigma}(\theta_0) + o_{\mathbb{P}}(1)$ and the influence function of $\hat{\boldsymbol{\Sigma}}_n$ is, by (2.124), the matrix with blocks $2\mathrm{IF}_{\boldsymbol{\Psi}} + \mathrm{IF}_b\boldsymbol{I}_q$, $\mathrm{IF}_{\boldsymbol{C}}\boldsymbol{D}_0$ and $\mathrm{IF}_b\boldsymbol{I}_{p-q}$. By (2.144)–(2.145) these are

$$\left(2\mathrm{IF}_{\boldsymbol{\Psi}} + \mathrm{IF}_b\boldsymbol{I}_q\right)_{ij} = \boldsymbol{y}_i\boldsymbol{y}_j - \delta_{ij}\tau_i, \qquad \left(\mathrm{IF}_{\boldsymbol{C}}\boldsymbol{D}_0\right)_{ki} = \sigma_i(\boldsymbol{W}_0)\frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)} = \boldsymbol{z}_k\boldsymbol{y}_i, \qquad \mathrm{IF}_b\boldsymbol{I}_{p-q}, \tag{2.149}$$

the diagonal case of the first identity being exactly the cancellation of IF_b in the second formula of (2.145). Now the influence function of $\boldsymbol{S}_n$, read in the same basis (2.121), is $\boldsymbol{J}_0^T(\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}_0)\boldsymbol{J}_0$, whose blocks are $\boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{T}_0$, $\boldsymbol{z}\boldsymbol{y}^T$ and $\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}$. The first two coincide with (2.149): *to first order the maximum likelihood fit reproduces the sample covariance matrix in the signal block and in the signal–noise block, and gains nothing there*. Only the trailing block differs, where $\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}$ is replaced by its average $\frac{\mathrm{tr}\left(\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}\right)}{p-q}\boldsymbol{I}_{p-q} = \mathrm{IF}_b\boldsymbol{I}_{p-q}$: the isotropy constraint of the model averages the $p-q$ noise directions, dividing the variance by $p-q$. This is the infinitesimal shadow of the exact identity (2.23) of Proposition 21, which says the same thing non-asymptotically: $\hat{\boldsymbol{\Sigma}}$ differs from $\boldsymbol{S}$ only through the trailing eigenvalues, which it replaces by their mean. It also explains why the leading block cannot be improved: by Proposition 23 the model imposes no constraint there at all.

Remark 69 (Cross-checks against classical principal component analysis)**.** Three consistency checks, each of which recovers a classical formula and none of which is used elsewhere.

1. Pooled noise eigenvalues. The variance $\frac{2\sigma_0^4}{p-q}$ of (2.139) is the classical asymptotic variance of the average of the $p-q$ sample eigenvalues attached to a population eigenvalue of multiplicity $p-q$; indeed $\hat{\sigma}_n^2 = \frac{1}{p-q}\sum_{j>q}\lambda_j(\boldsymbol{S}_n)$ exactly, by (2.20), and $\sum_{j>q}\lambda_j(\boldsymbol{S}_n) = \mathrm{tr}\left(\left(\boldsymbol{I}_p - \boldsymbol{P}_q(\boldsymbol{S}_n)\right)\boldsymbol{S}_n\right)$, in which $\boldsymbol{P}_q(\boldsymbol{S}_n)$ may be replaced by $\boldsymbol{P}_0$ at a cost of $O_{\mathbb{P}}(n^{-1})$, by Lemma 25 and part 5 of Lemma 15; the statistic that remains is $\frac{1}{p-q}\mathrm{tr}\left(\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right)\boldsymbol{S}_n\right)$, whose centred and $\sqrt{n}$-scaled version is exactly the average of the influence functions (2.148).

2. The leading block is not improved. By (2.134),

$$\mathrm{Var}\left(2(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ii} + Z_b\right) = 4\left[\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}\right] + 4\left[-\frac{\sigma_0^4}{p-q}\right] + \frac{2\sigma_0^4}{p-q} = 2\tau_i^2 = \mathrm{Var}\left(\left(\boldsymbol{U}_0^T\boldsymbol{X}\right)_i^2\right),$$

and likewise $\mathrm{Var}(2(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ij}) = \tau_i\tau_j = \mathrm{Var}(\boldsymbol{y}_i\boldsymbol{y}_j)$ for $i < j$ — the sample covariance variances for those coordinates, as Remark 68 requires.

3. Anderson's eigenvector variances. Suppose the $\sigma_i(\boldsymbol{W}_0)$ are simple. By part 2 of Lemma 61, the component of the aligned loading error orthogonal to the column space of $\boldsymbol{W}_0$ is $\boldsymbol{U}_0^{\perp}\boldsymbol{C}_n\boldsymbol{V}_0^T$, so the tilt of the estimated i-th direction out of the signal subspace is $\boldsymbol{U}_0^{\perp}\boldsymbol{C}_n\boldsymbol{e}_i/\sigma_i(\boldsymbol{W}_0)$, with limiting per-coordinate variance

$$\frac{1}{\sigma_i(\boldsymbol{W}_0)^2}\cdot\frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2} = \frac{\sigma_0^2\tau_i}{\left(\tau_i-\sigma_0^2\right)^2} = \frac{\lambda_i(\boldsymbol{\Sigma}_0)\lambda_j(\boldsymbol{\Sigma}_0)}{\left(\lambda_i(\boldsymbol{\Sigma}_0)-\lambda_j(\boldsymbol{\Sigma}_0)\right)^2} \quad \text{for } j > q,$$

which is Anderson's formula for the noise directions. When the $\sigma_i(\boldsymbol{W}_0)$ are repeated this eigenvector-by-eigenvector reading breaks down — individual eigenvectors are then not asymptotically normal — but nothing in Propositions 63 and 64 is affected: no formula in this section divides by a difference of singular values, and the only inverted quantities are τ_i, σ_0^2, $\sigma_i(\boldsymbol{W}_0)$ and $p-q$, all strictly positive under the standing assumptions $\mathrm{rank}(\boldsymbol{W}_0) = q$, $\sigma_0^2 > 0$ and $q < p$. This is the promised quantitative form of the third comment of Remark 39.

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